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FACOLTÀ DI SCIENZE E TECNOLOGIE

Corso di Laurea Magistrale in Matematica

CATEGORICAL GALOIS THEOREM AND APPLICATIONS TO MONOID EXTENSIONS

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Anno Accademico 2023/2024

Preface

Galois theory stands as one of the most profound foundational results in modern algebra, bridging the study of a priori distinct mathematical structures. It provides a framework that connects seemingly separate theories, such as the classical correspondence between group theory and field theory. The framework of Galois-type theories has been extended multiple times. An initial step in this generalization, primarily inspired by the work of Grothendieck, was the extension of the classical subfields-subgroups correspondence to arbitrary K -algebras and to infinite dimensional Galois extensions. The next major step replaced the base field K with a general commutative ring R , leading to the development of a Galois theory for rings. Magid, in [9], systematically developed Grothendieck's Galois theory in full generality for commutative rings. The final step, due to Janelidze [5], was a complete categorical generalization, inspired by the ring-theoretic setting.

This work aims to offer a shift in perspective on the classical and historical process of generalization. Despite the terminology, which maintains a clear connection to the classical Galois theory for fields, we approach and study the Categorical Galois Theorem independently, without explicit reference to the classical theory it generalizes. The categorical analogue of the fundamental theorem of Galois theory is presented as an equivalence of categories - specifically, between particular morphisms in an abstract category and a category of internal presheaves on a specific groupoid. The constructions in categorical Galois theory require the notions of admissible adjunction, which preserve pullbacks of a particular form, and effective descent morphisms, which are morphisms that allow one to move to a more "manageable extension" of an object and then "descend" back smoothly to the original object. For a fixed admissible adjunction, the specific groupoid is determined by the image through the adjunction of the kernel pair of a Galois effective descent morphism. Galois morphisms are also referred to as normal extensions or self-splitting extensions, recovering terminology from classical field theory. Only afterward we show that the classical results of Magid, Grothendieck and Galois himself emerge as specific cases of the categorical theorem, either by considering the algebraic setting of rings or by restricting it to the case of fields. In the latter case, the Galois groupoid is indeed a group, and the internal covariant presheaves on it reduce to a group action on a set.

The power of the categorical approach lies in its versatility: it allows the definition and study of Galois-type theories across different mathematical contexts. Notable examples include the study of central extensions of groups and the interpretation of the covering theory of locally connected topological spaces as a form of Galois theory. One of the objectives of this thesis is to contribute to the study of a Galois theory for monoids. This direction is primarily motivated by the work of Montoli, Rodolo and Van der Linden in [14], where a Galois theory for monoids is developed. We prove that the adjunction between the categories of monoids and groups, related to the Grothendieck group construction [10, 11, 12], is admissible in the sense of categorical Galois theory with respect to the class of surjective homomorphisms. A key result of

this approach is the explicit characterization of central extensions - those particular morphisms that establish the Galois equivalence - as special homogeneous surjections. In particular, our aim is to explicitly describe the Galois groupoid $Gal[\sigma]$ associated with a monoid extension σ . A fundamental feature of special homogeneous surjections, seen as special Schreier surjections, allows us to give this explicit description of $Gal[\sigma]$ in terms of a semidirect product construction. The algebraic structure of semidirect products of monoids, or equivalently, the study of monoid actions $M \rightarrow End(K)$, has been extensively investigated in [15]. It has been shown that such actions correspond to a specific class of split epimorphisms, later formalized as Schreier split epimorphisms in the more recent works [3, 4, 13]. Since normal extensions turn out to be exactly the special homogeneous surjections, we are able to describe the kernel pair of a normal extension $\sigma : M \twoheadrightarrow N$ as a semidirect product of monoids determined by its domain M and its kernel K . Moreover, this construction is preserved by the admissible adjunction: the Galois groupoid $Gal[\sigma]$ can be described as a semidirect product of groups determined by the Grothendieck group of M and K .

The thesis is organized as follows.

In Chapter 1, we introduce the categorical Galois theory in full generality. At first, introducing the notions of effective descent morphisms and internal groupoid and presheaves; then proving the categorical analogue of the fundamental theorem of Galois theory.

In Chapter 2, we apply the Categorical Galois Theorem to the category of rings. In particular, to the Pierce spectrum adjunction between the categories of rings and profinite spaces, that is admissible with respect to all the morphisms. Restricting our focus on fields, we fully recover the classical Galois correspondence.

Finally, in Chapter 3, we study the Galois structure for monoids given by the Grothendieck group construction. We prove that it is admissible with respect to the class of surjective homomorphisms and we characterize central and normal extensions as special homogeneous surjections. Moreover, we give an explicit description of the Galois groupoid as a semidirect product of groups completely determined by the kernel of the fixed monoid extension.

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Chapter 1

The Categorical Galois Theorem

In this chapter, we will prove the Categorical Galois Theorem, which forms the central focus of this thesis. This will lead to several applications discussed in later chapters, ranging from classical results to a Galois theory for monoids. Our approach is primarily based on the exposition presented in *Galois Theories* by Borceux and Janelidze [2], which serves as a foundational reference for both the concepts and the proofs we present. Before diving into the proof, we need to establish some foundational tools, which we will recall and define in the first two sections.

We begin in Section 1.1 with two key adjunctions that will play a crucial role throughout the main part of the theorem. In Section 1.2, we will introduce the concepts of internal categories, groupoids and presheaves. Finally, in Section 1.3, we will present and prove the theorem, relying on the tools developed in the previous sections.

1.1 Preliminary adjunctions

In the study of descent, adjunctions are useful because they formalize the interaction between categories of objects defined over different bases. To prepare for the categorical Galois theorem, we first examine two key adjunctions: $\Sigma_f \dashv f^*$, which describes the relationship between pullback and composition along a morphism, and $F_A \dashv G_A$, which arises naturally from a given adjunction $F \dashv G$. These adjunctions will provide the tools necessary for the theorem. We will not go into all the details, but enough to understand the main ideas we are working with.

We begin by studying the first one: $\Sigma_f \dashv f^*$ for a morphism $f : X \rightarrow Y$ in a category $\mathcal{C}$. This connects the categories $\mathcal{C}/X$ and $\mathcal{C}/Y$ capturing how objects over X and Y relate through the morphism f .

Let $f : X \rightarrow Y$ be a morphism in a category $\mathcal{C}$ with pullbacks. We define the "pullback along f " functor

$$f^* : \mathcal{C}/Y \longrightarrow \mathcal{C}/X, (B, b) \mapsto (X \times_Y B, p_X)$$

where the image of an object (B, b) is given by a chosen pullback:

$$\begin{array}{ccc} X \times_Y B & \xrightarrow{p_B} & B \\ \downarrow p_X & \lrcorner & \downarrow b \\ X & \xrightarrow{f} & Y \end{array}$$

Observation. Note that in defining the functor f^* , we have made a choice for a specific pullback for every (B, b) in $\mathcal{C}/Y$.

The action of f^* on morphisms is defined as follows: given an arrow $g : B \rightarrow C$ in $\mathcal{C}/Y$, $f^*(g)$ is of the form $1_X \times g$, as shown in the diagram below.

$$\begin{array}{ccccc}
 X \times_Y C & \xrightarrow{\quad} & C & & \\
 \downarrow \lrcorner & \searrow 1_X \times g & & \downarrow g & \\
 & X \times_Y B & \xrightarrow{\quad} & B & \\
 & \downarrow \lrcorner & & \downarrow b & \\
 X & \xrightarrow{1_X} & X & \xrightarrow{f} & Y
 \end{array}$$

We now wish to study the monadicity of this functor.

Proposition 1.1.1. *Let $\mathcal{C}$ be a category with pullbacks and $f : X \rightarrow Y$ a morphism in $\mathcal{C}$. The "composition with f " functor*

$$\Sigma_f : \mathcal{C}/X \longrightarrow \mathcal{C}/Y, (A, a) \mapsto (A, f \circ a)$$

is the left adjoint of f^ .*

Proof. For every object (A, a) in $\mathcal{C}/X$ and (B, b) in $\mathcal{C}/Y$ we aim to prove that there is a natural bijection

$$\text{Hom}_{\mathcal{C}/Y}(\Sigma_f(A, a), (B, b)) \cong \text{Hom}_{\mathcal{C}/X}((A, a), f^*(B, b)).$$

For every $h : (A, f \circ a) \rightarrow (B, b)$ in $\mathcal{C}/Y$ and $g : (A, a) \rightarrow (X \times_Y B, p_X)$ in $\mathcal{C}/X$ we define two maps between the hom-sets such that $h \mapsto \langle a, h \rangle$ and $g \mapsto p_B \circ g$, as described in the diagrams below.

$$\begin{array}{ccc}
 \begin{array}{ccccc}
 A & & & & \\
 \swarrow h & & & & \\
 & X \times_Y B & \xrightarrow{p_B} & B & \\
 \swarrow \langle a, h \rangle & \downarrow \lrcorner & & \downarrow b & \\
 a & X & \xrightarrow{f} & Y & \\
 \downarrow p_X & & & &
 \end{array}
 &
 &
 \begin{array}{ccccc}
 A & & & & \\
 \swarrow g & & & & \\
 & X \times_Y B & \xrightarrow{p_B} & B & \\
 \swarrow \langle a, p_B \circ g \rangle & \downarrow \lrcorner & & \downarrow b & \\
 a & X & \xrightarrow{f} & Y & \\
 \downarrow p_X & & & &
 \end{array}
 \end{array}$$

From the universal property of the pullback, we have that $p_B \circ \langle a, h \rangle = h$. Moreover $\langle a, p_B \circ g \rangle$ is the unique morphism for which the triangles commutes in the second diagram. Since g satisfies this property, we conclude that $g = \langle a, p_B \circ g \rangle$. Therefore, we have shown that the two maps are inverses of each other. $\square$

In this setting, the unit and counit of the adjunction are given by $\eta_{(A,a)} = \langle a, 1_A \rangle$ and $\varepsilon_{(B,b)} = p_B$. The next step is to describe the actual form of the monad and what it means to be monadic. We have the monad $\mathbb{T} = (T, \eta, \mu)$ with the composite functor

$$T := f^* \circ \Sigma_f : \mathcal{C}/X \longrightarrow \mathcal{C}/X$$

with the map over X that is $p_1 \circ (1_X \times a)$ as shown from the two pullback squares below.

$$\begin{array}{ccc} X \times_Y A & \xrightarrow{p_A} & A \\ \downarrow 1_X \times a & \lrcorner & \downarrow a \\ X \times_Y X & \xrightarrow{p_2} & X \\ \downarrow p_1 & \lrcorner & \downarrow f \\ X & \xrightarrow{f} & Y \end{array}$$

We deduce also that the unit and the multiplication of the monad are

$$\eta_{(A,a)} = \langle a, 1_A \rangle : A \longrightarrow X \times_Y A$$

and

$$\mu_{(A,a)} = 1_X \times p_A : X \times_Y X \times_Y A \longrightarrow X \times_Y A.$$

Let us describe now the comparison functor associated with this monad:

$$K^T : \mathcal{C}/Y \longrightarrow \mathcal{C}/X^T.$$

We deduce that it sends an object (B, b) of $\mathcal{C}/Y$ into algebras of the form $(X \times_Y B, p_X)$ with a structure morphism defined as

$$f^*(\varepsilon_{(B,b)}) = 1_X \times p_B : X \times_Y X \times_Y B \longrightarrow X \times_Y B$$

Definition 1.1.1. Let $\mathcal{C}$ a category with pullbacks. A morphism $f : X \rightarrow Y$ in $\mathcal{C}$ is an effective descent morphism when the functor "pullback along f "

$$f^* : \mathcal{C}/Y \longrightarrow \mathcal{C}/X$$

is monadic.

The monadicity of f^* implies that K^T is an equivalence of categories. This means that every $\mathbb{T}$ -algebra in $\mathcal{C}/X$ corresponds to an object in $\mathcal{C}/Y$. In other words, the local data over X , governed by the $\mathbb{T}$ -structure, can be reconstructed as global objects over Y .

Concretely, this property ensures the following: for every (A, a) in $\mathcal{C}/X$ equipped with a structure morphism $v : X \times_Y A \rightarrow A$, there exists an object (B, b) in $\mathcal{C}/Y$ such that the $\mathbb{T}$ -algebra $(X \times_Y B, 1_X \times p_B)$ is isomorphic to (A, v) . This is why such morphisms are called of effective descent: they establish a correspondence between local and global perspectives, generalizing the process of transitionning between different bases.

Now, let us start the second part of this section and move to the second adjunction, $F_A \dashv G_A$, which arises naturally from an adjunction $F \dashv G$.

Lemma 1.1.2. Consider an adjunction $\mathcal{A} \begin{matrix} \xrightarrow{F} \\ \perp \\ \xleftarrow{G} \end{matrix} \mathcal{B}$. Assume that $\mathcal{A}$ has pullbacks. For every $A \in \text{Ob}(\mathcal{A})$ the functor

$$F_A : \mathcal{A}/A \longrightarrow \mathcal{B}/F(A), (X, \alpha) \rightarrow (F(X), F(\alpha))$$

has a right adjoint functor

$$G_A : \mathcal{B}/F(A) \longrightarrow \mathcal{A}/A, (Y, \beta) \rightarrow (Z, \gamma)$$

where (Z, γ) is defined by a chosen pullback

$$\begin{array}{ccc} Z & \xrightarrow{\delta} & G(Y) \\ \downarrow \gamma & \lrcorner & \downarrow G(\beta) \\ A & \xrightarrow{\eta_A} & GF(A) \end{array}$$

in which η is the unit of the adjunction $F \dashv G$.

Observation. Observe that the functor G_A is related to the pullback along the unit η_A of the adjunction $F \dashv G$. Specifically, G_A is the composition of the restriction of G over $\mathcal{B}/F(A)$ and the functor $\eta_A^* : \mathcal{A}/GF(A) \longrightarrow \mathcal{A}/A$.

Proof. For every object (X, α) in $\mathcal{A}/A$ and (Y, β) in $\mathcal{B}/F(A)$, we aim to prove that there is a natural bijection

$$\text{Hom}_{\mathcal{B}/F(A)}(F_A(X, \alpha), (Y, \beta)) \cong \text{Hom}_{\mathcal{A}/A}((X, \alpha), G_A(Y, \beta)).$$

From the original adjunction $F \dashv G$, we have a natural bijection

$$\text{Hom}_{\mathcal{B}/F(A)}(F_A(X, \alpha), (Y, \beta)) \cong \text{Hom}_{\mathcal{A}/GF(A)}((X, \eta_A \circ \alpha), (G(Y), G(\beta))).$$

It is obvious now how we can use the previous result: if η is the unit of $F \dashv G$, from Proposition 1.1.1, the adjunction $\Sigma_{\eta_A} \dashv \eta_A^*$ ensures that

$$\text{Hom}_{\mathcal{A}/GF(A)}((X, \eta_A \circ \alpha), (G(Y), G(\beta))) \cong \text{Hom}_{\mathcal{A}/A}((X, \alpha), \eta_A^*(G(Y), G(\beta))).$$

With the observation made before we have that $\eta_A^*G(Y, \beta) = G_A(Y, \beta)$ establishing the desired isomorphism. $\square$

1.2 Internal groupoids and internal presheaves

Let us introduce now the notion of internal category and internal groupoid in a category with pullbacks. Before we define it, let us first introduce this topic informally.

Let S_0 and S_1 be objects in a category with pullbacks. If we think of S_0 as a class of objects and S_1 as a class of arrows, if they also have a "domain" and a "codomain" arrows and the "identity" arrow along with some properties expressed as commutativity of diagrams, we can in fact consider these data as a category whose *Ob* and *Arr* classes are objects of the original

category. The classic example will be any small category that could be seen as an internal category in the category **Set** of sets.

The concept of internal structure is the same if we consider a groupoid instead of just a category, so that an internal groupoid will be as a groupoid whose classes Ob and Arr are objects of any category with pullbacks.

Let us now state the formal definition of an internal groupoid. For more clarity, we will describe step by step the interpretation of every axiom that explains how it actually defines a groupoid.

Definition 1.2.1. Let $\mathcal{A}$ be a category with pullbacks. An internal groupoid in $\mathcal{A}$ consists in a diagram

$$\begin{array}{ccccc}
 & & & \xrightarrow{d_0} & \\
 S_1 \times_{S_0} S_1 & \xrightarrow{m} & S_1 & \xleftarrow{n} & S_0 \\
 & & \downarrow \tau & & \downarrow d_1
 \end{array}$$

with S_0 and S_1 objects of $\mathcal{A}$ that represent (respectively) the objects and the arrows of the internal category, d_0 and d_1 represent (respectively) the domain and codomain of an arrow, n the identities over the objects, m the composition of two arrows and τ that represents the inversion of an arrow. We also require the following conditions:

(G1) $S_1 \times_{S_0} S_1$ is defined by a pullback

$$\begin{array}{ccc}
 S_1 \times_{S_0} S_1 & \xrightarrow{p_0} & S_1 \\
 \downarrow p_1 & \lrcorner & \downarrow d_1 \\
 S_1 & \xrightarrow{d_0} & S_0
 \end{array}$$

so that $S_1 \times_{S_0} S_1$ is defined to represent composable pairs of arrows, where the codomain of the first one is equal to the domain of the second one.

(G2) The triangles

$$\begin{array}{ccc}
 S_0 & \xrightarrow{n} & S_1 \\
 \parallel & & \downarrow d_0 \\
 & 1_{S_0} & S_0
 \end{array}
 \qquad
 \begin{array}{ccc}
 S_0 & \xrightarrow{n} & S_1 \\
 \parallel & & \downarrow d_1 \\
 & 1_{S_0} & S_0
 \end{array}$$

are commutative to define the identity morphism n so that the domain and codomain of the identity over an object are the object itself.

(G3) The squares

$$\begin{array}{ccc}
 S_1 \times_{S_0} S_1 & \xrightarrow{m} & S_1 \\
 \downarrow p_0 & & \downarrow d_0 \\
 S_1 & \xrightarrow{d_0} & S_0
 \end{array}
 \qquad
 \begin{array}{ccc}
 S_1 \times_{S_0} S_1 & \xrightarrow{m} & S_1 \\
 \downarrow p_1 & & \downarrow d_1 \\
 S_1 & \xrightarrow{d_1} & S_0
 \end{array}$$

are commutative to define the composition morphism m so that the domain of the composition is indeed the domain of the first component and dually the codomain is the one of the second component.

(G4) The triangles

$$\begin{array}{ccc}
 S_1 & \xrightarrow{\langle n \circ d_1, 1_{S_1} \rangle} & S_1 \times_{S_0} S_1 \\
 \searrow 1_{S_1} & & \downarrow m \\
 & & S_1
 \end{array}
 \qquad
 \begin{array}{ccc}
 S_1 & \xrightarrow{\langle 1_{S_1}, n \circ d_0 \rangle} & S_1 \times_{S_0} S_1 \\
 \searrow 1_{S_1} & & \downarrow m \\
 & & S_1
 \end{array}$$

are commutative. With this we ensure that the identity is the unit of the composition.

(G5) Considering the pullback in the first diagram

$$\begin{array}{ccc}
 S_3 & \xrightarrow{\pi_0} & S_1 \times_{S_0} S_1 \\
 \pi_1 \downarrow \lrcorner & & \downarrow p_1 \\
 S_1 \times_{S_0} S_1 & \xrightarrow{p_0} & S_1
 \end{array}
 \qquad
 \begin{array}{ccc}
 S_3 & \xrightarrow{\langle p_0 \circ \pi_0, m \circ \pi_1 \rangle} & S_1 \times_{S_0} S_1 \\
 \downarrow \langle m \circ \pi_0, p_1 \circ \pi_1 \rangle & & \downarrow m \\
 S_1 \times_{S_0} S_1 & \xrightarrow{m} & S_1
 \end{array}$$

the second square is commutative. This axiom introduces S_3 , which can be interpreted as pairs of pairs of composable arrows where the second and third arrows coincide ($p_1 \circ \pi_0 = p_0 \circ \pi_1$). Consequently, these pairs can also be viewed as triplets of composable arrows, with the commutativity of the second square requiring the associativity of the composition.

(G6) The triangles

$$\begin{array}{ccc}
 S_1 & \xrightarrow{\tau} & S_1 \\
 & \searrow d_1 & \downarrow d_0 \\
 & & S_0
 \end{array}
 \qquad
 \begin{array}{ccc}
 S_1 & \xrightarrow{\tau} & S_1 \\
 & \searrow d_0 & \downarrow d_1 \\
 & & S_0
 \end{array}$$

are commutative to define the inverse morphism τ so that the inverse of an arrow has switched domain and codomain.

(G7) The squares

$$\begin{array}{ccc}
 S_1 & \xrightarrow{\langle 1_{S_1}, \tau \rangle} & S_1 \times_{S_0} S_1 \\
 d_1 \downarrow & & \downarrow m \\
 S_0 & \xrightarrow{n} & S_1
 \end{array}
 \qquad
 \begin{array}{ccc}
 S_1 & \xrightarrow{\langle \tau, 1_{S_1} \rangle} & S_1 \times_{S_0} S_1 \\
 d_0 \downarrow & & \downarrow m \\
 S_0 & \xrightarrow{n} & S_1
 \end{array}$$

are commutative to ensure that the inverse of an arrow is indeed the one that composed with the original one (in both ways) gives the identity over the domain or over the codomain.

Observation. Note that we require τ , **(G6)** and **(G7)** only to define an internal groupoid. Without τ and these two axioms we have just the definition of an internal category.

Example 1.2.2. Let us consider a small category $\mathcal{S}$. We have that $Ob(\mathcal{S})$ and $Arr(\mathcal{S})$ are sets. It is quite intuitive that we can see $\mathcal{S}$ as an internal category in **Set**.

$$\{(g, f) \in Arr(\mathcal{S})^2 : dom(g) = cod(f)\} \xrightarrow{- \circ -} Arr(\mathcal{S}) \begin{array}{c} \xrightarrow{dom(_)} \\ \xleftarrow{1} \\ \xrightarrow{cod(_)} \end{array} Ob(\mathcal{S})$$

Of course it satisfies all the axioms **(G1)** - **(G5)** thanks to the definition of $\mathcal{S}$ as a category:

(G1) The set of composable arrows $S_2 := \{(g, f) \in Arr(\mathcal{S})^2 : dom(g) = cod(f)\}$ is the pullback in **Set** of $dom(_)$ and $cod(_)$.

(G2) $dom(1_A) = A$ and $cod(1_A) = A$ for all $A \in Ob(\mathcal{S})$.

(G3) $dom(g \circ f) = dom(f)$ and $cod(g \circ f) = cod(g)$ for all composable pair (f, g) .

(G4) $1_B \circ f = f$ and $f \circ 1_A = f$ for all $f : A \rightarrow B$ in $Arr(\mathcal{S})$.

(G5) The pullback S_3 is $\{((h, g), (g, f)) \in S_2^2\}$ or the set of composable triplets and then we have the requirement from the second diagram: the associativity $h \circ (g \circ f) = (h \circ g) \circ f$.

If then $\mathcal{S}$ is a groupoid, i.e. all morphisms are isomorphisms, then for each $f \in Arr(\mathcal{S})$ we have $f^{-1} \in Arr(\mathcal{S})$ such that $dom(f^{-1}) = cod(f)$ and $cod(f^{-1}) = dom(f)$, as **(G6)** states, and $f \circ f^{-1} = 1_{cod(f)}$ and $f^{-1} \circ f = 1_{dom(f)}$ as **(G7)** requires.

Definition 1.2.3. Let $\mathcal{A}$ be a category with pullbacks and $\mathcal{S}$ an internal category in $\mathcal{A}$. An internal covariant presheaf on the internal category $\mathcal{S}$ is a triple (P, p, π) where

- (i) P is an object of $\mathcal{A}$
- (ii) $p : P \rightarrow S_0$ is a morphism in $\mathcal{A}$
- (iii) $\pi : S_1 \times_{S_0} P \rightarrow P$ is a morphism in $\mathcal{A}$ such that the following diagram commutes:

$$\begin{array}{ccccc} S_1 \times_{S_0} S_1 \times_{S_0} P & \xrightarrow{1_{S_1} \times \pi} & S_1 \times_{S_0} P & \xleftarrow{\langle n \circ p, 1_P \rangle} & P \\ m \times 1_P \downarrow & & \downarrow \pi & \nearrow & \\ S_1 \times_{S_0} P & \xrightarrow{\pi} & P & & \\ \alpha_1 \downarrow & & \downarrow p & & \\ S_1 & \xrightarrow{d_1} & S_0 & & \end{array} \quad (1.1)$$

where both $S_1 \times_{S_0} P$ are defined by the pullback of p along d_0 , and $S_1 \times_{S_0} S_1 \times_{S_0} P$ is defined by the pullback of p along $d_0 \circ m$.

Example 1.2.4. To get an intuition of this notion, let us describe the setting with $\mathcal{A} = \mathbf{Set}$ and $\mathcal{S}$ a small category. In Example 1.2.2, we have seen $\mathcal{S}$ as an internal category in $\mathbf{Set}$. We have $S_0 := \text{Ob}(\mathcal{S})$ and $S_1 := \text{Arr}(\mathcal{S})$ and m and n , respectively, the composition and the identity map in $\mathcal{S}$. We wish to show how Definition 1.2.3 gives us a classical covariant presheaf on the small category $\mathcal{S}$.

We wish to construct a functor $\mathcal{P} : \mathcal{S} \rightarrow \mathbf{Set}$ from an internal covariant presheaf (P, p, π) . Fixing the map $p : P \rightarrow S_0$ is equivalent to defining the family $(p^{-1}(s))_{s \in S_0}$, so that we can define $\mathcal{P}(s) := p^{-1}(s)$. In this setting, one has

$$S_1 \times_{S_0} P = \{(f, a) \in S_1 \times P : p(a) = \text{dom}(f)\} = \{(f, a) \in S_1 \times P : a \in p^{-1}(\text{dom}(f))\}.$$

Fixing the map $\pi : S_1 \times_{S_0} P \rightarrow P$ is equivalent to defining, for each pair (f, a) such that $a \in \mathcal{P}(\text{dom}(f))$, an element $\pi(f, a) \in P$. The commutativity of the bottom square in Diagram (1.1) shows that $p(\pi(f, a)) = \text{cod}(f)$, so that $\pi(f, a) \in p^{-1}(\text{cod}(f)) = \mathcal{P}(\text{cod}(f))$. Thus, we can define a map $\mathcal{P}(f) : \mathcal{P}(\text{dom}(f)) \rightarrow \mathcal{P}(\text{cod}(f))$ such that $\mathcal{P}(f)(a) = \pi(f, a)$. It remains to prove the axioms for the compatibility of $\mathcal{P}$ with compositions and identities. The commutativity of the top square in Diagram (1.1) gives us the composition axiom:

$$\mathcal{P}(m(g, f))(a) = \pi(m(g, f), a) = \pi(g, \pi(f, a)) = \mathcal{P}(g)(\mathcal{P}(f)(a)).$$

Moreover, for every $s \in S_0$ and $a \in p^{-1}(s) = \mathcal{P}(s)$, one states the identity axiom from the commutativity of the triangle in Diagram (1.1):

$$\mathcal{P}(1_s)(a) = \pi(1_s, a) = a = 1_{\mathcal{P}(s)}(a).$$

Definition 1.2.5. Let $\mathcal{A}$ be a category with pullbacks and $\mathcal{S}$ an internal category in $\mathcal{A}$. Given two internal covariant presheaves (P, p, π) and (P', p', π') on $\mathcal{S}$, an internal natural transformation between them

$$\gamma : (P, p, \pi) \Rightarrow (P', p', \pi')$$

is a morphism $\gamma : P \rightarrow P'$ such that the following diagram commutes:

$$\begin{array}{ccccc} S_1 \times_{S_0} P & \xrightarrow{\pi} & P & & \\ 1_{S_1} \times \gamma \downarrow & & \downarrow \gamma & \searrow p & \\ S_1 \times_{S_0} P' & \xrightarrow{\pi'} & P' & \xrightarrow{p'} & S_0 \end{array} \quad (1.2)$$

Example 1.2.6. Let us continue our example with $\mathcal{S}$ a small category seen as an internal category in $\mathbf{Set}$. An internal natural transformation between two internal covariant presheaves is indeed a natural transformation between the functors defined in Example 1.2.4.

Note that for every $s \in S_0$, a map $\gamma : P \rightarrow P'$ such that $p = p' \circ \gamma$ splits into a family of maps $\gamma_s : p^{-1}(s) \rightarrow p'^{-1}(s)$. Moreover, the commutativity of the square in Diagram (1.2) gives us the naturality axiom: for every $f : s \rightarrow r$ and every $a \in \mathcal{P}(s) = p^{-1}(s)$, one has:

$$\gamma_r(\mathcal{P}(f)(a)) = \gamma(\pi(f, a)) = \pi'(f, \gamma(a)) = \mathcal{P}'(f)(\gamma_s(a)).$$

We will denote by $\mathcal{A}^{\mathcal{S}}$ the category of internal covariant presheaves on $\mathcal{S}$ with internal natural transformations as arrows.

Proposition 1.2.1. *Let $\mathcal{A}$ be a category with pullbacks and $\mathcal{S}$ an internal category in $\mathcal{A}$. The forgetful functor*

$$\mathcal{A}^{\mathcal{S}} \longrightarrow \mathcal{A}/S_0, (P, p, \pi) \mapsto (P, p)$$

is monadic. Moreover, the functorial part of the monad is the composite

$$\mathcal{A}/S_0 \xrightarrow{d_0^*} \mathcal{A}/S_1 \xrightarrow{\Sigma_{d_1}} \mathcal{A}/S_0$$

where d_0^ is the "pullback along d_0 " functor and Σ_{d_1} is the "composition with d_1 " functor.*

Proof. Let us call T the composite $\Sigma_{d_1} \circ d_0^*$ and show that it is part of a monad on $\mathcal{A}/S_0$. For every (A, a) in $\mathcal{A}/S_0$, define $\eta_{(A,a)} := \langle n \circ a, 1_A \rangle$ and $\mu_{(A,a)} := m \times 1_A$. One has that the axioms for (T, η, μ) to be a monad on $\mathcal{A}/S_0$ arise from the unit axiom of n and the associativity of m (axioms **(G4)** and **(G5)** of Definition 1.2.1).

An algebra for this monad is then a pair (A, a) in $\mathcal{A}/S_0$ along with a structure morphism $v : (S_1 \times_{S_0} A, d_1 \circ p_{S_1}) \longrightarrow (A, a)$ such that the following diagram commutes:

$$\begin{array}{ccccc} S_1 \times_{S_0} S_1 \times_{S_0} A & \xrightarrow{1_{S_1} \times v} & S_1 \times_{S_0} A & \xleftarrow{\langle n \circ a, 1_A \rangle} & A \\ \downarrow m \times 1_A & & \downarrow v & \nearrow & \\ S_1 \times_{S_0} A & \xrightarrow{v} & A & & \\ \downarrow p_{S_1} & & \downarrow a & & \\ S_1 & \xrightarrow{d_1} & S_0 & & \end{array}$$

It is obvious that the requirements for a triple (A, a, v) to be an algebra for the monad (T, η, μ) coincide with the requirements of Definition 1.2.3 for (A, a, v) to be an internal covariant presheaf on $\mathcal{S}$. $\square$

Let us now introduce a specific kind of internal groupoid in a category with pullbacks that will be the one we will consider in the next section.

Example 1.2.7. Let $\mathcal{A}$ be a category with pullbacks and $\sigma : S \rightarrow R$ a morphism in $\mathcal{A}$.

Consider the kernel pair $S \times_R S \xrightleftharpoons[p_2]{p_1} S \xrightarrow{\sigma} R$ of σ . From the universal property of the pullback, one has the diagonal map $\Delta := \langle 1_S, 1_S \rangle : S \rightarrow S \times_R S$ and the twisting isomorphism $\tau := \langle p_2, p_1 \rangle : S \times_R S \rightarrow S \times_R S$. Moreover, consider the pullback

$$\begin{array}{ccc} (S \times_R S) \times_S (S \times_R S) & \xrightarrow{p_3} & S \times_R S \\ \downarrow p_4 & \lrcorner & \downarrow p_2 \\ S \times_R S & \xrightarrow{p_1} & S \end{array}$$

and denote $\pi_1 := p_1 \circ p_3$ and $\pi_4 := p_2 \circ p_4$. One has the unique morphism

$$\langle \pi_1, \pi_4 \rangle : (S \times_R S) \times_S (S \times_R S) \rightarrow S \times_R S.$$

It is just a computation to show that these choices satisfy all the axioms of Definition 1.2.1. Thus, we have the following groupoid $K(\sigma)$, induced by the kernel pair of $\sigma : S \rightarrow R$.

$$(S \times_R S) \times_S (S \times_R S) \xrightarrow{\langle \pi_1, \pi_4 \rangle} S \times_R S \begin{array}{c} \xleftarrow{\Delta} S \\ \xrightarrow{p_1} S \\ \xrightarrow{p_2} S \end{array}$$

τ

Observation. This is just a reformulation of the fact that the kernel pair of σ is an equivalence relation on S . Let us show that in the case $\mathcal{A} = \mathbf{Set}$. In this setting, one has that $S \times_R S$ is the set of pairs (s, r) such that $\sigma(s) = \sigma(r)$ and

$$\langle \pi_1, \pi_4 \rangle((s, r), (r, h)) = (s, h)$$

is just the projection on the first and fourth element. The groupoid structure indeed expresses the reflexivity, symmetry, and transitivity of the relation $\equiv_\sigma$ induced by σ :

$$s \equiv_\sigma r \iff \sigma(s) = \sigma(r),$$

through, respectively, the maps Δ , τ , and $\langle \pi_1, \pi_4 \rangle$.

1.3 The Galois Theorem

With all the needed prerequisites recalled, let us introduce the actual Galois theorem. We will need to restrict our work to some "good enough" adjunctions and, in that setting, to use the previous instruments to prove the main theorem.

Definition 1.3.1. Let $\mathcal{A}$ be a category. A class $\overline{\mathcal{A}}$ of arrows is admissible when

- (i) every isomorphism is in $\overline{\mathcal{A}}$;
- (ii) $\overline{\mathcal{A}}$ is closed under composition;

(iii) if $a, b \in \overline{\mathcal{A}}$, considering a pullback

$$\begin{array}{ccc} \bullet & \xrightarrow{c} & \bullet \\ d \downarrow & \lrcorner & \downarrow b \\ \bullet & \xrightarrow{a} & \bullet \end{array}, \text{ one has also } c, d \in \overline{\mathcal{A}}.$$

For an object A in $\mathcal{A}$, we will write $\overline{\mathcal{A}}/A$ for the full subcategory of $\mathcal{A}/A$ in which the objects are admissible arrows. So, pairs (X, f) , as in $\mathcal{A}/A$, where $f : X \rightarrow A \in \overline{\mathcal{A}}$.

Definition 1.3.2. A relatively admissible adjunction consists in

(i) an adjunction $\mathcal{A} \begin{array}{c} \xrightarrow{\mathcal{S}} \\ \perp \\ \xleftarrow{\mathcal{C}} \end{array} \mathcal{P}$;

(ii) two admissible classes of arrows $\overline{\mathcal{A}}$ and $\overline{\mathcal{P}}$ such that

- for all $f \in \text{Arr}(\mathcal{A})$, $f \in \overline{\mathcal{A}} \implies \mathcal{S}(f) \in \overline{\mathcal{P}}$
- for all $g \in \text{Arr}(\mathcal{P})$, $g \in \overline{\mathcal{P}} \implies \mathcal{C}(g) \in \overline{\mathcal{A}}$
- for all $A \in \text{Ob}(\mathcal{A})$, $\eta_A \in \overline{\mathcal{A}}$, where η is the unit of $\mathcal{S} \dashv \mathcal{C}$
- for all $X \in \text{Ob}(\mathcal{P})$, $\varepsilon_X \in \overline{\mathcal{P}}$, where ε is the counit of $\mathcal{S} \dashv \mathcal{C}$.

We will write $(\mathcal{A}, \overline{\mathcal{A}}) \overset{\mathcal{S}}{\underset{\mathcal{C}}{\perp}} (\mathcal{P}, \overline{\mathcal{P}})$ for a relatively admissible adjunction.

The conditions specified in Definitions 1.3.1 and 1.3.2 clearly lead to the following two lemmas.

Lemma 1.3.1. *Let $\mathcal{A}$ be an admissible class of arrows in a category $\mathcal{A}$ with pullbacks. For every morphism $\sigma : S \rightarrow R$ in $\overline{\mathcal{A}}$, the adjunction $\Sigma_\sigma \dashv \sigma^*$ of Proposition 1.1.1 restricts to an adjunction*

$$\overline{\mathcal{A}}/R \overset{\sigma^*}{\underset{\Sigma_\sigma}{\dashv}} \overline{\mathcal{A}}/S$$

Proof. Just note from Definition 1.3.1 that $\overline{\mathcal{A}}$ is closed under composition and pullbacks. If $g : X \rightarrow S$ is in $\overline{\mathcal{A}}$, the composition $\sigma \circ g$ is again in $\overline{\mathcal{A}}$. Similarly, if $f : X \rightarrow R$ is in $\overline{\mathcal{A}}$, the pullback along σ gives us that $p_S : S \times_R X \rightarrow S$ is again in $\overline{\mathcal{A}}$. $\square$

Lemma 1.3.2. *Let $(\mathcal{A}, \overline{\mathcal{A}}) \overset{\mathcal{S}}{\underset{\mathcal{C}}{\perp}} (\mathcal{P}, \overline{\mathcal{P}})$ be a relatively admissible adjunction where $\mathcal{A}$ is a category with pullbacks. For every $A \in \text{Ob}(\mathcal{A})$ the adjunction between $\mathcal{A}/A$ and $\mathcal{P}/\mathcal{S}(A)$ obtained by Lemma 1.1.2 restricts to an adjunction*

$$\overline{\mathcal{A}}/A \overset{\mathcal{S}_A}{\underset{\mathcal{C}_A}{\dashv}} \overline{\mathcal{P}}/\mathcal{S}(A)$$

Proof. Recall that $\mathcal{S}_A$ acts as $\mathcal{S}$ on the slice category and $\mathcal{C}_A$ is the composition of $\mathcal{C}$ and η_A^* as seen in Lemma 1.1.2. For every $f \in \overline{\mathcal{A}}$ and $g \in \overline{\mathcal{P}}$, Definition 1.3.2 give us that $\mathcal{S}(f) \in \overline{\mathcal{P}}$, $\mathcal{C}(g) \in \overline{\mathcal{A}}$ and $\eta_A \in \overline{\mathcal{A}}$. Again from Definition 1.3.1, the pullback of $\mathcal{C}(g)$ along η_A is in the admissible class $\overline{\mathcal{A}}$. $\square$

We have established that the adjunctions from Section 1.1 can be restricted to the admissible classes. Now, we introduce a specific property for morphisms in $\overline{\mathcal{A}}$ that will play a crucial role in the Galois Theorem. Specifically, we consider morphisms that establish a meaningful link between the two adjunctions involved. To achieve this, we will require the pullback functor to be monadic, ensuring a smooth transition when shifting perspectives between objects over R and objects over S . Moreover, our focus will be on objects in $\overline{\mathcal{A}}/R$ whose pullback is in a full subcategory of $\overline{\mathcal{A}}/S$, which is equivalent to $\overline{\mathcal{P}}/\mathcal{S}(S)$ through the adjunction $\mathcal{S}_S \dashv \mathcal{C}_S$.

Definition 1.3.3. Let $\overline{\mathcal{A}}$ be an admissible class of arrows in a category $\mathcal{A}$ with pullbacks. An arrow $\sigma : S \rightarrow R$ is an effective descent morphism relatively to $\overline{\mathcal{A}}$ if

- (i) $\sigma \in \overline{\mathcal{A}}$;
- (ii) the functor $\sigma^* : \overline{\mathcal{A}}/R \longrightarrow \overline{\mathcal{A}}/S$ is monadic.

Definition 1.3.4. Let $(\mathcal{A}, \overline{\mathcal{A}}) \begin{array}{c} \xrightarrow{\mathcal{S}} \\ \perp \\ \xleftarrow{\mathcal{C}} \end{array} (\mathcal{P}, \overline{\mathcal{P}})$ be a relatively admissible adjunction, where $\mathcal{A}$ is a category with pullbacks. An object $(A, a) \in \text{Ob}(\overline{\mathcal{A}}/R)$ is split by a morphism $\sigma : S \rightarrow R$ of $\overline{\mathcal{A}}$ when the unit

$$\eta_{\sigma^*(A, a)}^S : \sigma^*(A, a) \longrightarrow \mathcal{C}_S \mathcal{S}_S(\sigma^*(A, a))$$

of $\mathcal{S}_S \dashv \mathcal{C}_S$ is an isomorphism at the object $\sigma^*(A, a)$.

We will write $\text{Split}_R(\sigma)$ for the full subcategory of $\overline{\mathcal{A}}/R$ of those objects split by $\sigma : S \rightarrow R$.

Definition 1.3.5. Let $(\mathcal{A}, \overline{\mathcal{A}}) \begin{array}{c} \xrightarrow{\mathcal{S}} \\ \perp \\ \xleftarrow{\mathcal{C}} \end{array} (\mathcal{P}, \overline{\mathcal{P}})$ be a relatively admissible adjunction where $\mathcal{A}$ and $\mathcal{P}$ are categories with pullbacks. A morphism $\sigma : S \rightarrow R$ is of relative Galois descent with respect to these data when

- (i) σ is an effective descent morphism relatively to $\overline{\mathcal{A}}$;
- (ii) the counit ε^S of the adjunction $\mathcal{S}_S \dashv \mathcal{C}_S$ is an isomorphism at every object;
- (iii) the object $\Sigma_\sigma \mathcal{C}_S(X, f)$ is split by σ , for every (X, f) object of $\overline{\mathcal{P}}/\mathcal{S}(S)$.

From now on and for the rest of this chapter, we will be in the conditions of Definition 1.3.5. Let us highlight the key components of the setup: we fix a morphism $\sigma : S \rightarrow R$ of relative Galois descent with respect to a relatively admissible adjunction between categories with pullbacks. We have the following two adjunctions that arise from these data:

$$\begin{array}{ccc} \overline{\mathcal{A}}/R & \begin{array}{c} \xrightarrow{\sigma^*} \\ \top \\ \xleftarrow{\Sigma_\sigma} \end{array} & \overline{\mathcal{A}}/S \\ & & \begin{array}{c} \xrightarrow{\mathcal{S}_S} \\ \perp \\ \xleftarrow{\mathcal{C}_S} \end{array} \end{array} \overline{\mathcal{P}}/\mathcal{S}(S) \quad (1.3)$$

We will denote by η^S and ε^S the unit and counit of the adjunction $\mathcal{S}_S \dashv \mathcal{C}_S$.

Lemma 1.3.3. *The following conditions are equivalent for an object (A, a) of $\overline{\mathcal{A}}/S$:*

1. (A, a) is split by $1_S : S \rightarrow S$.
2. $\eta_{(A, a)}^S : (A, a) \longrightarrow \mathcal{C}_S \mathcal{S}_S(A, a)$ is an isomorphism.
3. $(A, a) \cong \mathcal{C}_S(X, f)$ for some object (X, f) of $\overline{\mathcal{P}}/\mathcal{S}(S)$.

Proof. 1. $\iff$ 2. It is just Definition 1.3.4 if we recall that $(A, a) \cong 1_S^*(A, a)$.

2. $\Rightarrow$ 3. Just choose $(X, f) := \mathcal{S}_S(A, a)$; then $\eta_{(A, a)}^S : (A, a) \longrightarrow \mathcal{C}_S(X, f)$ is the desired isomorphism.

3. $\Rightarrow$ 2. For every (A, a) in $\overline{\mathcal{A}}/S$ there is (X, f) in $\overline{\mathcal{P}}/\mathcal{S}(S)$ such that $(A, a) \cong \mathcal{C}_S(X, f)$. From Definition 1.3.5, one has that $\varepsilon_{(X, f)}^S$ is an isomorphism and so is $\mathcal{C}_S(\varepsilon_{(X, f)}^S)$. Using the triangular identity

$$\mathcal{C}_S(\varepsilon_{(X, f)}^S) \circ \eta_{\mathcal{C}_S(X, f)}^S = 1_{\mathcal{C}_S(X, f)},$$

we deduce that also η^S is an isomorphism $\mathcal{C}_S(X, f) \cong (A, a)$. □

Corollary 1.3.3.1. *The following conditions are equivalent for an object (A, a) of $\overline{\mathcal{A}}/R$:*

1. (A, a) is split by $\sigma : S \rightarrow R$.
2. $\sigma^*(A, a)$ is split by $1_S : S \rightarrow S$.
3. $\sigma^*(A, a) \cong \mathcal{C}_S(X, f)$ for some object (X, f) of $\overline{\mathcal{P}}/\mathcal{S}(S)$.

Lemma 1.3.4. *Consider the adjunction $\Sigma_\sigma \dashv \sigma^*$ in Diagram (1.3); it restricts to an adjunction*

$$\begin{array}{ccc} & \sigma^* & \\ \text{Split}_R(\sigma) & \xrightarrow{\quad} & \text{Split}_S(1_S) \\ & \Sigma_\sigma & \end{array} \quad .$$

Moreover, the functor

$$\sigma^* : \text{Split}_R(\sigma) \longrightarrow \text{Split}_S(1_S)$$

is monadic.

Proof. If (A, a) is split by $1_S : S \rightarrow S$, by Lemma 1.3.3 we have $(A, a) \cong \mathcal{C}_S(X, f)$ for some (X, f) in $\overline{\mathcal{P}}/\mathcal{S}(S)$. By Definition 1.3.5, $\Sigma_\sigma \mathcal{C}_S(X, f) \cong \Sigma_\sigma(A, a)$ is split by σ . On the other hand, by Corollary 1.3.3.1, if (A, a) is split by σ , then $\sigma^*(A, a)$ is split by 1_S , so that the restriction is well defined.

We will prove the monadicity with Beck's criterion (see [8], Chapter V): since $\sigma^* : \overline{\mathcal{A}}/R \longrightarrow \overline{\mathcal{A}}/S$ is monadic, it reflects isomorphisms, thus the same holds for its restriction to the full subcategory of split objects. Moreover, consider two morphisms $(B, b) \xrightleftharpoons[u]{u} (A, a)$ in $\text{Split}_R(\sigma)$ such that the following is a split coequalizer of $\sigma^*(u)$ and $\sigma^*(v)$ in $\text{Split}_S(1_S)$:

$$\begin{array}{ccccc} & \xleftarrow{t} & & \xleftarrow{r} & \\ & \sigma^*(u) & & h & \\ \sigma^*(B, b) & \xrightleftharpoons[\sigma^*(v)]{\quad} & \sigma^*(A, a) & \xrightarrow{\quad} & (Q, q) \end{array}$$

This split coequalizer is also a split coequalizer in $\overline{\mathcal{A}}/S$, so that the Beck's criterion in the case of $\sigma^* : \overline{\mathcal{A}}/R \longrightarrow \overline{\mathcal{A}}/S$ implies the existence of a coequalizer (C, c) of u and v in $\overline{\mathcal{A}}/R$ which is preserved by σ^* ; thus, $\sigma^*(C, c) \cong (Q, q)$ and by Corollary 1.3.3.1 we obtain that (C, c) is indeed in $\text{Split}_R(\sigma)$. □

Lemma 1.3.5. *Consider the adjunction $\mathcal{S}_S \dashv \mathcal{C}_S$ in Diagram (1.3); it restricts to an equivalence of categories*

$$\text{Split}_S(1_S) \approx \overline{\mathcal{P}}/\mathcal{S}(S)$$

Proof. If (X, f) is an object in $\overline{\mathcal{P}}/\mathcal{S}(S)$, by Lemma 1.3.3 we have that $\mathcal{C}_S(X, f)$ is split by 1_S , so that the restriction is well defined. Moreover, η^S is an isomorphism on split objects and the counit ε^S is an isomorphism from Definition 1.3.5. $\square$

Corollary 1.3.5.1. *The functor*

$$\mathcal{S}_S \circ \sigma^* : \text{Split}_R(\sigma) \longrightarrow \overline{\mathcal{P}}/\mathcal{S}(S)$$

is monadic.

Proof. It is the composite of an equivalence of categories with a monadic functor as shown in Lemma 1.3.4 and Lemma 1.3.5. $\square$

The relative Galois descent property of σ allowed us to restrict our work on split objects while maintaining the monadicity of σ^* . Thus, we can describe the setting with the following diagram:

$$\begin{array}{ccccc}
 & \xrightarrow{\sigma^*} & & & \\
 \overline{\mathcal{A}}/R & \xrightarrow{\quad \top \quad} & \overline{\mathcal{A}}/S & \xrightarrow{\mathcal{S}_S} & \overline{\mathcal{P}}/\mathcal{S}(S) \\
 & \xleftarrow{\Sigma_\sigma} & & \nwarrow \mathcal{C}_S & \\
 & & & & \nearrow \approx \\
 & \xrightarrow{\sigma^*} & & & \\
 \text{Split}_R(\sigma) & \xrightarrow{\quad \top \quad} & \text{Split}_S(1_S) & \xrightarrow{\mathcal{S}_S} & \overline{\mathcal{P}}/\mathcal{S}(S) \\
 & \xleftarrow{\Sigma_\sigma} & & &
 \end{array} \tag{1.4}$$

Moreover, the monadicity of $\mathcal{S}_S \circ \sigma^*$ ensure an equivalence of categories between $\text{Split}_R(\sigma)$ and the algebras over $\overline{\mathcal{P}}/\mathcal{S}(S)$. Let us describe the structure of this monad: the functorial part of the monad is the composite:

$$T^\sigma = \mathcal{S}_S \circ \sigma^* \circ \Sigma_\sigma \circ \mathcal{C}_S : \overline{\mathcal{P}}/\mathcal{S}(S) \longrightarrow \overline{\mathcal{P}}/\mathcal{S}(S),$$

it is just the image under $\mathcal{S}_S$ of the composite $\sigma^* \circ \Sigma_\sigma$ over objects $\mathcal{C}_S(X, f)$. Recall that η^S and ε^S are isomorphisms on split objects, so that $\mathcal{C}_S \circ \mathcal{S}_S(A, a) \cong (A, a)$ and $(X, f) \cong \mathcal{S}_S \circ \mathcal{C}_S(X, f)$; we can exploit these isomorphisms to describe unit and multiplication in terms of the unit and multiplication of the monad seen in Section 1.1: $\eta_{(A, a)} = \langle a, 1_A \rangle$ and $\mu_{(A, a)} = 1_S \times p_A$. For every (X, f) in $\overline{\mathcal{P}}/\mathcal{S}(S)$, one has:

$$\eta_{(X, f)}^\sigma : (X, f) \xrightarrow{\varepsilon_{(X, f)}^S} \mathcal{S}_S \circ \mathcal{C}_S(X, f) \xrightarrow{\mathcal{S}_S(\eta_{\mathcal{C}_S(X, f)})} \mathcal{S}_S \circ \sigma^* \circ \Sigma_\sigma \circ \mathcal{C}_S(X, f)$$

and

$$\mu_{(X, f)}^\sigma : T^\sigma \circ T^\sigma(X, f) \xrightarrow{\mathcal{S}_S \sigma^* \Sigma_\sigma (\eta_{\sigma^* \Sigma_\sigma \mathcal{C}_S(X, f)}^S)} \mathcal{S}_S \circ \sigma^* \circ \Sigma_\sigma \circ \sigma^* \circ \Sigma_\sigma \circ \mathcal{C}_S(X, f) \xrightarrow{\mathcal{S}_S(\mu_{\mathcal{C}_S(X, f)})} T^\sigma(X, f).$$

Concretely, up to a composition with isomorphisms ε^S and η^S , the unit $\eta_{(X, f)}^\sigma$ and the multiplication $\mu_{(X, f)}^\sigma$ are just the images under $\mathcal{S}_S$ of η and μ over objects $\mathcal{C}_S(X, f)$. If we denote $(A, a) := \mathcal{C}_S(X, f)$:

$$\eta_{(X, f)}^\sigma = \mathcal{S}_S(\langle a, 1_A \rangle) \circ \varepsilon_{(X, f)}^S$$

and

$$\mu_{(X,f)}^\sigma = \mathcal{S}_S(1_S \times p_A) \circ \mathcal{S}_S \sigma^* \Sigma_\sigma(\eta_{(S \times_R A, p_S)}^S).$$

We will denote by $\mathbb{T}^\sigma$ the monad $(T^\sigma, \eta^\sigma, \mu^\sigma)$ associated with the adjunction $\Sigma_\sigma \circ \mathcal{C}_S \dashv \mathcal{C}_S \circ \sigma^*$. Let us now apply the framework introduced in Section 1.2 to define the Galois groupoid associated with the morphism σ . This groupoid is constructed specifically to capture the structure of σ , and the Galois Theorem will show that the category of internal covariant presheaves on this groupoid is equivalent to the category $Split_R(\sigma)$ of objects split by σ .

Lemma 1.3.6. *The functor $S : \mathcal{A} \longrightarrow \mathcal{P}$ transforms the groupoid $K(\sigma)$ in $\mathcal{A}$ generated by the kernel pair of $\sigma : S \rightarrow R$, as described in Example 1.2.7, into a groupoid in $\overline{\mathcal{P}}$.*

Proof. Let us first recall the structure of $K(\sigma)$:

$$(S \times_R S) \times_S (S \times_R S) \xrightarrow{\langle \pi_1, \pi_4 \rangle} S \times_R S \begin{array}{c} \xrightarrow{p_1} S \\ \xleftarrow{\Delta} S \\ \xrightarrow{p_2} S \end{array}$$

where

$$\begin{array}{ccc}
(S \times_R S) \times_S (S \times_R S) & \xrightarrow{p_3} & S \times_R S \\
\downarrow p_4 & \lrcorner & \downarrow p_2 \\
S \times_R S & \xrightarrow{p_1} & S
\end{array}$$

and $\pi_1 := p_1 \circ p_3$ and $\pi_4 := p_2 \circ p_4$.

Note that $\sigma \in \overline{\mathcal{A}}$ implies that all the arrows involved are in the admissible class $\overline{\mathcal{A}}$ too. Thus, we can see the last pullback as the "object part" of a pullback in $\overline{\mathcal{A}}/S$ considering the projections as morphisms over S . Let us prove that it lies in fact in $Split_S(1_S)$: it is just a computation to show that $(S, 1_S)$ is split by 1_S ; moreover, $S \times_R S \cong \sigma^* \circ \Sigma_\sigma(S, 1_S)$ and by Lemma 1.3.4 it is an object split by 1_S . The same argument applies for $(S \times_R S) \times_S (S \times_R S) \cong S \times_R S \times_R S$, and it can be proved by induction for any other iterated pullback of $(S, 1_S)$.

Applying the equivalence $\mathcal{S}_S : Split_S(1_S) \longrightarrow \overline{\mathcal{P}}/\mathcal{S}(S)$, we get a pullback in $\overline{\mathcal{P}}/\mathcal{S}(S)$:

$$\begin{array}{ccc}
\mathcal{S}_S((S \times_R S) \times_S (S \times_R S)) & \xrightarrow{\mathcal{S}_S(p_3)} & \mathcal{S}_S(S \times_R S) \\
\mathcal{S}_S(p_4) \downarrow \lrcorner & & \downarrow \mathcal{S}_S(p_2) \\
\mathcal{S}_S(S \times_R S) & \xrightarrow{\mathcal{S}_S(p_1)} & \mathcal{S}_S(S)
\end{array}$$

Since pullbacks in $\overline{\mathcal{P}}/\mathcal{S}(S)$ are computed as in $\overline{\mathcal{P}}$, we get that

$$\mathcal{S}((S \times_R S) \times_S (S \times_R S)) \cong \mathcal{S}(S \times_R S) \times_{\mathcal{S}(S)} \mathcal{S}(S \times_R S).$$

An analogous argument applies to prove

$$\mathcal{S}((S \times_R S) \times_S (S \times_R S) \times_S (S \times_R S)) \cong \mathcal{S}(S \times_R S) \times_{\mathcal{S}(S)} \mathcal{S}(S \times_R S) \times_{\mathcal{S}(S)} \mathcal{S}(S \times_R S).$$

The axioms for $\mathcal{S}(K(\sigma))$ to be an internal groupoid are expressed by commutativity of some diagrams involving the previous two pullbacks. Since $\mathcal{S}$ preserves them and the commutativity of diagrams, as any functor, one has that $\mathcal{S}(K(\sigma))$ is indeed an internal groupoid in $\overline{\mathcal{P}}$. $\square$

Definition 1.3.6. We will call Galois groupoid of σ the groupoid $Gal[\sigma] := \mathcal{S}(K(\sigma))$ described in Lemma 1.3.6:

$$\begin{array}{ccc} & \mathcal{S}(p_1) & \\ & \curvearrowright & \\ \mathcal{S}(S \times_R S) \times_{\mathcal{S}(S)} \mathcal{S}(S \times_R S) & \xrightarrow{\langle \mathcal{S}(\pi_1), \mathcal{S}(\pi_4) \rangle} \mathcal{S}(S \times_R S) & \xleftarrow{\mathcal{S}(\Delta)} \mathcal{S}(S) \\ & \curvearrowleft \mathcal{S}(\tau) & \curvearrowright \mathcal{S}(p_2) \end{array} \quad (1.5)$$

We are set to prove the categorical Galois Theorem. Let us note that in our setting we established two monadic functors on $\overline{\mathcal{P}}/\mathcal{S}(S)$: the pullback functor along σ composed with the equivalence $\mathcal{S}_S$, as shown in Corollary 1.3.5.1, and the forgetful functor from the internal covariant presheaf category $\overline{\mathcal{P}}^{Gal[\sigma]}$, as shown in Proposition 1.2.1. We will prove that the two monads are equivalent, establishing in this way an equivalence of categories through their respective categories of algebras.

Theorem 1.3.7. Let $(\mathcal{A}, \overline{\mathcal{A}}) \overset{\mathcal{S}}{\perp} \overset{\mathcal{C}}{\leftarrow} (\mathcal{P}, \overline{\mathcal{P}})$ be a relatively admissible adjunction where $\mathcal{A}$ and $\mathcal{P}$ are categories with pullbacks. Let $\sigma : S \rightarrow R$ be of relative Galois descent with respect to these data. There exists an equivalence of categories

$$Split_R(\sigma) \approx \overline{\mathcal{P}}^{Gal[\sigma]}$$

between the category of those objects in $\overline{\mathcal{A}}/R$ that are split by σ and the category of internal covariant presheaves (P, p, π) on the internal groupoid $Gal[\sigma]$ in $\mathcal{P}$, in which $p \in \overline{\mathcal{P}}$.

Proof. As said before, we have two monads on $\overline{\mathcal{P}}/\mathcal{S}(S)$:

$$(T = \Sigma_{\mathcal{S}(p_2)} \circ \mathcal{S}(p_1)^*, \langle \mathcal{S}(\Delta) \circ f, 1_X \rangle, \langle \mathcal{S}(\pi_1), \mathcal{S}(\pi_4) \rangle \times 1_X),$$

given in Proposition 1.2.1 applied to the Galois groupoid of σ , and

$$(T^\sigma = \mathcal{S}_S \circ \sigma^* \circ \Sigma_\sigma \circ \mathcal{C}_S, \eta^\sigma, \mu^\sigma).$$

Putting $(A, a) = \mathcal{C}_S(X, f)$, we have that $\sigma^* \circ \Sigma_\sigma(A, a) \cong \Sigma_{p_2} \circ p_1^*(A, a)$, as shown in the two pullback diagrams below, and applying the equivalence $\mathcal{S}_S$ the pullbacks are preserved.

$$\begin{array}{ccccc} S \times_R A & \longrightarrow & A & & \mathcal{S}_S(S \times_R A) & \longrightarrow & \mathcal{S}_S(A) \cong X \\ \downarrow a_1 & \lrcorner & \downarrow a & & \downarrow \mathcal{S}(a_1) & \lrcorner & \downarrow \mathcal{S}(a) = f \\ S \times_R S & \xrightarrow{p_1} & S & \xrightarrow{\mathcal{S}_S} & \mathcal{S}_S(S \times_R S) & \xrightarrow{\mathcal{S}(p_1)} & \mathcal{S}_S(S) \\ \downarrow p_2 & \lrcorner & \downarrow \sigma & & \downarrow \mathcal{S}(p_2) & & \\ S & \xrightarrow{\sigma} & R & & \mathcal{S}_S(S) & & \end{array}$$

Thus, we have

$$(\mathcal{S}_S(S \times_R A), \mathcal{S}(p_2) \circ \mathcal{S}(a_1)) \cong (\mathcal{S}_S(S \times_R S) \times_{\mathcal{S}_S(S)} \mathcal{S}_S(A), \mathcal{S}(p_2) \circ \mathcal{S}(a_1))$$

and, recalling that $\mathcal{S}_S(A) = \mathcal{S}_S \mathcal{C}_S(X, f) \cong (X, f)$, the functorial parts of the two monads are isomorphic:

$$T^\sigma(X, f) = \mathcal{S}_S \circ \sigma^* \circ \Sigma_\sigma(A, a) \cong \Sigma_{\mathcal{S}(p_2)} \circ \mathcal{S}(p_1)^*(X, f) = T(X, f).$$

Similarly, one has

$$\mathcal{S}_S(S \times_R S \times_R A) \cong \mathcal{S}_S(S \times_R S) \times_{\mathcal{S}_S(S)} \mathcal{S}_S(S \times_R S) \times_{\mathcal{S}_S(S)} \mathcal{S}_S(A).$$

Moreover, for all (X, f) in $\overline{\mathcal{P}}/\mathcal{S}(S)$, we have shown that, up to isomorphisms, $\eta_{(X, f)}^\sigma$ is $\mathcal{S}_S(\eta_{(A, a)})$ and $\mu_{(X, f)}^\sigma$ is $\mathcal{S}_S(\mu_{(A, a)})$ where $(A, a) = \mathcal{C}_S(X, f)$ and η and μ are the unit and the multiplication of the monad $(\sigma^* \circ \Sigma_\sigma, \eta, \mu)$ from Section 1.1 where we have shown that $\eta_{(A, a)} = \langle a, 1_A \rangle$ and $\mu_{(A, a)} = 1_S \times p_A = p_1 \times 1_A$. It is straightforward to show that, up to the isomorphisms above, the unit and multiplication coincide:

$$\begin{array}{ccc} (X, f) & \xrightarrow{\langle \mathcal{S}(\Delta) \circ f, 1_X \rangle} & \mathcal{S}_S(S \times_R S) \times_{\mathcal{S}_S(S)} X \\ \cong \updownarrow & & \updownarrow \cong \\ \mathcal{S}_S(A, a) & \xrightarrow{\mathcal{S}_S(\langle a, 1_A \rangle)} & \mathcal{S}_S(S \times_R A) \end{array}$$

$$\begin{array}{ccc} \mathcal{S}_S(S \times_R S) \times_{\mathcal{S}_S(S)} \mathcal{S}_S(S \times_R S) \times_{\mathcal{S}_S(S)} X & \xrightarrow{\langle \mathcal{S}(\pi_1), \mathcal{S}(\pi_4) \rangle \times 1_X} & \mathcal{S}_S(S \times_R S) \times_{\mathcal{S}_S(S)} X \\ \cong \updownarrow & & \updownarrow \cong \\ \mathcal{S}_S(S \times_R S \times_R A) & \xrightarrow{\mathcal{S}_S(p_1 \times 1_A)} & \mathcal{S}_S(S \times_R A) \end{array}$$

The desired equivalence is then given via their respective categories of algebras:

$$Split_R(\sigma) \approx \overline{\mathcal{P}}/\mathcal{S}(S)^{T^\sigma} \approx \overline{\mathcal{P}}/\mathcal{S}(S)^T \approx \overline{\mathcal{P}}^{Gal[\sigma]}.$$

□

Observation. A key observation about Theorem 1.3.7 is that the core of its proof relies on two fundamental results: the isomorphism $\sigma^* \circ \Sigma_\sigma(A, a) \cong \Sigma_{p_2} \circ p_1^*(A, a)$ and the equivalence $Split_S(1_S) \approx \overline{\mathcal{P}}/\mathcal{S}(S)$. These two steps form the backbone of the argument.

Chapter 2

From Categorical to Classical Galois theory

In this chapter, we apply Theorem 1.3.7 to a fundamental case: the classical setting that originally inspired its formulation. The theorem is called "Galois" for a reason, and here we explore why, demonstrating that it contains and generalizes the well-known Galois correspondence. Our approach follows a natural progression: in Section 2.1, we need to recall some preliminary results that will be instrumental in the following sections. Next, in Section 2.2 we establish the appropriate concrete algebraic setting by considering an adjunction involving the category of R -algebras and the category of profinite spaces, and a ring homomorphism $\sigma : R \rightarrow S$. Finally, in Section 2.3, we specialize further to the field case, where $\sigma : K \hookrightarrow L$ is a Galois extension of fields. In this setting, we show that the categorical notion of splitting corresponds exactly to the classical definition of a field being split by an extension, and that the internal Galois groupoid recovers the classical Galois group. This analysis will confirm that the classical Galois theory is fully contained within the categorical framework developed in Chapter 1. Once again, our approach is primarily based on the exposition presented in *Galois Theories* by Borceux and Janelidze [2]. Before proceeding, we note that throughout this chapter we will assume that all rings and algebras are commutative and have a unit.

2.1 Preliminary results

In the first place, we need to recall some preliminary results that will be instrumental in the following sections. When proving these results is not in the focus of this thesis, we will state them without any proof and direct the reader to the appropriate references for detailed proofs. To begin, we introduce the Pierce spectrum functor between the category **Ring** of rings and the category **Prof** of profinite spaces.

Definition 2.1.1. A topological space is profinite when it is the projective limit of finite discrete topological spaces. Equivalent conditions are to be a compact, Hausdorff, totally disconnected space or a compact, Hausdorff space such that the clopen subsets form a base for the topology (see [9], Proposition 2.4, 2.5).

Definition 2.1.2. Let B be a Boolean algebra. A proper filter in B is a subset $F \subset B$ such that

1. $1 \in F$;
2. $x \in F$ and $y \in F$, then $x \wedge y \in F$;

3. $x \in F$ and $x \leq y$, then $y \in F$;

4. $0 \notin F$;

An ultrafilter in B is a maximal element in the poset of proper filters in B ordered by inclusion. We will write $\text{Spec}(B)$ for the set of ultrafilters in B .

Definition 2.1.3. Let B be a Boolean algebra. For every $b \in B$, the subsets

$$U_b = \{F \in \text{Spec}(B) : b \in F\}$$

constitute a basis of clopens for a topology $\mathcal{T}$ on $\text{Spec}(B)$ (Lemma A.1.1). The topological space $(\text{Spec}(B), \mathcal{T})$ is called the spectrum of the Boolean algebra B .

We now introduce functors between these objects, whose composition will play a key role in Section 2.2. First, we consider those establishing the Stone duality theorem, which is discussed in more detail in [2], Section 4.1, and briefly summarized in Appendix A.1, while a much more extensive treatment can be found in [7].

Theorem 2.1.1 (Stone duality). *The functors*

$$\text{Spec} : \mathbf{Bool}^{\text{op}} \longrightarrow \mathbf{Prof}$$

$$\text{Clopen} : \mathbf{Prof} \longrightarrow \mathbf{Bool}^{\text{op}}$$

$$\begin{array}{ccc} B & \longrightarrow & \text{Spec}(B) \\ f \downarrow & & \uparrow f^{-1} \\ B' & \longrightarrow & \text{Spec}(B') \end{array}$$

$$\begin{array}{ccc} X & \longrightarrow & \text{Clopen}(X) \\ g \downarrow & & \uparrow g^{-1} \\ X' & \longrightarrow & \text{Clopen}(X') \end{array}$$

form an equivalence of categories. Here, the arrows are represented as in $\mathbf{Bool}$ and not in $\mathbf{Bool}^{\text{op}}$, $\text{Spec}(B)$ is the spectrum of the Boolean algebra B and we denoted with $\text{Clopen}(X)$ the set of clopens of X , which has a natural structure of Boolean algebra (Lemma A.1.3).

Another functor that we introduce gives us a link between the categories of rings and of Boolean algebras, allowing us to construct a contravariant functor from $\mathbf{Ring}$ to $\mathbf{Prof}$.

Proposition 2.1.2 ([2], Proposition 4.2.1). *Let R be a ring. The set of idempotents of R , with the operations*

$$e \wedge e' := ee',$$

$$e \vee e' := e + e' - ee',$$

$$e^c := 1 - e,$$

constitutes a Boolean algebra. Moreover, if $f : R \rightarrow R'$ is a ring homomorphism, then f maps idempotents of R onto idempotents of R' and preserves the mentioned operations.

Definition 2.1.4. We will write $\text{Idemp} : \mathbf{Ring} \longrightarrow \mathbf{Bool}$ for the functor that sends a ring R to the Boolean algebra of its idempotents and a morphism to its restriction. The Pierce spectrum functor

$$\text{Sp} : \mathbf{Ring}^{\text{op}} \longrightarrow \mathbf{Prof}$$

is given by the composition between $Idemp$ on the opposite categories and $Spec$. Let us explicitly state how it acts also on morphisms: for any ring homomorphism $f : A \rightarrow B$,

$$Sp(f) : Sp(B) \longrightarrow Sp(A) \\ F \longmapsto f^{-1}(F) = \{a \in Idemp(A) : f(a) \in F\}$$

since $Idemp$ acts trivially on morphisms.

In Section 2.2, we will also describe more concretely the structure of internal covariant presheaves in **Prof**. In doing so, it will be useful to recall the notion of G -sets.

Definition 2.1.5. Let G be a group. A (left) G -set is a set X provided with a (left) action of G

$$G \times X \longrightarrow X, (g, x) \mapsto g \cdot x$$

that satisfies the axioms $1 \cdot x = x$ and $g \cdot (g' \cdot x) = (gg') \cdot x$, for all $x \in X$ and $g, g' \in G$. A morphism $f : X \rightarrow Y$ between G -sets is a map which preserves the action of G , that is, $f(g \cdot x) = g \cdot f(x)$ for all $x \in X$ and $g \in G$.

We will write **G -Set** for the category of (left) G -sets and their morphism, and **G -Set_f** for the full subcategory of finite (left) G -sets.

Note that G is indeed a G -set with the canonical action of (left) multiplication. In Section 2.3, we will need a relation between the two different points of view on G : as a group or as a G -set.

Proposition 2.1.3. *Let G be a group. There exists a bijection between the subgroups H of G and the surjections $p : G \twoheadrightarrow Q$ of G -sets.*

Proof. Let us exhibit the bijection. With every $H \leq G$ is associated the equivalence relation

$$x \equiv y \iff x^{-1}y \in H;$$

consider the quotient $p_H : G \twoheadrightarrow G/H$, with G/H provided with the canonical structure of a G -set $g \cdot [x] = [gx]$, for every $g, x \in G$. p_H is clearly a morphism of G -sets and note that

$$p_H(g) = p_H(1) \iff [g] = [1] \iff g \in H.$$

Now, for every surjection $p : G \twoheadrightarrow Q$ of G -sets we have that

$$p(xy) = x \cdot p(y)$$

for every $x, y \in G$. Consider $H_p = \{g \in G : p(g) = p(1)\}$. It is a subgroup of G since obviously $1 \in H_p$ and

$$p(xy) = x \cdot p(y) = x \cdot p(1) = p(x) = p(1), \\ p(x^{-1}) = x^{-1} \cdot p(1) = x^{-1} \cdot p(x) = p(1).$$

for all $x, y \in H_p$. The two maps are inverses of each other: obviously $H_{p_H} = H$; conversely, we need to show that

$$p(x) = p(y) \iff [x] = [y] \text{ in } G/H_p \iff x^{-1}y \in H_p \iff p(x^{-1}y) = p(1)$$

for all $x, y \in G$. Indeed, if $p(x) = p(y)$,

$$p(x^{-1}y) = x^{-1} \cdot p(y) = x^{-1} \cdot p(x) = p(x^{-1}x) = p(1).$$

On the other hand, if $p(x^{-1}y) = p(1)$,

$$p(x) = p(x1) = x \cdot p(1) = x \cdot p(x^{-1}y) = p(xx^{-1}y) = p(y).$$

□

Before moving on to the next section, we recollect some essential properties of field extensions. In particular, we need to recall that field extensions are faithfully flat and to establish a key property in the relation between a field extension $K \hookrightarrow L$ and a K -algebra A . These results will be useful in Section 2.3.

Definition 2.1.6. A module M over a ring R is faithfully flat when the functor

$$M \otimes_R - : \mathbf{Mod}_R \rightarrow \mathbf{Mod}_R$$

preserves and reflects exact sequences.

Proposition 2.1.4. *Let $\sigma : K \hookrightarrow L$ be a field extension; then L is faithfully flat as a K -module.*

Proof. It follows directly from the fact that every K -module is free and that the tensor product of free modules is again a free module. $\square$

Definition 2.1.7. Let $\sigma : K \hookrightarrow L$ be a field extension. A K -algebra A is split by L when

1. for every $a \in A$, there exists a non-zero polynomial $p(X) \in K[X]$ such that $p(a) = 0$. Among all such polynomials, there is a unique one of minimal degree with leading coefficient 1, which we call the minimal polynomial of a over K ;
2. for every $a \in A$, the minimal polynomial of a over K factors entirely in $L[X]$ in polynomials of degree 1 with distinct roots.

We will write $\text{Split}_K(L)$ for the full subcategory of $\mathbf{K-Alg}$ of those objects split by L .

Proposition 2.1.5. *Let $\sigma : K \hookrightarrow L$ be an extension of fields and A a K -algebra of finite dimension n . Then A is split by L if and only if there is an isomorphism of L -algebras*

$$L \otimes_K A \cong L^n. \quad (2.1)$$

Proof. Let $a \in A$ have minimal polynomial $p(X) \in K[X]$ of degree n . Since A is split by L , $p(X)$ factors in $L[X]$ in n distinct polynomials of degree 1 and we obtain (2.1) for the K -algebra $K(a)$:

$$L \otimes_K K(a) \cong L \otimes_K \frac{K[X]}{\langle p(X) \rangle} \cong \frac{L[X]}{\langle p(X) \rangle} \cong L^n.$$

Now, consider the algebra $A_1 \cdot A_2$ generated by A_1 and A_2 , where both satisfy (2.1) and their dimension over K are respectively n_1 and n_2 . The K -algebra $A_1 \cdot A_2$ can be viewed as a quotient

$$A_1 \otimes_K A_2 \twoheadrightarrow A_1 \cdot A_2, \text{ where } a_1 \otimes a_2 \mapsto a_1 \cdot a_2.$$

Since $L \otimes_K -$ has a right adjoint, it preserves quotients and we obtain the quotient

$$L \otimes_K (A_1 \otimes_K A_2) \twoheadrightarrow L \otimes_K (A_1 \cdot A_2).$$

Putting $n = n_1 n_2$, by the assumption on A_1 and A_2 , we get

$$L^n \cong L^{n_1} \otimes_K A_2 \cong (L \otimes_K A_1) \otimes_K A_2 \cong L \otimes_K (A_1 \otimes_K A_2).$$

Let us consider the kernel I of the last quotient: it is well known that ideals of L^n are of the form

$$I = \{(l_i) \in L^n : l_j = 0, \forall j \in J\} \text{ for some } J \subset \{1, \dots, n\}$$

and $I \cong L^d$, where d is given by the cardinality of J . We get

$$L \otimes_K (A_1 \cdot A_2) \cong L^n / I \cong L^{n-d}.$$

Thus, we obtained that (2.1) holds also for the K -algebra $A_1 \cdot A_2$.

We have that A is a finitely generated K -algebra, and so it is generated by a finite number of K -subalgebras $K(a_i)$, for some $a_i \in A$, each of them satisfying (2.1). Applying the previous argument finitely many times, we obtain the thesis:

$$L \otimes_K A \cong L^n.$$

On the other hand, let $a \in A$ have a minimal polynomial $p(X) \in K[X]$ of degree n . Recall that the roots of $p(X)$ in L are in bijection with $\text{Hom}_K(K[X]/\langle p(X) \rangle, L)$. If $L \otimes_K A \cong L^n$, then

$$n = \#\text{Hom}_L(L^n, L) = \#\text{Hom}_L(L \otimes_K A, L) = \#\text{Hom}_K(A, L) = \#\text{Hom}_K\left(\frac{K[X]}{\langle p(X) \rangle}, L\right),$$

so that $p(X)$ splits in n factors of degree 1 in $L[X]$ and A is a K -algebra split by L . $\square$

2.2 Galois theory of commutative rings

In this section, we apply Theorem 1.3.7 to an adjunction involving the Pierce spectrum functor defined in Definition 2.1.4 on the category of R -algebras. In particular, we will prove the existence of a right adjoint and we will state the theorem in this algebraic setting. Since we will work with the opposite categories of rings, we will always underline if a given morphism is considered as an arrow in the opposite category or if it is considered as a ring homomorphism.

Proposition 2.2.1. *Let R be a ring. The Pierce spectrum functor*

$$Sp : \mathbf{R}\text{-Alg}^{\text{op}} \longrightarrow \mathbf{Prof}$$

admits as right adjoint the functor

$$\mathcal{C}(-, R) : \mathbf{Prof} \longrightarrow \mathbf{R}\text{-Alg}^{\text{op}}$$

where $\mathcal{C}(X, R)$ denotes the ring of continuous maps and R is provided with the discrete topology. The functor $\mathcal{C}(-, R)$ acts on the arrows of $\mathbf{Prof}$ by composition.

Proof. We will construct the corresponding unit and counit and prove the required triangular identities. Let us construct the counit

$$\varepsilon_X : Sp(\mathcal{C}(X, R)) \longrightarrow X$$

for a profinite space X . By Stone duality, this reduces to constructing a homomorphism of Boolean algebras

$$\tilde{\varepsilon}_X : \text{Clopen}(X) \longrightarrow \text{Idemp}(\mathcal{C}(X, R)).$$

For every $U \in \text{Clopen}(X)$, we define

$$f_U : X \longrightarrow R, f_U(x) = \begin{cases} 1 & \text{if } x \in U, \\ 0 & \text{if } x \notin U. \end{cases}$$

It is clearly an idempotent continuous map and the association $U \mapsto f_U$ is a homomorphism of Boolean algebras. Moreover, $\tilde{\varepsilon}$ is natural: if $h : X \rightarrow Y$ is a continuous map, for every $V \in Clopen(Y)$ and $x \in X$,

$$f_{h^{-1}(V)}(x) = 1 \iff x \in h^{-1}(V) \iff h(x) \in V \iff f_V(h(x)) = 1$$

from which the commutativity of the diagram

$$\begin{array}{ccc} Clopen(X) & \xrightarrow{\tilde{\varepsilon}_X} & Idemp(\mathcal{C}(X, R)) \\ \uparrow h^{-1} & & \uparrow - \circ h \\ Clopen(Y) & \xrightarrow{\tilde{\varepsilon}_Y} & Idemp(\mathcal{C}(Y, R)) \end{array}$$

Note that it is expressed in **Bool** and not in **Bool**^{op}.

Next, let us construct the unit

$$\eta_A : A \longrightarrow \mathcal{C}(Sp(A), R)$$

for an R -algebra A . Since we are working in **R-Alg**^{op}, we need a homomorphism of R -algebras

$$\eta_A : \mathcal{C}(Sp(A), R) \longrightarrow A.$$

For every continuous map $f : Sp(A) \rightarrow R$, we shall prove the existence of a finite partition $Sp(A) = U_{e_1} \cup \dots \cup U_{e_n}$ of $Sp(A)$ into clopens U_{e_i} , with $e_i \in A$ idempotent, such that f is constant on each U_{e_i} with different values $r_i \in R$. We define

$$\eta_A(f) = \sum_{i=1}^n r_i e_i.$$

We have a partition $Sp(A) = \bigcup_{r \in R} f^{-1}(r)$ and, since R is discrete, each $f^{-1}(r)$ is a clopen. Since $Sp(A)$ is compact, we can extract a finite subpartition of $Sp(A)$

$$Sp(A) = f^{-1}(r_1) \cup \dots \cup f^{-1}(r_n)$$

and each $f^{-1}(r_i)$ is compact in $Sp(A)$, so it can be written as a finite union of elements of the basis U_e and it is easy to check that

$$f^{-1}(r_i) = \bigcup_{j=1}^m U_{e_j} = U_{e_1 \vee \dots \vee e_m},$$

so that we can fix for every $i = 1, \dots, n$ a unique $e_i \in Idemp(A)$ such that

$$Sp(A) = f^{-1}(r_1) \cup \dots \cup f^{-1}(r_n) \text{ and } f^{-1}(r_i) = U_{e_i}.$$

Let us show that the definition $\eta_A(f) = \sum_{i=1}^n r_i e_i$ does not depend on the construction we have made by showing it is the only partition with this properties. Consider another partition $Sp(A) = \bigcup_{j=1}^m U_{e_j}$ with the same properties. The fact that f is constant on each piece of the partition implies that for every $j = 1, \dots, m$, there exists an $i = 1, \dots, n$ such that

$$U_{e_j} \subset f^{-1}(r_i).$$

Moreover, if it is a proper subset, then there exists $k \neq j$ such that

$$U_{e_k} \subset f^{-1}(r_i) \setminus U_{e_j},$$

contradicting the condition that f is constant on each piece with different values. It is just a computation to show that η_A is a homomorphism of R -algebras. Let us show the naturality: if $h : A \rightarrow B$ is a homomorphism of R -algebras, consider any $f \in \mathcal{C}(Sp(A), R)$; we need to establish that

$$h(\eta_A(f)) = h\left(\sum_{i=1}^n r_i e_i\right) = \sum_{i=1}^n r_i h(e_i) = \eta_B(f \circ Sp(h)),$$

in order to obtain the commutativity of the diagram

$$\begin{array}{ccc} \mathcal{C}(Sp(A), R) & \xrightarrow{\eta_A} & A \\ \downarrow - \circ Sp(h) & & \downarrow h \\ \mathcal{C}(Sp(B), R) & \xrightarrow{\eta_B} & B \end{array}$$

It is sufficient to prove that for every $i = 1, \dots, n$ and for every $F \in U_{h(e_i)}$,

$$f \circ Sp(h)(F) = r_i.$$

But we have

$$U_{h(e_i)} = \{F \in Sp(B) : e_i \in h^{-1}(F)\} = \{F \in Sp(B) : e_i \in Sp(h)(F)\},$$

so that $Sp(h)(F) \in U_{e_i}$ and by definition $f(Sp(h)(F)) = r_i$.

We must now prove the triangular identities of the adjunction. The first one is the commutativity of the left triangle below,

$$\begin{array}{ccc} Sp(A) & \xrightarrow{Sp(\eta_A)} & Sp(\mathcal{C}(Sp(A), R)) \\ & \searrow & \downarrow \varepsilon_{Sp(A)} \\ & & Sp(A) \end{array} \quad \begin{array}{ccc} Idemp(A) & \xleftarrow{\eta_A} & Idemp(\mathcal{C}(Sp(A), R)) \\ & \searrow & \uparrow \tilde{\varepsilon}_{Sp(A)} \\ & & Idemp(A) \end{array}$$

which is equivalent to the commutativity of the right triangle, identifying the clopens of $Sp(A)$ with the idempotents of A , via $Clopen(Sp(A)) \cong Idemp(A)$. An idempotent $e \in A$ is mapped by $\tilde{\varepsilon}_{Sp(A)}$ to f_e , from which

$$\eta_A(f_e) = 1e + 0(1 - e) = e.$$

The second triangular identity is the commutativity of

$$\begin{array}{ccc} \mathcal{C}(X, R) & \xrightarrow{- \circ \varepsilon_X} & \mathcal{C}(Sp(\mathcal{C}(X, R)), R) \\ & \searrow & \downarrow \eta_{\mathcal{C}(X, R)} \\ & & \mathcal{C}(X, R) \end{array}$$

Choose $g \in \mathcal{C}(X, R)$ and consider the composite $g \circ \varepsilon_X$. As before, $Sp(\mathcal{C}(X, R))$ can be written as a finite partition of clopens

$$Sp(\mathcal{C}(X, R)) = (g \circ \varepsilon_X)^{-1}(r_1) \cup \cdots \cup (g \circ \varepsilon_X)^{-1}(r_n).$$

and for every $i = 1, \dots, n$,

$$(g \circ \varepsilon_X)^{-1}(r_i) = \varepsilon_X^{-1}(g^{-1}(r_i)) = U_{\tilde{\varepsilon}_X(g^{-1}(r_i))} = U_{f_{g^{-1}(r_i)}},$$

so that

$$\eta_{\mathcal{C}(X, R)}(g \circ \varepsilon_X) = \sum_{i=1}^n r_i f_{g^{-1}(r_i)}.$$

By definition of $f_{g^{-1}(r_i)}$, we have for every $x \in X$

$$f_{g^{-1}(r_i)}(x) = 1 \iff x \in g^{-1}(r_i) \iff g(x) = r_i,$$

obtaining

$$g = \sum_{i=1}^n r_i f_{g^{-1}(r_i)} = \eta_{\mathcal{C}(X, R)}(g \circ \varepsilon_X).$$

□

To apply the categorical Galois theorem in this setting, we consider as class of admissible arrows all morphisms in both categories. This choice ensures that the adjunction $Sp \dashv \mathcal{C}(-, R)$ is relatively admissible. Moreover, observe that both $\mathbf{R}\text{-alg}^{\text{op}}$ and $\mathbf{Prof}$ admit pullbacks.

Theorem 2.2.2 (Galois theorem of rings). *Consider the adjunction $\mathbf{R}\text{-Alg}^{\text{op}} \begin{array}{c} \xrightarrow{Sp} \\ \perp \\ \xleftarrow{\mathcal{C}(-, R)} \end{array} \mathbf{Prof}$. Let*

$\sigma : S \rightarrow R$ *be of relative Galois descent with respect to these data. There exists an equivalence of categories*

$$Split_R(\sigma)^{\text{op}} \approx \mathbf{Prof}^{Gal[\sigma]}$$

between the dual of the category of R -algebras split by σ and the category of internal covariant presheaves (P, p, π) on the internal groupoid $Gal[\sigma]$ in the category $\mathbf{Prof}$ of profinite spaces.

Let us analyse more concretely some of the elements involved. First of all, let us recall the meaning of the assumption for $\sigma : S \rightarrow R$ to be a morphism of relative Galois descent in this setting. We ask for a ring homomorphism $\sigma : R \rightarrow S$ that satisfies the following:

1. the pullback functor along σ in $\mathbf{R}\text{-alg}^{\text{op}}$ is monadic;
2. the counit ε^S of the adjunction $Sp_S \dashv \mathcal{C}(-, R)_S$ is an isomorphism at every object (X, f) of $\mathbf{Prof}/Sp(S)$. More explicitly, writing $\mathcal{C}_S$ for $\mathcal{C}(-, R)_S$, we ask for the counit ε^S

$$Sp(\mathcal{C}_S(X, f)) \cong X$$

to be an homeomorphism for every profinite space X along with a continuous map $f : X \rightarrow Sp(S)$;

3. the object $\mathcal{C}_S(X, f)$, seen as an R -algebra, is split by σ , for every (X, f) in $\mathbf{Prof}/Sp(S)$. More explicitly, let us clarify what it means for an R -algebra A to be split by σ : we are asking for the unit η^S of the adjunction $Sp_S \dashv \mathcal{C}(-, R)_S$

$$S \otimes_R A \cong \mathcal{C}_S(Sp(S \otimes_R A)).$$

to be an isomorphism of S -algebras on $S \otimes_R A$.

These last two conditions will be analyzed more deeply in the next part of this section, where we will gain a clearer understanding of the functor $\mathcal{C}(-, R)_S$ under one more assumption on S . Note that the pullback along σ in $\mathbf{R-Alg}^{\text{op}}$ is the pushout in $\mathbf{R-Alg}$. Conversely, the functor Σ_σ is indeed a forgetful functor: an S -algebra A is a ring provided with a ring homomorphism $\rho_A : S \rightarrow A$, and the composition with the ring homomorphism $\sigma : R \rightarrow S$ provides to A the structure of an R -algebra. From now on we will omit the functor Σ_σ since it simply represents the process of viewing an S -algebra as an R -algebra. For the sake of clarity, we will indicate explicitly when we regard an object in this way rather than introducing additional notation, keeping the presentation more readable. Now, let us show the functors involved as in Diagram 1.4, recalling that for every R -algebra S , one has an isomorphism of categories $\mathbf{R-Alg}^{\text{op}}/S \cong \mathbf{S-Alg}^{\text{op}}$.

$$\begin{array}{ccccc}
 & & \sigma^* & & \\
 & & \curvearrowright & & \\
 \mathbf{R-Alg}^{\text{op}} & & & \mathbf{S-Alg}^{\text{op}} & \xrightarrow{Sp_S} \mathbf{Prof}/Sp(S) \\
 & \nwarrow & & \nearrow & \downarrow \lrcorner \\
 & & \mathcal{C}(-, R)_S & & \\
 & \nwarrow & & \nearrow & \downarrow \simeq \\
 & & \sigma^* & & \\
 Split_R(\sigma)^{\text{op}} & & & Split_S(1_S)^{\text{op}} & \xrightarrow{Sp_S} \mathbf{Prof}/Sp(S)
 \end{array} \quad (2.2)$$

The localized adjunction $Sp_S \dashv \mathcal{C}(-, R)_S$ is given as in Section 1.1 of Chapter 1; explicitly, the left adjoint Sp_S acts as Sp on the category of S -algebras

$$A \mapsto (Sp(A), Sp(\rho_A))$$

for an S -algebra $\rho_A : S \rightarrow A$. The right adjoint $\mathcal{C}(-, R)_S$ has the following expression that will be clarified further under one more assumption: recalling that $\mathcal{C}(-, R)_S$ is the composition of $\mathcal{C}(-, R)$ and the pullback functor along the unit η_S of $Sp \dashv \mathcal{C}(-, R)$, one has

$$\begin{array}{ccc}
 \mathcal{C}(X, R)_S := S \otimes_{\mathcal{C}(Sp(S), R)} \mathcal{C}(X, R) & \xrightarrow{\quad} & \mathcal{C}(X, R) \\
 \downarrow \lrcorner & & \downarrow - \circ f \\
 S & \xrightarrow{\eta_S} & \mathcal{C}(Sp(S), R)
 \end{array} \quad (2.3)$$

for every profinite space X along with a morphism $f : X \rightarrow Sp(S)$. Note that we have expressed Diagram 2.3 following the theoretic definition, so as a pullback in $\mathbf{R-Alg}^{\text{op}}$.

Since our goal is to specialize this setting to the case of fields extensions, we now examine certain properties that arise specifically when S has exactly two idempotent elements, namely

0 and 1. It is a strong condition to assume on a ring, but whenever a result remains valid without this assumption we will explicitly state it. We will only provide proofs under this hypothesis. In particular, under the idempotents assumption, we get a clearer form for the functor $\mathcal{C}(-, R)_S$. Moreover, we will show that the requirement on the counit ε^S to be an isomorphism is unnecessary, as it is always satisfied, even without this assumption.

Proposition 2.2.3. *Let R be a ring and S an R -algebra with exactly two idempotent elements. The functor*

$$\mathcal{C}(-, S) : \mathbf{Prof} \longrightarrow \mathbf{S-Alg}^{\text{op}}$$

is full and faithful. Moreover, $Sp(S)$ is a singleton, and there is an isomorphism between the functor

$$\mathcal{C}(-, S) : \mathbf{Prof} \longrightarrow \mathbf{S-Alg}^{\text{op}}$$

and

$$\mathcal{C}(-, R)_S : \mathbf{Prof}/Sp(S) \longrightarrow \mathbf{S-Alg}^{\text{op}},$$

establishing an isomorphism of S -algebras

$$S \otimes_R \mathcal{C}(X, R) \cong \mathcal{C}(X, S)$$

for every profinite space X .

Proof. To prove that $\mathcal{C}(-, S)$ is full and faithful, we show that $\varepsilon_X : Sp(\mathcal{C}(X, S)) \rightarrow X$, the counit of $Sp \dashv \mathcal{C}(-, S)$ described in Proposition 2.2.1, is a homeomorphism for every profinite space X . Since it is a continuous map between Hausdorff spaces, it suffices to prove its bijectivity. If $X = \emptyset$, then $Sp(\mathcal{C}(X, S)) = \emptyset$ and obviously ε_X is a homeomorphism. Otherwise, we show that $\tilde{\varepsilon}_X : Clopen(X) \rightarrow Idemp(\mathcal{C}(X, S))$ is an isomorphism of Boolean algebras by constructing an inverse. An idempotent continuous map $f : X \rightarrow S$ takes values on idempotent of S , so only 0 and 1. Since S is discrete, the preimage of 1 is a clopen of X and the preimage of 0 is its complement. Therefore, the map $f \mapsto f^{-1}(1)$ is the inverse of $\tilde{\varepsilon}_X$, and ε_X is bijective for every profinite space X . Moreover, considering $X = \{*\}$, we get $\mathcal{C}(X, S) \cong S$ and so

$$Sp(S) \cong Sp(\mathcal{C}(X, S)) \cong X = \{*\}.$$

We could also have computed directly

$$Sp(S) = Spec(Idemp(S)) = Spec(\{0, 1\}) = \{\{1\}\},$$

since $\{1\}$ is the only ultrafilter in the Boolean algebra $\{0, 1\}$.

Finally, since $Sp(S) = \{*\}$ is the terminal object in $\mathbf{Prof}$, it is clear that $\mathbf{Prof}/Sp(S) \cong \mathbf{Prof}$ and $Sp_S = Sp : \mathbf{S-Alg}^{\text{op}} \longrightarrow \mathbf{Prof}$. So that there is an isomorphism between the two right adjoints $\mathcal{C}(-, S)$ and $\mathcal{C}(-, R)_S$ of Sp . Explicitly, Diagram 2.3 is in the form

$$\begin{array}{ccc} \mathcal{C}(X, S) \cong \mathcal{C}(X, R)_S & \xrightarrow{\quad} & \mathcal{C}(X, R) \\ \downarrow \lrcorner & & \downarrow \\ S & \xrightarrow{\eta_S} & \mathcal{C}(Sp(S), R) \cong R \end{array}$$

establishing the desired isomorphism

$$\mathcal{C}(X, S) \cong S \otimes_R \mathcal{C}(X, R)$$

for every profinite space X . □

Actually, to prove that the counit ε^S is an isomorphism, it is not needed that S has exactly two idempotents. In general, the functor $\mathcal{C}(-, R)_S$ has a different explicit form as a Hom-functor, considering the Pierce structural space, and it can be shown that it remains full and faithful. Some details can be found in Appendix A.2.

Proposition 2.2.4 ([2], Theorem 4.3.6). *Let R be a ring. For every R -algebra S , the functor*

$$\mathcal{C}(-, R)_S : \mathbf{Prof}/Sp(S) \longrightarrow \mathbf{S-Alg}^{\text{op}}$$

is full and faithful.

In Proposition 2.2.3 we have shown that it is not necessary to ask for the counit ε^S to be an isomorphism. Moreover, with this new assumption on S , we got a clearer view on the functor $\mathcal{C}(-, R)_S$ and we can proceed to exhibit once again Diagram 2.2 and the conditions on the ring homomorphism $\sigma : R \rightarrow S$ in the case of S having exactly two idempotent elements. We obtained the following:

$$\begin{array}{ccccc}
 & & S \otimes_R - & & \\
 & \swarrow & & \searrow & \\
 \mathbf{R-Alg}^{\text{op}} & & \top & & \mathbf{S-Alg}^{\text{op}} \\
 & \nwarrow & & \nearrow & \\
 & & S \otimes_R - & & \\
 \text{Split}_R(\sigma)^{\text{op}} & & \top & & \text{Split}_S(1_S)^{\text{op}} \\
 & \nwarrow & & \nearrow & \\
 & & S \otimes_R - & & \\
 & \swarrow & & \searrow & \\
 & & \mathcal{C}(-, S) & & \\
 & \nwarrow & & \nearrow & \\
 & & \approx & & \\
 & \swarrow & & \searrow & \\
 & & \text{Prof} & &
 \end{array}
 \quad (2.4)$$

Regarding the assumption on the ring homomorphism $\sigma : R \rightarrow S$, we need only two conditions:

1. the functor $S \otimes_R - : \mathbf{R-Alg}^{\text{op}} \longrightarrow \mathbf{S-Alg}^{\text{op}}$ is monadic;
2. the object $\mathcal{C}(X, S)$, seen as an R -algebra, is split by σ , for every profinite space X . Explicitly, the splitting condition for an R -algebra A is given by the unit η of the adjunction $Sp \dashv \mathcal{C}(-, S)$, described in Proposition 2.2.1, to be an isomorphism of S -algebras on $S \otimes_R A$:

$$S \otimes_R A \cong \mathcal{C}(Sp(S \otimes_R A), S).$$

Finally, let us show another property that arises when S has exactly two idempotent elements regarding the Galois groupoid:

Proposition 2.2.5. *In the setting above consider the Galois groupoid $\text{Gal}[\sigma]$ of σ . Suppose S has exactly two idempotent elements; then $\text{Gal}[\sigma]$ is a profinite group. Moreover, there is an isomorphism of categories*

$$\mathbf{Prof}^{\text{Gal}[\sigma]} \cong \text{Gal}[\sigma]\text{-}\mathbf{Prof}$$

between the category of internal covariant presheaves on $\text{Gal}[\sigma]$ in $\mathbf{Prof}$ and the category of profinite spaces provided with a continuous group action of $\text{Gal}[\sigma]$.

Proof. Let us recall Definition 1.3.6 in this setting:

$$\begin{array}{ccc}
& & \text{Sp}(p_1) \\
& & \curvearrowright \\
\text{Sp}(S \otimes_R S) \otimes_{\text{Sp}(S)} \text{Sp}(S \otimes_R S) & \xrightarrow{\langle \text{Sp}(\pi_1), \text{Sp}(\pi_4) \rangle} & \text{Sp}(S \otimes_R S) \xleftarrow{\text{Sp}(\Delta)} \text{Sp}(S) \\
& \text{Sp}(\tau) \curvearrowleft & \text{Sp}(p_2) \curvearrowright
\end{array}$$

By Proposition 2.2.3, $\text{Sp}(S) = \{*\}$ is a singleton and so the Galois groupoid has only one object; it is isomorphic to an internal group in **Prof**, given by $\text{Sp}(S \otimes_R S)$, that is a profinite group. Moreover, an internal covariant presheaf (P, p, π) on $\text{Gal}[\sigma]$ in **Prof** is a profinite space P , along with the unique map $p : P \rightarrow \{*\}$ and a continuous map

$$\pi : \text{Sp}(S \otimes_R S) \times P \rightarrow P.$$

The axioms in Definition 1.2.3 correspond exactly to the axioms for π to be a group action on P of $\text{Gal}[\sigma] \cong \text{Sp}(S \otimes_R S)$. Furthermore, an internal natural transformation $\gamma : P \Rightarrow P'$ in **Prof** is a continuous map $\gamma : P \rightarrow P'$ and the axioms in Definition 1.2.5 are exactly the requirement that γ is compatible with the actions π and π' . $\square$

To conclude, we can state Theorem 2.2.2 once again with the idempotents condition considered on S and explicitly stating the assumption and the conclusion as analysed in this section.

Theorem 2.2.6. *Let $\sigma : R \rightarrow S$ be a ring homomorphism such that*

1. *S has exactly two idempotent elements;*
2. *the functor $S \otimes_R - : \mathbf{R}\text{-Alg}^{\text{op}} \rightarrow \mathbf{S}\text{-Alg}^{\text{op}}$ is monadic;*
3. *for every profinite space X , the unit η of the adjunction $\text{Sp} \dashv \mathcal{C}(-, S)$ is an isomorphism on $S \otimes_R \mathcal{C}(X, S)$; explicitly*

$$S \otimes_R \mathcal{C}(X, S) \cong \mathcal{C}(\text{Sp}(S \otimes_R \mathcal{C}(X, S)), S).$$

Then there is an equivalence of categories given by

$$\text{Sp}(S \otimes_R -) : \text{Split}_R(\sigma)^{\text{op}} \longrightarrow \text{Gal}[\sigma]\text{-Prof}$$

between the opposite category of R -algebras split by σ and the category of profinite spaces along with a continuous group action of $\text{Gal}[\sigma] \cong \text{Sp}(S \otimes_R S)$.

2.3 The field case: recovering the classical Galois correspondence

In this section, we further specialize to the field case. Specifically, we consider the same setting developed in Section 2.2, now with $R = K$ and $S = L$, where $\sigma : K \hookrightarrow L$ is a field extension. In particular, as assumed at the end of the previous section, L has exactly two idempotent elements. The purpose of this section is to recover completely the classical theories that inspired the categorical theorem of Chapter 1 and to derive the classical Galois correspondence in the context of a finite Galois extension of fields. To do so, we first establish that a Galois extension of fields satisfies the assumption on the morphism $\sigma : K \hookrightarrow L$ required by Theorem 2.2.6.

Definition 2.3.1. Let $\sigma : K \hookrightarrow L$ be a field extension. σ is Galois when

1. for every $l \in L$, there exists a non-zero polynomial $p(X) \in K[X]$ such that $p(l) = 0$. Among all such polynomials, there is a unique one of minimal degree with leading coefficient 1, which we call the minimal polynomial of l over K ;
2. for every $l \in L$, the minimal polynomial of l over K factors entirely in $L[X]$ in polynomials of degree 1 with distinct roots.

It follows from Definition 2.1.7 that:

$$\sigma : K \hookrightarrow L \text{ is Galois} \iff L \text{ is a } K\text{-algebra split by } L.$$

Lemma 2.3.1. *Let $\sigma : K \hookrightarrow L$ be a Galois extension of fields. L is a filtered colimit of its finite dimensional Galois subextension $K \hookrightarrow M \hookrightarrow L$. Analogously, every K -algebra A split by L is a filtered colimit of its finite dimensional subalgebras $B \subset A$, each of them being itself split by a finite dimensional Galois subextension of L .*

Proof. If $l \in L$ has a minimal polynomial $p(X) \in K[X]$ with roots $l_1, \dots, l_n \in L$, then

$$l \in K(l_1, \dots, l_n) \hookrightarrow L$$

where $K \hookrightarrow K(l_1, \dots, l_n)$ is a finite dimensional Galois extension, so that L is the set-theoretical union of its finite dimensional Galois subextension. Moreover, it is filtered since the subalgebra generated by two finite dimensional Galois extension is still a finite dimensional Galois extension of K . A similar argument applies in the second case: if $a \in A$ has a minimal polynomial $p(X) \in K[X]$ with roots $a_1, \dots, a_n \in L$, then $a \in B$, where B is the finite K -algebra generated by $a_1, \dots, a_n$. Furthermore, B is split by the finite dimensional Galois subextension $K \hookrightarrow K(a_1, \dots, a_n) \hookrightarrow L$. $\square$

Observation. We have given a profinite structure to $\text{Hom}_K(A, L)$ for every K -algebra A split by L . Indeed, recall that the functor $\text{Hom}_K(-, -) : \mathbf{K}\text{-Alg}^{\text{op}} \times \mathbf{K}\text{-alg} \longrightarrow \mathbf{Set}$ preserves limits in both arguments, so that

$$\text{Hom}_K(A, L) \cong \text{Hom}_K(\text{colim}_B B, L) \cong \lim_B \text{Hom}_K(B, L).$$

Moreover, $\text{Hom}_K(B, L) \cong \text{Hom}_K(B, M)$ for some finite dimensional subextension $K \hookrightarrow M \hookrightarrow L$ and so it is a finite set.

Lemma 2.3.2. *The functor $Sp : \mathbf{K}\text{-Alg}^{\text{op}} \longrightarrow \mathbf{Prof}$ preserves cofiltered limits.*

Proof. Applying Stone duality, we need to prove that the functor

$$Idemp : \mathbf{K}\text{-Alg} \longrightarrow \mathbf{Bool}$$

preserves filtered colimits. Both categories are algebraic, thus colimits are computed as in $\mathbf{Set}$, and $Idemp$ is the factorization, through $\mathbf{Bool}$, of the representable functor

$$\text{Hom}(K[X]/\langle X^2 \rangle, -) : \mathbf{K}\text{-Alg} \longrightarrow \mathbf{Set}.$$

This representable functor preserves filtered colimits, because $K[X]/\langle X^2 \rangle$ is finitely presented as a K -algebra (see [1]). $\square$

Before moving on, let us underline that in general, since $Sp : \mathbf{K}\text{-}\mathbf{Alg}^{\text{op}} \longrightarrow \mathbf{Prof}$ is a left adjoint, it preserves colimits. Explicitly, Sp transforms limits in $\mathbf{K}\text{-}\mathbf{Alg}$ into colimits in $\mathbf{Prof}$ since it is a left adjoint, and filtered colimits in $\mathbf{K}\text{-}\mathbf{Alg}$ into cofiltered limits in $\mathbf{Prof}$ because of the previous Lemma.

Proposition 2.3.3. *Let $\sigma : K \hookrightarrow L$ be a Galois extension of fields. Then $\sigma : L \rightarrow K$ is a morphism in $\mathbf{K}\text{-}\mathbf{Alg}^{\text{op}}$ of relative Galois descent.*

Proof. We have seen that to be of relative Galois descent, we need to establish the following:

1. the functor $L \otimes_K - : \mathbf{K}\text{-}\mathbf{Alg}^{\text{op}} \longrightarrow \mathbf{L}\text{-}\mathbf{Alg}^{\text{op}}$ is monadic;
2. the object $\mathcal{C}(X, L)$, seen as an K -algebra, is split by σ , for every profinite space X . Explicitly,

$$L \otimes_K \mathcal{C}(X, L) \cong \mathcal{C}(Sp(L \otimes_K \mathcal{C}(X, L)), L).$$

To prove the first condition, recall that L is faithfully flat as a K -module, as stated in Proposition 2.1.4. Since the forgetful functors $\mathbf{K}\text{-}\mathbf{Alg} \longrightarrow \mathbf{Mod}_K$ and $\mathbf{L}\text{-}\mathbf{Alg} \longrightarrow \mathbf{Mod}_L$ preserve and reflect exact sequences, the restriction of the functor $L \otimes_K -$ to the categories of algebras preserves and reflects exact sequences. Moreover, since $\mathbf{K}\text{-}\mathbf{Alg}$ and $\mathbf{L}\text{-}\mathbf{Alg}$ have equalizers computed as in $\mathbf{Mod}_K$ and $\mathbf{Mod}_L$, the functor

$$L \otimes_K - : \mathbf{K}\text{-}\mathbf{Alg} \longrightarrow \mathbf{L}\text{-}\mathbf{Alg}$$

reflects isomorphisms and preserves equalizers. The monadicity follows from Beck's criterion. About the second condition, we will prove that for every K -algebra A split by L and for every profinite space X , the morphism

$$A \otimes_K \mathcal{C}(X, L) \longrightarrow \mathcal{C}(Sp(A \otimes_K \mathcal{C}(X, L)), L) \tag{2.5}$$

is an isomorphism of L -algebras. Since σ is a Galois extension, L is split by L as a K -algebra, and putting $A = L$ we get the desired isomorphism. By Lemma 2.3.1, we have

$$A \otimes_K L \cong (\text{colim}_B B) \otimes_K (\text{colim}_M M) \cong \text{colim}_{(B,M)} B \otimes_K M$$

where B and M are finite dimensional over K , and by Proposition 2.2.3

$$\mathcal{C}(X, L) \cong L \otimes_K \mathcal{C}(X, K).$$

Note that the previous isomorphism holds even for the finite extension $K \hookrightarrow M$. Recall that $\mathcal{C}(-, K) : \mathbf{Prof} \longrightarrow \mathbf{K}\text{-}\mathbf{Alg}^{\text{op}}$ transforms limits in $\mathbf{Prof}$ into colimits in $\mathbf{K}\text{-}\mathbf{Alg}$, since it is a right adjoint. Now, suppose (2.5) holds in the finite dimensional case, we have the following chain of

isomorphisms:

$$\begin{aligned}
A \otimes_K \mathcal{C}(X, L) &\cong A \otimes_K L \otimes_K \mathcal{C}(X, K) \\
&\cong \operatorname{colim}_{(B, M)} B \otimes_K M \otimes_K \mathcal{C}(X, K) \\
&\cong \operatorname{colim}_{(B, M)} B \otimes_K \mathcal{C}(X, M) \\
(*) &\cong \operatorname{colim}_{(B, M)} \mathcal{C}(Sp(B \otimes_K \mathcal{C}(X, M)), M) \\
&\cong \operatorname{colim}_{(B, M)} M \otimes_K \mathcal{C}(Sp(B \otimes_K \mathcal{C}(X, M)), K) \\
&\cong L \otimes_K \operatorname{colim}_{(B, M)} \mathcal{C}(Sp(B \otimes_K M \otimes_K \mathcal{C}(X, K)), K) \\
&\cong L \otimes_K \mathcal{C}(\lim_{(B, M)} Sp(B \otimes_K M \otimes_K \mathcal{C}(X, K)), K) \\
(\text{Lemma 2.3.2}) &\cong L \otimes_K \mathcal{C}(Sp(\operatorname{colim}_{(B, M)} B \otimes_K M \otimes_K \mathcal{C}(X, K)), K) \\
&\cong L \otimes_K \mathcal{C}(Sp(A \otimes_K L \otimes_K \mathcal{C}(X, K)), K) \\
&\cong L \otimes_K \mathcal{C}(Sp(A \otimes_K \mathcal{C}(X, L)), K) \\
&\cong \mathcal{C}(Sp(A \otimes_K \mathcal{C}(X, L)), L).
\end{aligned}$$

It remains to prove the argument (*): it is (2.5) in the finite dimensional case. In this setting, $B \otimes_K M \cong M^n$ as in Proposition 2.1.5, where n is the dimension of B as a K -algebra. We have the following:

$$\begin{aligned}
B \otimes_K \mathcal{C}(X, M) &\cong B \otimes_K M \otimes_K \mathcal{C}(X, K) \\
&\cong M^n \otimes_K \mathcal{C}(X, K) \\
&\cong (M \otimes_K \mathcal{C}(X, K))^n \\
&\cong \mathcal{C}(X, M)^n = \prod_n \mathcal{C}(X, M) \\
&\cong \mathcal{C}(\prod_n X, M) \\
(**) &\cong \mathcal{C}(\prod_n Sp(\mathcal{C}(X, M)), M) \\
&\cong \mathcal{C}(Sp(\prod_n \mathcal{C}(X, M)), M) \\
&\cong \mathcal{C}(Sp(B \otimes_K \mathcal{C}(X, M)), M)
\end{aligned}$$

where we used Proposition 2.2.3 for the isomorphism $X \cong Sp(\mathcal{C}(X, M))$ in the argument (**). $\square$

Observation. Note that the first part of the proof required only that L is faithfully flat as a K -module. Indeed, the monadicity of the pullback functor holds also in the case of rings, if S is faithfully flat as an R -module. More generally, in a Barr-exact category, the pullback functor σ^* is monadic if and only if σ is a regular epimorphism and, in an algebraic category, it is the case if and only if σ is a surjection.

We have proved that Theorem 2.2.6 applies for a Galois extension of fields. Now, let us proceed to establish the equivalence between $Split_K(L)$ and $Split_K(\sigma)$, and to gain a clearer view of the functor that realizes the Galois equivalence.

Proposition 2.3.4. *Let $\sigma : K \hookrightarrow L$ be a Galois extension of fields. There is an isomorphism of categories*

$$\text{Split}_K(L) \cong \text{Split}_K(\sigma)$$

between the full subcategory of K -algebras split by L and the full subcategory of K -algebras split by σ .

Proof. It is sufficient to prove that for every K -algebra A

$$A \text{ is split by } \sigma \iff A \text{ is split by } L.$$

In proposition 2.3.3 we have established that for every K -algebra A split by L and for every profinite space X

$$A \otimes_K \mathcal{C}(X, L) \longrightarrow \mathcal{C}(\text{Sp}(A \otimes_K \mathcal{C}(X, L)), L) \quad (2.5)$$

is an isomorphism of L -algebras. Putting $X = \{*\}$ gives us $\mathcal{C}(X, L) \cong L$. Thus,

$$A \otimes_K L \cong \mathcal{C}(\text{Sp}(A \otimes_K L), L)$$

establishing that A is also split by σ . On the other hand, if A is split by σ , by Corollary 1.3.3.1, there exists a profinite space X such that

$$L \otimes_K A \cong \mathcal{C}(X, L) \cong L \otimes_K \mathcal{C}(X, K).$$

It is sufficient to prove for every profinite space X , that $\mathcal{C}(X, K)$ is a K -algebra split by L . Write $X = \lim_{i \in I} X_i$ as a cofiltered limit of finite discrete spaces. For every $i \in I$, X_i and K are discrete spaces, so that

$$\mathcal{C}(X_i, K) = \text{Hom}(X_i, K) \cong K^{n_i},$$

where $n_i = \#X_i$. By Proposition 2.1.5, $\mathcal{C}(X_i, K)$ is split by L for every $i \in I$. Moreover,

$$\mathcal{C}(X, K) = \mathcal{C}(\lim_{i \in I} X_i, K) \cong \text{colim}_{i \in I} \mathcal{C}(X_i, K)$$

and it is a filtered colimit in **K-Alg**, so that it is computed as in **Set**. Let us show that a filtered colimit A of K -algebras A_i split by L remains split by L : each $a \in A$ has the form $[a_i]$ for some $i \in I$ and $a_i \in A_i$; the minimal polynomial $p(X)$ of a over K is a factor of the minimal polynomial $p_i(X)$ of a_i over K . Since A_i is split by L , p_i factors in $L[X]$ into different factors of degree 1, so that $p(X)$ also factors in $L[X]$ in polynomials of degree 1 with distinct roots. $\square$

Proposition 2.3.5. *Let $\sigma : K \hookrightarrow L$ be a Galois extension of fields. The functor*

$$\text{Sp}(L \otimes_K -) : \text{Split}_K(L)^{\text{op}} \longrightarrow \mathbf{Prof}$$

is isomorphic to the functor

$$\text{Hom}_K(-, L) : \text{Split}_K(L)^{\text{op}} \longrightarrow \mathbf{Prof}.$$

Proof. When the K -algebra A is finite dimensional, we know $\text{Hom}_K(A, L) \cong \text{Hom}_K(A, M)$ for some finite subextension $K \hookrightarrow M \hookrightarrow L$ and this isomorphism is clearly natural in A . Moreover, $\text{Hom}_K(A, M)$ is a finite discrete space of cardinality n , where n is the dimension of A as a K -algebra. From Proposition 2.1.5, $L \otimes_K A \cong L^n$ and this isomorphism is again natural in A . Since

L has only 0 and 1 as idempotents, one has also that $Sp(L \otimes_K A) \cong Sp(L^n)$ is a finite discrete space of cardinality n , establishing a natural isomorphism

$$Sp(L \otimes_K A) \cong \text{Hom}_K(A, L)$$

for a finite dimensional K -algebra A . Now, when A is infinite dimensional, $A \cong \text{colim}_B B$ in $\mathbf{K}\text{-Alg}$, where B runs through the finite dimensional subalgebras of A . Applying 2.3.2, we obtain

$$\begin{aligned} Sp(L \otimes_K A) &\cong Sp(\text{colim}_B L \otimes_K B) \cong \lim_B Sp(L \otimes_K B) \\ &\cong \lim_B \text{Hom}_K(B, L) \cong \text{Hom}_K(\text{colim}_B B, L) \cong \text{Hom}_K(A, L). \end{aligned}$$

□

Recall that $\sigma : K \hookrightarrow L$ is Galois if and only if L is split by itself as a K -algebra, thus we obtain a clearer form of the Galois group:

$$Gal[\sigma] \cong Sp(L \otimes_K L) \cong \text{Hom}_K(L, L) = \text{Aut}_K(L) =: Gal[L : K].$$

This last two arguments are exactly the relation between the classical Galois theory of fields and the categorical Galois theory developed in this thesis. We moved from the categorical Theorem 1.3.7 to Theorem 2.2.2, fixing an adjunction and a morphism in the category of rings. When we looked into the special case of fields (actually, rings with exactly two idempotents), we got Theorem 2.2.6: an equivalence of categories

$$Sp(L \otimes_K -) : \text{Split}_K(\sigma)^{\text{op}} \longrightarrow Gal[\sigma]\text{-Prof}$$

with the Galois group presented as $Sp(L \otimes_K L)$. Now, with Propositions 2.3.4 and 2.3.5, we get that

$$\text{Hom}_K(-, L) : \text{Split}_K(L)^{\text{op}} \longrightarrow Gal[L : K]\text{-Prof}$$

is the same equivalence of categories, but this time in a full field-theoretical version. Finally, we can state the Galois theorem with the classical notation.

Theorem 2.3.6 (Galois theorem of fields). *Let $\sigma : K \hookrightarrow L$ be a Galois extension of fields. There exists an equivalence of categories*

$$\text{Hom}_K(-, L) : \text{Split}_K(L)^{\text{op}} \longrightarrow Gal[L : K]\text{-Prof}$$

between the dual of the category of K -algebras split by L and the category of profinite spaces provided with a continuous group action of $Gal[L : K] \cong \text{Aut}_K(L)$.

If we restrict our work on the finite dimensional case, recall that a finite profinite space is a finite discrete space, so that we obtain the following:

Theorem 2.3.7. *Let $\sigma : K \hookrightarrow L$ be a finite Galois extension of fields. There exists an equivalence of categories*

$$\text{Hom}_K(-, L) : \text{Split}_K(L)_f^{\text{op}} \longrightarrow Gal[L : K]\text{-Set}_f$$

between the dual of the category of finite dimensional K -algebras split by L and the category of finite $Gal[L : K]$ -sets, where $Gal[L : K] \cong \text{Aut}_K(L)$.

To conclude the chapter, let us consider the special case of a finite dimensional K -algebra split by L that it is also a field. In this scenario, the previous equivalence gives us a contravariant isomorphism between the lattice of subobjects M

$$K \hookrightarrow M \hookrightarrow L$$

in $\text{Split}_K(L)_f$ and the lattice of quotients

$$\text{Gal}[L : K] \cong \text{Hom}_K(L, L) \twoheadrightarrow \text{Hom}_K(M, L) \twoheadrightarrow \text{Hom}_K(K, L) = \{\sigma\}$$

in $\text{Gal}[L : K]\text{-Set}_f$. By Proposition 2.1.3, there is a bijection between the quotients

$$\text{Gal}[L : K] \twoheadrightarrow \text{Hom}_K(M, L)$$

and the subgroups of those $g \in \text{Aut}_K(L)$ that are the identity when restricted to M . In this way, we have established a contravariant isomorphism between the subfields $K \hookrightarrow M \hookrightarrow L$ and the subgroups $\text{Aut}_M(L) \leq \text{Aut}_K(L)$ retrieving the classical Galois correspondence.

Theorem 2.3.8 (Classical Galois correspondence). *Let $\sigma : K \hookrightarrow L$ be a finite Galois extension of fields. There is a contravariant isomorphism*

$$\{M : K \hookrightarrow M \hookrightarrow L\} \longleftrightarrow \{G : G \leq \text{Aut}_K(L)\}$$

between the intermediate field extensions $K \hookrightarrow M \hookrightarrow L$ and the subgroups G of $\text{Aut}_K(L)$. Explicitly, it is given by

$$M \longmapsto \text{Aut}_M(L)$$

and

$$G \longmapsto \text{Fix}(G) = \{l \in L : g(l) = l, \forall g \in G\}$$

Chapter 3

Galois theory for monoids

In this chapter, we apply the work of Chapter 1 to an adjunction between the category of monoids and the category of groups. In particular, in Section 3.1, we restrict our work to an admissible class of arrows in order to satisfy some conditions of Theorem 1.3.7: we prove that the adjunction is admissible with respect to surjections in both categories and that we cannot consider all morphisms as we did in Chapter 2. In Section 3.2, we introduce the notions of trivial, central, and normal extensions related to a Galois structure; these turns out to be just a reformulation of the categorical notion of splitting we gave in Section 1.3. In Section 3.3, we characterize the extensions with respect to the monoid Galois structure considered in Section 3.1, and describe the Galois groupoid in that setting. In particular, central extensions are precisely the so-called special homogeneous surjections, and the Galois groupoid $Gal[\sigma]$ of σ is uniquely determined by a single group: the kernel of σ . Before starting, we underline that the work in this chapter is primarily based on the recent paper *A Galois theory for monoids* by Montoli, Rodelo and Van der Linden [14]. Throughout this chapter, we restrict ourselves to Barr-exact categories, and we will always take canonical kernels and pullbacks when considering the algebraic setting of monoids or groups.

3.1 Galois structure for monoids

In this section, we develop the basis for a Galois theory for monoids. We define an adjunction between the category of monoids and the category of groups and will prove that it is admissible with respect to a specific class of admissible arrows. In particular, in an analysis of the hypothesis of Theorem 1.3.7, one can see that there are two possible points of view: the first one is to consider a larger class of admissible arrows $\overline{\mathcal{A}}$ (for example, the class of all morphisms satisfies Definition 1.3.1) and to consider a morphism $\sigma : S \rightarrow R$ that meets all the requirements in Definition 1.3.5; the second one is to restrict our work into a smaller class of admissible arrows, such that the requirement $\sigma \in \overline{\mathcal{A}}$ gives us already some conditions to be of relative Galois descent. In this way, we will first consider only the adjunction and the admissible classes of arrows, in order to describe the morphisms involved in Theorem 1.3.7 in later sections.

Definition 3.1.1. A Galois structure $\Gamma = (\mathcal{A}, \mathcal{P}, \mathcal{S}, \mathcal{C}, \eta, \varepsilon, \overline{\mathcal{A}}, \overline{\mathcal{P}})$ consists of an adjunction

$$\begin{array}{ccc} & \mathcal{S} & \\ \mathcal{A} & \xrightarrow{\quad} & \mathcal{P} \\ & \mathcal{C} & \end{array} \quad \begin{array}{c} \perp \\ \leftarrow \end{array}$$

with unit $\eta : 1_{\mathcal{A}} \Rightarrow \mathcal{CS}$ and counit $\varepsilon : \mathcal{SC} \Rightarrow 1_{\mathcal{P}}$ between categories $\mathcal{A}$ and $\mathcal{P}$, as well as classes of morphisms $\overline{\mathcal{A}}$ in $\mathcal{A}$ and $\overline{\mathcal{P}}$ in $\mathcal{P}$ such that:

1. $\overline{\mathcal{A}}$ and $\overline{\mathcal{P}}$ contain all isomorphisms;
2. $\overline{\mathcal{A}}$ and $\overline{\mathcal{P}}$ are pullback-stable
3. $\overline{\mathcal{A}}$ and $\overline{\mathcal{P}}$ are closed under composition
4. $\mathcal{S}(\overline{\mathcal{A}}) \subset \overline{\mathcal{P}}$;
5. $\mathcal{C}(\overline{\mathcal{P}}) \subset \overline{\mathcal{A}}$.

Observation. Note that a Galois structure is what we called a relatively admissible adjunction in Definition 1.3.2. The only differences are the requirements for the unit and the counit to be in the admissible classes, but in general it is not a necessary condition to establish a Galois correspondence. Moreover, in the algebraic setting where we will work there are no differences with the results of Chapter 1, even without this assumption.

Considering Definition 1.3.5, one can note that the requirement for the counit of the induced adjunction to be an isomorphism is not exactly on the morphism $\sigma : S \rightarrow R$, but it is a request on the adjunction induced on the domain S of σ . In this new point of view, we do not want it to be a condition on the morphism, but to be a condition on the adjunction: in this way, we define an admissible Galois structure.

Definition 3.1.2. Let $\Gamma = (\mathcal{A}, \mathcal{P}, \mathcal{S}, \mathcal{C}, \eta, \varepsilon, \overline{\mathcal{A}}, \overline{\mathcal{P}})$ be a Galois structure. Γ is said to be admissible when for every $S \in \text{Ob}(\mathcal{A})$, the induced functor

$$\mathcal{C}_S : \overline{\mathcal{P}}/\mathcal{S}(S) \longrightarrow \overline{\mathcal{A}}/S$$

is full and faithful.

Example 3.1.3. In Section 2.2 we studied the Galois structure

$$(\mathbf{R}\text{-}\mathbf{Alg}^{\text{op}}, \mathbf{Prof}, Sp, \mathcal{C}(-, R), \eta, \varepsilon, \overline{\mathbf{R}\text{-}\mathbf{Alg}^{\text{op}}}, \overline{\mathbf{Prof}}),$$

with $\overline{\mathbf{R}\text{-}\mathbf{Alg}^{\text{op}}}$ and $\overline{\mathbf{Prof}}$ the classes of all arrows in their respective categories, and we stated that it is admissible in Proposition 2.2.4. Moreover, when we restricted ourselves to field extensions in Section 2.3, the monadicity of the pullback functor automatically followed from the intrinsic properties of the considered morphism, without the need to impose it as an explicit assumption.

Let us now define a Galois structure for monoids. We will restrict ourselves to specific classes of arrows so that the only requirement on a morphism $\sigma : S \rightarrow R$ will be the third in Definition 1.3.5. Explicitly, we will work in a Galois structure where every morphism has a monadic pullback functor and we will prove that the Galois structure considered is admissible.

Definition 3.1.4. Let $(M, \cdot, 1)$ be a monoid. The Grothendieck group (or group completion) of M is given by a group $Gp(M)$ together with a monoid homomorphism $M \rightarrow Gp(M)$ which is universal with respect to monoid homomorphisms from M to groups. More precisely,

$$Gp(M) = \frac{GpF(M)}{N(M)},$$

where $GpF(M)$ denotes the free group on M and $N(M)$ the normal subgroup generated by elements of the form $[m_1][m_2][m_1 \cdot m_2]^{-1}$. This gives us that the concatenation of words is

compatible with the monoid structure on M . Thus, an arbitrary element in $Gp(M)$ can be represented by a word of the form

$$[m_1][m_2]^{-1}[m_3][m_4]^{-1} \cdots [m_n]^{\iota(n)} \text{ or } [m_1]^{-1}[m_2][m_3]^{-1}[m_4] \cdots [m_n]^{\iota(n)},$$

where $\iota(n) = \pm 1$, $n \in \mathbb{N}$, $m_1, \dots, m_n \in M$ and no further cancellation is possible.

Observation. In order to be completely explicit, let us note the difference between $GpF(M)$ and $Gp(M)$. We will denote a class in $GpF(M)$ with $[]$ and a class in the quotient $Gp(M)$ with $\overline{[]}$. For example, fix $m_1, m_2 \in M$ and $m_3 := m_1 \cdot m_2$. One has

$$[m_1][m_2] \neq [m_3], \text{ but } \overline{[m_1][m_2]} = \overline{[m_3]},$$

since $[m_1][m_2]$ and $[m_3]$ are simply different words with letters in M . In the construction of the free group, the monoid structure on M is not considered. What we ask for is the compatibility between the concatenation of words and the operation in M . This also implies

$$\overline{[m]^{-1}} = \overline{[m^{-1}]},$$

for every invertible $m \in M^*$. Thus, if $M = G$ is a group, $Gp(G) \cong G$ because an arbitrary element of $Gp(G)$ can be represented as $[g]$, for a single $g \in G$.

Definition 3.1.5. Let **Mon** and **Grp** represent the categories of monoids and groups, respectively. The group completion of a monoid determines an adjunction

$$\begin{array}{ccc} & Gp & \\ \text{Mon} & \xrightarrow{\quad} & \text{Grp} \\ & U & \end{array} \quad \perp$$

where U is the forgetful functor. To simplify the notation, we write $Gp(M)$ instead of $U(Gp(M))$ when referring to the monoid structure of $Gp(M)$.

The counit is $\varepsilon = 1_{\mathbf{Grp}}$ and the unit is defined by

$$\eta_M : M \longrightarrow Gp(M), m \longmapsto \overline{[m]}.$$

Recall that in a Barr-exact category regular epimorphisms are exactly the effective descent morphisms (see [6]) and in an algebraic category regular epimorphisms are exactly the surjections. If we restrict ourselves to the classes of surjections in **Mon** and **Grp**, we get at once that every surjection $\sigma : M \twoheadrightarrow N$ has a monadic pullback functor. Obviously, in the considered categories, these classes satisfy the conditions in Definition 3.1.1. We consider the Galois structure

$$\Gamma = (\mathbf{Mon}, \mathbf{Grp}, Gp, U, \eta, 1_{\mathbf{Grp}}, \mathcal{E}, \mathcal{F})$$

with $\mathcal{E}$ and $\mathcal{F}$ the classes of surjections in **Mon** and **Grp** respectively.

We will prove that this Galois structure is admissible, but before we do so, let us explicitly exhibit the induced adjunction by a monoid M

$$\begin{array}{ccc} & Gp_M & \\ \mathcal{E}/M & \xrightarrow{\quad} & \mathcal{F}/Gp(M), \\ & U_M & \end{array} \quad \perp$$

where we write $\mathcal{E}/M$ and $\mathcal{F}/Gp(M)$ for the full subcategories of admissible arrows as we did in Chapter 1. Here Gp_M is the restriction of Gp , and U_M sends (A, f) to the pullback in **Mon**

$$\begin{array}{ccc} M \times_{Gp(M)} A & \xrightarrow{\pi_A} & A \\ \downarrow U_M(f) & \lrcorner & \downarrow f \\ M & \xrightarrow{\eta_M} & Gp(M) \end{array}$$

of f along η_M .

Observation. Note that in the proof of Lemma 1.3.2 we used the fact that the unit is admissible to prove that $U_M(f)$ is admissible, but in this setting regular epimorphisms are preserved by any pullback.

Given a morphism $\alpha : (A, f) \rightarrow (B, g)$ in $\mathcal{F}/Gp(M)$, its image $U_M(\alpha)$ is given by the universal property of the pullback:

$$\begin{array}{ccccc} M \times_{Gp(M)} A & \xrightarrow{\pi_A} & A & & \\ \downarrow U_M(f) & \searrow U_M(\alpha) & \downarrow & \searrow \alpha & \\ & M \times_{Gp(M)} B & \xrightarrow{\pi_B} & B & \\ & \downarrow U_M(g) & \downarrow f & \searrow g & \\ M & \xrightarrow{\eta_M} & Gp(M) & & \end{array} \quad (3.1)$$

Proposition 3.1.1. *The Galois structure $\Gamma = (\mathbf{Mon}, \mathbf{Grp}, Gp, U, \eta, 1_{\mathbf{Grp}}, \mathcal{E}, \mathcal{F})$ is admissible.*

Proof. We split the proof into three lemmas. In Lemma 3.1.2 we prove that U_M is faithful, in Lemma 3.1.3 we prove that U_M is full if M is a free monoid and finally, in Lemma 3.1.4 we extend the proof to the general case. $\square$

Lemma 3.1.2. *The functor U_M is faithful for all monoids M .*

Proof. Consider two morphisms $\alpha, \beta : (A, f) \rightarrow (B, g)$ such that $U_M(\alpha) = U_M(\beta)$. For any $a \in A$, the class $f(a)$ in $Gp(M)$ is represented by a word of the form

$$[m_1][m_2]^{-1}[m_3][m_4]^{-1} \cdots [m_n]^{\iota(n)} \text{ or } [m_1]^{-1}[m_2][m_3]^{-1}[m_4] \cdots [m_n]^{\iota(n)},$$

where no further cancellation is possible. We prove $\alpha(a) = \beta(a)$ by induction on the length n of the word that represents $f(a)$.

If $n = 1$ and $f(a) = \overline{[m]}$, then $(m, a) \in M \times_{Gp(M)} A$ and $U_M(\alpha)(m, a) = U_M(\beta)(m, a)$ implies

$$\alpha(a) = \pi_B(U_M(\alpha)(m, a)) = \pi_B(U_M(\beta)(m, a)) = \beta(a).$$

If $n = 1$ and $f(a) = \overline{[m]^{-1}}$, then $f(a^{-1}) = \overline{[m]}$ and from the previous case $\alpha(a^{-1}) = \beta(a^{-1})$; hence $\alpha(a) = \beta(a)$.

If $n > 1$, $f(a)$ is represented by the concatenation of a word of length one and a word of length $n - 1$. By the surjectivity of f , there exists $a_1 \in A$ such that $f(a_1) = \overline{[m_1]}$ (or $\overline{[m_1]}^{-1}$ in the second case); hence $f(a_1^{-1}a)$ is represented by the second part of the representation of $f(a)$ of length $n - 1$. By the induction hypothesis, $\alpha(a_1) = \beta(a_1)$ and $\alpha(a_1^{-1}a) = \beta(a_1^{-1}a)$. Then

$$\alpha(a) = \alpha(a_1)\alpha(a_1^{-1}a) = \beta(a_1)\beta(a_1^{-1}a) = \beta(a).$$

□

Lemma 3.1.3. *The functor U_M is full for all free monoids M .*

Proof. Since M is free, the group $Gp(M)$ is free and coincide with $GpF(M)$. Consider a monoid homomorphism

$$\gamma : (M \times_{Gp(M)} A, U_M(f)) \longrightarrow (M \times_{Gp(M)} B, U_M(g)).$$

We define a group homomorphism $\alpha : (A, f) \rightarrow (B, g)$ in $\mathcal{F}/Gp(M)$ as follows. For any $a \in A$, if $f(a) = [m]$, then $(m, a) \in M \times_{Gp(M)} A$ and we define

$$\alpha(a) := \pi_B(\gamma(m, a)).$$

If $f(a) = [m]^{-1}$, then $f(a^{-1}) = [m]$ and we define

$$\alpha(a) := \pi_B(\gamma(m, a^{-1}))^{-1}.$$

In the general case, $f(a)$ is a concatenation of n words of length one $[m_i]^{\iota(i)}$, with $\iota(i) = \pm 1$. By surjectivity of f , there exist $a_1, \dots, a_n \in A$ such that $f(a_i) = [m_i]$ for every $i = 1, \dots, n$; hence, we have a decomposition $a = a_1 a_2^{-1} \dots a_n^{\iota(n)}$ or $a = a_1^{-1} a_2 \dots a_n^{\iota(n)}$. Note that each of these a_i is an element in the image of π_A since $(m_i, a_i) \in M \times_{Gp(M)} A$ for every $i = 1, \dots, n$. We can take the minimum n for which there is a decomposition of a into a product of elements in the image of π_A , and define

$$\alpha(a) := \alpha(a_1)\alpha(a_2)^{-1} \dots \alpha(a_n)^{\iota(n)}$$

or

$$\alpha(a) = \alpha(a_1)^{-1}\alpha(a_2) \dots \alpha(a_n)^{\iota(n)}.$$

We will consider only the first case in the decomposition; the second case can be studied similarly. To prove that α is a group homomorphism, it is sufficient to show that it is well defined. If $a = x_1 x_2^{-1} \dots x_k^{\iota(k)}$ is a second decomposition of a into elements in the image of π_A , then there are $l_i \in M$ such that $f(x_i) = [l_i]$, and we need to prove that

$$\alpha(a_1)\alpha(a_2)^{-1} \dots \alpha(a_n)^{\iota(n)} = \alpha(x_1)\alpha(x_2)^{-1} \dots \alpha(x_k)^{\iota(k)}.$$

Since $Gp(M)$ is a free group, n is the minimum length and f is a group homomorphism, considering the image

$$[m_1][m_2]^{-1} \dots [m_n]^{\iota(n)} = f(a_1 a_2^{-1} \dots a_n^{\iota(n)}) = f(a) = f(x_1 x_2^{-1} \dots x_k^{\iota(k)}) = [l_1][l_2]^{-1} \dots [l_k]^{\iota(k)},$$

we get that $k = n$ and $l_i = m_i$ for every $i = 1, \dots, n$. In this way, we need to prove the result for decompositions of length n mapping down to the same word in $Gp(M)$ through f . We do this by induction on n .

If $n = 1$ and $f(a_1) = f(x_1) = [m_1]$, then obviously

$$\alpha(a_1) = \pi_B(\gamma(m_1, a_1)) = \pi_B(\gamma(m_1, a)) = \pi_B(\gamma(m_1, x_1)) = \alpha(x_1).$$

More generally, if two elements $b, c \in A$ are such that $f(b) = f(c) = [m]$, then $f(c^{-1}b) = [1]$, so

$$\alpha(b) = \pi_B(\gamma(m, b)) = \pi_B(\gamma(m, c))\pi_B(\gamma(1, c^{-1}b)) = \alpha(c)\alpha(c^{-1}b),$$

which implies that

$$\alpha(c^{-1}b) = \alpha(c)^{-1}\alpha(b).$$

If $n = 2$ consider $a = a_1a_2^{-1} = x_1x_2^{-1}$ and $f(a_i) = f(x_i) = [m_i]$. Then $x_1^{-1}a_1 = x_2^{-1}a_2$, and

$$\alpha(x_1^{-1}a_1) = \alpha(x_1)^{-1}\alpha(a_1) = \alpha(x_2)^{-1}\alpha(a_2) = \alpha(x_2^{-1}a_2)$$

by the formula above. Reordering the terms, we get:

$$\alpha(a_1)\alpha(a_2)^{-1} = \alpha(x_1)\alpha(x_2)^{-1}.$$

If $n = 3$ consider $a = a_1a_2^{-1}a_3 = x_1x_2^{-1}x_3$ and $f(a_i) = f(x_i) = [m_i]$. Multiplying by x_1^{-1} on the left, we get

$$x_1^{-1}a_1a_2^{-1}a_3 = (a_2a_1^{-1}x_1)^{-1}a_3 = x_2^{-1}x_3,$$

and they both map to $[m_2]^{-1}[m_3]$, so that

$$\alpha(a_2a_1^{-1}x_1)^{-1}\alpha(a_3) = \alpha(x_2)^{-1}\alpha(x_3)$$

by the case $n = 2$. Similarly, multiplying by a_3^{-1} on the right, we get

$$\alpha(a_1)\alpha(a_2)^{-1} = \alpha(x_1)\alpha(a_3x_3^{-1}x_2)^{-1}.$$

Moreover, $a_2a_1^{-1}x_1$ and $a_3x_3^{-1}x_2$ both map to $[m_2]$ and we get

$$\alpha(a_2a_1^{-1}x_1) = \alpha(a_3x_3^{-1}x_2)$$

by the case $n = 1$. As a consequence of these last three equations, we obtain

$$\alpha(x_1)^{-1}\alpha(a_1)\alpha(a_2)^{-1} = \alpha(a_3x_3^{-1}x_2)^{-1} = \alpha(a_2a_1^{-1}x_1)^{-1} = \alpha(x_2)^{-1}\alpha(x_3)\alpha(a_3)^{-1}.$$

Reordering the terms, we get:

$$\alpha(a_1)\alpha(a_2)^{-1}\alpha(a_3) = \alpha(x_1)\alpha(x_2)^{-1}\alpha(x_3).$$

If $n \geq 4$ consider $a = a_1a_2^{-1}a_3 \cdots a_n^{\iota(n)} = x_1x_2^{-1}x_3 \cdots x_n^{\iota(n)}$ and $f(a_i) = f(x_i) = [m_i]$. Multiplying by x_1^{-1} on the left, we get

$$x_1^{-1}a_1a_2^{-1}a_3 \cdots a_n^{\iota(n)} = (a_2a_1^{-1}x_1)^{-1}a_3 \cdots a_n^{\iota(n)} = x_2^{-1}x_3 \cdots x_n^{\iota(n)},$$

and they both map to $[m_2]^{-1}[m_3] \cdots [m_n]^{\iota(n)}$, so that

$$\alpha(a_2a_1^{-1}x_1)^{-1}\alpha(a_3) \cdots \alpha(a_n)^{\iota(n)} = \alpha(x_2)^{-1}\alpha(x_3) \cdots \alpha(x_n)^{\iota(n)}$$

by the induction hypothesis. In the case $n = 3$ we have shown that

$$\alpha(x_1)^{-1}\alpha(a_1)\alpha(a_2)^{-1} = \alpha(a_2a_1^{-1}x_1)^{-1}.$$

Multiplying again by $\alpha(x_1)$ on the left, we get

$$\alpha(a_1)\alpha(a_2)^{-1}\alpha(a_3) \cdots \alpha(a_n)^{\iota(n)} = \alpha(x_1)\alpha(x_2)^{-1}\alpha(x_3) \cdots \alpha(x_n)^{\iota(n)}.$$

□

Lemma 3.1.4. *The functor U_M is full for all monoids M .*

Proof. Consider a monoid homomorphism

$$\gamma : (M \times_{Gp(M)} A, U_M(f)) \longrightarrow (M \times_{Gp(M)} B, U_M(g)).$$

We cover the monoid M with the free monoid $F(M)$ on M , then apply the Grothendieck group functor and consider the kernels to obtain the following diagram:

$$\begin{array}{ccccc} Ker(r_M) & \hookrightarrow & F(M) & \xrightarrow{r_M} & M \\ \downarrow & & \downarrow \eta_{F(M)} & & \downarrow \eta_M \\ N(M) & \hookrightarrow & GpF(M) & \xrightarrow{q_M} & Gp(M) \end{array}$$

where r_M sends a word in $F(M)$ to the product of its letters in M . We pull back $U_M(f)$, $U_M(g)$ and γ along the surjection r_M and we obtain the following:

$$\begin{array}{ccccccc} F(M) \times_{Gp(M)} A & \xrightarrow{r_M \times_{Gp(M)} 1_A} & M \times_{Gp(M)} A & \xrightarrow{\pi_A} & A & & \\ \downarrow & \searrow r_M^*(\gamma) & \downarrow & \searrow \gamma & \downarrow & & \\ & F(M) \times_{Gp(M)} B & \xrightarrow{r_M \times_{Gp(M)} 1_B} & M \times_{Gp(M)} B & \xrightarrow{\pi_B} & B & \\ & \downarrow & \downarrow & \downarrow & \downarrow & & \\ r_M^*(U_M(f)) & \swarrow r_M^*(U_M(g)) & U_M(f) & \swarrow U_M(g) & f & \searrow g & \\ F(M) & \xrightarrow{r_M} & M & \xrightarrow{\eta_M} & Gp(M). & & \end{array}$$

Since $\eta_M r_M = q_M \eta_{F(M)}$, the pullback of f and g along $\eta_M r_M$ can also be obtained by taking the pullbacks along $q_M \eta_{F(M)}$, obtaining

$$\begin{array}{ccccccc} F(M) \times_{Gp(M)} A & \xrightarrow{\eta_{F(M)} \times_{Gp(M)} 1_A} & GpF(M) \times_{Gp(M)} A & \xrightarrow{p_A} & A & & \\ \downarrow & \searrow r_M^*(\gamma) & \downarrow & \searrow & \downarrow & & \\ & F(M) \times_{Gp(M)} B & \xrightarrow{\eta_{F(M)} \times_{Gp(M)} 1_B} & GpF(M) \times_{Gp(M)} B & \xrightarrow{p_B} & B & \\ & \downarrow & \downarrow & \downarrow & \downarrow & & \\ r_M^*(U_M(f)) & \swarrow r_M^*(U_M(g)) & q_M^*(f) & \swarrow q_M^*(g) & f & \searrow g & \\ F(M) & \xrightarrow{\eta_{F(M)}} & GpF(M) & \xrightarrow{q_M} & Gp(M). & & \end{array}$$

Since $U_{F(M)}$ is full by Lemma 3.1.3, there is a morphism

$$\beta : (GpF(M) \times_{Gp(M)} A, q_M^*(f)) \longrightarrow (GpF(M) \times_{Gp(M)} B, q_M^*(g))$$

such that $U_{F(M)}(\beta) = r_M^*(\gamma)$. We define the desired morphism $\alpha : A \rightarrow B$ by the universal property of p_A as a cokernel of its kernel. Explicitly, if we show that $\beta(Ker(p_A)) \subset Ker(p_B)$, there exists a unique $\alpha : (A, f) \rightarrow (B, g)$ such that $p_B \circ \beta = \alpha \circ p_A$. We now prove that β keeps

elements in the kernel fixed. Note that the kernels of p_A and p_B are given by $N(M) \times_{Gp(M)} \{1\} \cong N(M)$; since $N(M)$ is generated by words $[m_1][m_2][m_1 \cdot m_2]^{-1}$ as a normal subgroup of $GpF(M)$, it suffices to prove that

$$\beta([m_1][m_2][m_1 \cdot m_2]^{-1}, 1_A) = ([m_1][m_2][m_1 \cdot m_2]^{-1}, 1_B).$$

By the surjectivity of f , there exists $a \in A$ such that $f(a) = \overline{[m_1][m_2]} = \overline{[m_1 \cdot m_2]}$, so that $([m_1][m_2], a), ([m_1 \cdot m_2], a) \in F(M) \times_{Gp(M)} A$ and

$$(r_M \times_{Gp(M)} 1_A)([m_1][m_2], a) = (m_1 \cdot m_2, a) = (r_M \times_{Gp(M)} 1_A)([m_1 \cdot m_2], a).$$

For some $b \in B$ we have $\gamma(m_1 \cdot m_2, a) = (m_1 \cdot m_2, b)$, so that

$$r_M^*(\gamma)([m_1][m_2], a) = ([m_1][m_2], b)$$

and

$$r_M^*(\gamma)([m_1 \cdot m_2], a) = ([m_1 \cdot m_2], b)$$

by the commutativity of the second diagram. On the other hand,

$$\begin{aligned} \beta([m_1][m_2], a) &= \beta \circ (\eta_{F(M)} \times_{Gp(M)} 1_A)([m_1][m_2], a) \\ &= (\eta_{F(M)} \times_{Gp(M)} 1_B) \circ r_M^*(\gamma)([m_1][m_2], a) \\ &= r_M^*(\gamma)([m_1][m_2], a) = ([m_1][m_2], b) \end{aligned}$$

and

$$\begin{aligned} \beta([m_1 \cdot m_2], a) &= \beta \circ (\eta_{F(M)} \times_{Gp(M)} 1_A)([m_1 \cdot m_2], a) \\ &= (\eta_{F(M)} \times_{Gp(M)} 1_B) \circ r_M^*(\gamma)([m_1 \cdot m_2], a) \\ &= r_M^*(\gamma)([m_1 \cdot m_2], a) = ([m_1 \cdot m_2], b) \end{aligned}$$

by the commutativity of the third diagram. Since β is a group homomorphism, we obtain

$$\begin{aligned} \beta([m_1][m_2][m_1 \cdot m_2]^{-1}, 1_A) &= \beta([m_1][m_2][m_1 \cdot m_2]^{-1}, aa^{-1}) \\ &= \beta([m_1][m_2], a)\beta([m_1 \cdot m_2], a)^{-1} \\ &= ([m_1][m_2], b)([m_1 \cdot m_2], b)^{-1} \\ &= ([m_1][m_2][m_1 \cdot m_2]^{-1}, bb^{-1}) = ([m_1][m_2][m_1 \cdot m_2]^{-1}, 1_B). \end{aligned}$$

To conclude the proof we show explicitly that $U_M(\alpha) = \gamma$. Since $U_M(\alpha)$ is defined by the universal property of $M \times_{Gp(M)} B$ as a pullback, it is sufficient to show that $\pi_B \circ \gamma = \alpha \circ \pi_A$. Fix $(m, a) \in M \times_{Gp(M)} A$ and the element $b \in B$ such that $\gamma(m, a) = (m, b)$. We need to show that $\alpha(a) = b$. As we did in the previous argument, we obtain that $\beta([m], a) = ([m], b)$. Since α is defined by the universal property of p_A as a cokernel, we have that $p_B \circ \beta = \alpha \circ p_A$. Thus,

$$\alpha(a) = p_B \circ \beta([m], a) = b.$$

□

Example 3.1.6. Let us show that we cannot have admissibility if we consider all the arrows in **Mon** and **Grp** as we did in Chapter 2. Take $M = \mathbb{N}^2$; then $Gp(\mathbb{N}^2) \cong \mathbb{Z}^2$. Fix (A, f) as the inclusion of the subgroup $A = \{(a, -a) : a \in \mathbb{Z}\}$ in $\mathbb{Z}^2$ and (B, g) as the projection on the first two components $\mathbb{Z}^3 \twoheadrightarrow \mathbb{Z}^2$. Note that f is not a surjection: this example shows that $U_{\mathbb{N}^2}$ is not faithful. It is easy to check that $\mathbb{N}^2 \times_{\mathbb{Z}^2} A = 0$ and $\mathbb{N}^2 \times_{\mathbb{Z}^2} B \cong \mathbb{Z}$, so that any two different morphisms $A \rightarrow B$ are mapped to the unique arrow $0 \rightarrow \mathbb{Z}$ through $U_{\mathbb{N}^2}$.

3.2 Trivial, central and normal extensions

In this section, we move our focus to the morphisms that are related to a Galois structure. In particular, in Definition 1.3.4, we defined the notion of an object in a slice category split by a morphism, relative to the adjunction considered. Explicitly, with a fixed morphism $\sigma : M \rightarrow N$ and under the assumption that the induced adjunction has a full and faithful right adjoint, this definition was made to exhibit two full subcategories: $Split_M(1_M)$ of those objects that make the induced adjunction an equivalence, as in Lemma 1.3.5; and $Split_N(\sigma)$ of those objects whose pullback along σ is split by 1_M . As in Section 3.1, we want to give notions that are possibly not related to a fixed morphism. In doing so, we first define some subclasses of arrows; then we prove that they are those considered in Chapter 1. Finally, we can state Theorem 1.3.7 with these new notation.

Definition 3.2.1. Let $\Gamma = (\mathcal{A}, \mathcal{P}, \mathcal{S}, \mathcal{C}, \eta, \varepsilon, \overline{\mathcal{A}}, \overline{\mathcal{P}})$ be a Galois structure. A trivial extension is a morphism $f : A \rightarrow M$ in $\overline{\mathcal{A}}$ such that the square

$$\begin{array}{ccc} A & \xrightarrow{\eta_A} & \mathcal{CS}(A) \\ \downarrow f & \lrcorner & \downarrow \mathcal{CS}(f) \\ M & \xrightarrow{\eta_M} & \mathcal{CS}(M) \end{array}$$

is a pullback.

We will write $Triv(M)$ for the full subcategory of $\overline{\mathcal{A}}/M$ of trivial extensions of M .

Definition 3.2.2. Let $\Gamma = (\mathcal{A}, \mathcal{P}, \mathcal{S}, \mathcal{C}, \eta, \varepsilon, \overline{\mathcal{A}}, \overline{\mathcal{P}})$ be a Galois structure. A central extension is a morphism $f : A \rightarrow N$ in $\overline{\mathcal{A}}$ whose pullback $\sigma^*(f) : M \times_N A \rightarrow M$ is a trivial extension for some morphism $\sigma : M \rightarrow N$ in $\overline{\mathcal{A}}$.

We will write $Centr(N)$ for the full subcategory of $\overline{\mathcal{A}}/N$ of central extensions of N , and $Centr(N, \sigma)$ for the full subcategory of central extensions of N that satisfy the definition with a fixed $\sigma : M \rightarrow N$.

Definition 3.2.3. Let $\Gamma = (\mathcal{A}, \mathcal{P}, \mathcal{S}, \mathcal{C}, \eta, \varepsilon, \overline{\mathcal{A}}, \overline{\mathcal{P}})$ be a Galois structure. A normal extension is a morphism $\sigma : M \rightarrow N$ in $\overline{\mathcal{A}}$ such that $\sigma^*(\sigma) : M \times_N M \rightarrow M$ is a trivial extension. Explicitly, σ is a normal extension if its kernel pair projections are trivial extensions.

We will write $Norm(N)$ for the full subcategory of $\overline{\mathcal{A}}/N$ of normal extensions of N .

Observation. It is clear by definition that every normal extension is also a central extension:

$$\sigma \in Norm(N) \iff \sigma \in Centr(N, \sigma) \implies \sigma \in Centr(N).$$

Let us show the relations between these definitions and the subcategories $Split_M(1_M)$ and $Split_N(\sigma)$ considered in Section 1.3, for a $\sigma : M \rightarrow N$.

Proposition 3.2.1. Let $\Gamma = (\mathcal{A}, \mathcal{P}, \mathcal{S}, \mathcal{C}, \eta, \varepsilon, \overline{\mathcal{A}}, \overline{\mathcal{P}})$ be a Galois structure and fix $M \in Ob(\mathcal{A})$; then

$$Triv(M) = Split_M(1_M).$$

Proof. Fix $f : A \rightarrow M$ and consider η^M the unit of the induced adjunction $\mathcal{S}_M \dashv \mathcal{C}_M$. One has the following commutative diagram defining the unit η^M of the induced adjunction:

$$\begin{array}{ccccc}
A & & \xrightarrow{\eta_A} & & \mathcal{CS}(A) \\
& \searrow \eta_{(A,f)}^M & & \searrow \pi_{\mathcal{CS}(A)} & \\
& \mathcal{C}_M \mathcal{S}(A) & \xrightarrow{\quad} & \mathcal{CS}(A) & \\
& \downarrow \mathcal{C}_M(\mathcal{S}(f)) & \lrcorner & \downarrow \mathcal{CS}(f) & \\
& M & \xrightarrow{\eta_M} & \mathcal{CS}(M). & \\
& \uparrow f & & &
\end{array}$$

By Lemma 1.3.3, f is split by 1_M if and only if $\eta_{(A,f)}^M$ is an isomorphism. That is, if and only if f is a trivial extension. $\square$

From the previous result it is clearer that we could have defined trivial extensions of M as the image of the induced functor $\mathcal{C}_M : \overline{\mathcal{P}}/\mathcal{S}(M) \rightarrow \overline{\mathcal{A}}/M$. As we did in Section 1.3, when we have the admissibility of the structure, we obtain an equivalence of categories.

Corollary 3.2.1.1. *Let $\Gamma = (\mathcal{A}, \mathcal{P}, \mathcal{S}, \mathcal{C}, \eta, \varepsilon, \overline{\mathcal{A}}, \overline{\mathcal{P}})$ be an admissible Galois structure and fix $M \in \text{Ob}(\mathcal{A})$. The induced adjunction $\mathcal{S}_M \dashv \mathcal{C}_M$ restricts to an equivalence of categories*

$$\text{Triv}(M) \approx \overline{\mathcal{P}}/\mathcal{S}(M).$$

Proof. As in Lemma 1.3.5: note that the admissibility of the structure was crucial to establish the equivalence. $\square$

Corollary 3.2.1.2. *Let $\Gamma = (\mathcal{A}, \mathcal{P}, \mathcal{S}, \mathcal{C}, \eta, \varepsilon, \overline{\mathcal{A}}, \overline{\mathcal{P}})$ be a Galois structure and fix a morphism $\sigma : M \rightarrow N$ in $\overline{\mathcal{A}}$; then*

$$\text{Centr}(N, \sigma) = \text{Split}_N(\sigma).$$

Proof. It follows directly from Proposition 3.2.1 and Corollary 1.3.3.1. $\square$

Observation. The previous results show us that any trivial extension $f : A \rightarrow M$ is a central extension choosing $\sigma = 1_M$:

$$\text{Triv}(M) = \text{Centr}(M, 1_M) \subseteq \text{Centr}(M).$$

It remains to show how normal extensions are related with the work done in Section 1.3. The following result will be useful:

Proposition 3.2.2. *Let $\Gamma = (\mathcal{A}, \mathcal{P}, \mathcal{S}, \mathcal{C}, \eta, \varepsilon, \overline{\mathcal{A}}, \overline{\mathcal{P}})$ be an admissible Galois structure; then the functor $\mathcal{S} : \mathcal{A} \rightarrow \mathcal{P}$ preserves pullbacks along trivial extensions. In particular, trivial extensions are pullback-stable and every trivial extension is a normal extension.*

Proof. Fix a trivial extension $f : A \rightarrow M$ and a morphism $g : B \rightarrow M$. Consider the pullback of f along g and the pullback defining f as a trivial extension:

$$\begin{array}{ccccc}
C & \xrightarrow{p_A} & A & \xrightarrow{\eta_A} & \mathcal{CS}(A) \\
\downarrow p_B & \lrcorner & \downarrow f & \lrcorner & \downarrow \mathcal{CS}(f) \\
B & \xrightarrow{g} & M & \xrightarrow{\eta_M} & \mathcal{CS}(M).
\end{array}$$

We need to prove that $\mathcal{S}(C)$ is a pullback of $\mathcal{S}(f)$ along $\mathcal{S}(g)$. We have that the exterior of the previous diagram is a pullback and equivalently, using $\eta_M \circ g = \mathcal{CS}(g) \circ \eta_B$, the exterior of

$$\begin{array}{ccccc} C & \xrightarrow{\eta_C} & \mathcal{CS}(C) & \xrightarrow{\mathcal{CS}(p_A)} & \mathcal{CS}(A) \\ p_B \downarrow & & \downarrow \mathcal{CS}(p_B) & & \downarrow \mathcal{CS}(f) \\ B & \xrightarrow{\eta_B} & \mathcal{CS}(B) & \xrightarrow{\mathcal{CS}(g)} & \mathcal{CS}(M). \end{array}$$

is a pullback. Fix X as a pullback of $\mathcal{S}(f)$ along $\mathcal{S}(g)$. We need to prove that the comparison morphism $h = \langle \mathcal{S}(p_A), \mathcal{S}(p_B) \rangle : \mathcal{S}(C) \rightarrow X$ is an isomorphism. Since the functor $\mathcal{C}_B$ is full and faithful, it suffices to prove that $\mathcal{C}_B(h)$ is an isomorphism. One has

$$\begin{array}{ccccccc} C & \xrightarrow{\eta_C} & \mathcal{CS}(C) & \xrightarrow{\mathcal{CS}(p_A)} & \mathcal{CS}(A) & & \\ & \searrow & \downarrow & \searrow \mathcal{C}(h) & \downarrow & \searrow & \\ & C & \xrightarrow{\quad} & \mathcal{C}(X) & \xrightarrow{\mathcal{C}(a)} & \mathcal{CS}(A) & \\ & \swarrow & \downarrow \mathcal{CS}(p_B) & \swarrow \mathcal{C}(b) & \downarrow \mathcal{CS}(f) & \swarrow & \\ B & \xrightarrow{\eta_B} & \mathcal{CS}(B) & \xrightarrow{\mathcal{CS}(g)} & \mathcal{CS}(M) & & \end{array}$$

where the exterior is a pullback and the front square on the right is a pullback since $\mathcal{C}$ preserves limits as a right adjoint. In this way, also the front square on the left is a pullback and

$$1_C = \eta_B^* \mathcal{C}(h) = \mathcal{C}_B(h).$$

In particular, p_B is a trivial extension of B and taking $g = f$ one has that every trivial extension is a normal extension. $\square$

Observation. In this way, for every $M \in \text{Ob}(\mathcal{A})$, we established that

$$\text{Triv}(M) \subseteq \text{Norm}(M) \subseteq \text{Centr}(M).$$

Suppose that the Galois structure Γ is admissible and fix $\sigma : M \rightarrow N$ in $\overline{\mathcal{A}}$; in Lemma 1.3.4 we proved that the functor Σ_σ restricts to the subcategories of split objects, but it was actually assumed on σ in Definition 1.3.5 through the property we now recall here:

- (iii) the object $\Sigma_\sigma \mathcal{C}_M(X, f)$ is split by σ , for every (X, f) object of $\overline{\mathcal{P}}/\mathcal{S}(M)$.

By Corollary 3.2.1.1 and Corollary 3.2.1.2, we can write this property in terms of trivial and central extensions:

$$\sigma \circ g \in \text{Centr}(N, \sigma) \text{ for all } g \in \text{Triv}(M). \quad (\text{P})$$

Proposition 3.2.3. *Let $\Gamma = (\mathcal{A}, \mathcal{P}, \mathcal{S}, \mathcal{C}, \eta, \varepsilon, \overline{\mathcal{A}}, \overline{\mathcal{P}})$ be an admissible Galois structure. A morphism $\sigma : M \rightarrow N$ is a normal extension if and only if σ has the Property (P).*

Proof. Fix $g \in \text{Triv}(M)$. We have the following:

$$\begin{array}{ccc}
M \times_N A & \xrightarrow{\quad} & A \\
\downarrow p_1^*(g) & \lrcorner & \downarrow g \\
M \times_N M & \xrightarrow{p_1} & M \\
\downarrow p_2 & \lrcorner & \downarrow \sigma \\
M & \xrightarrow{\sigma} & N.
\end{array}$$

If σ is a normal extension, p_1 and p_2 are trivial; and we need to prove that $p_2 \circ p_1^*(g)$ is trivial. Obviously, the composition of trivial extensions is trivial and $p_1^*(g)$ is trivial by Proposition 3.2.2. On the other hand, take $g = 1_M$; then Property (P) ensures that $\sigma \in \text{Centr}(N, \sigma)$, so that σ is a normal extension. $\square$

Example 3.2.4. In Section 2.3, we actually studied trivial, central, and normal extensions in that scenario. In Proposition 2.3.4, we have shown that the K -algebras split by L (in the classical sense of Definition 2.1.7) are exactly the central extensions of K . Moreover, in Proposition 2.3.3, we proved that a Galois field extension $\sigma : K \hookrightarrow L$ is a normal extension of K . Note that σ is Galois if and only if it splits itself.

To conclude this section, let us state Theorem 1.3.7 exploiting these new notations.

Theorem 3.2.4. *Let $\Gamma = (\mathcal{A}, \mathcal{P}, \mathcal{S}, \mathcal{C}, \eta, \varepsilon, \overline{\mathcal{A}}, \overline{\mathcal{P}})$ be an admissible Galois structure. Suppose that $\sigma : M \rightarrow N$ is a normal extension that has a monadic pullback functor. There exists an equivalence of categories*

$$\text{Centr}(N, \sigma) \approx \overline{\mathcal{P}}^{\text{Gal}[\sigma]}$$

between the central extensions that are split by σ and the internal covariant presheaves (P, p, π) on the internal groupoid $\text{Gal}[\sigma]$, where $\text{Gal}[\sigma]$ is given as in Definition 1.3.6.

3.3 Special Schreier and special homogeneous surjections

In this section, we want to describe trivial, central and normal extensions with respect to the admissible Galois structure for monoids defined in Section 3.1

$$\Gamma = (\mathbf{Mon}, \mathbf{Grp}, Gp, U, \eta, 1_{\mathbf{Grp}}, \mathcal{E}, \mathcal{F})$$

where $\mathcal{E}$ and $\mathcal{F}$ are the classes of surjections in $\mathbf{Mon}$ and $\mathbf{Grp}$ respectively. It turns out that central and normal extensions coincide and are special homogeneous surjections. We will see that the notion of special homogeneous surjection is given referring to an underlying split epimorphism of sets; this characterization shows that it is not necessary, and special homogeneous surjections, along with its related notions, can be described from the point of view of categorical Galois theory.

Definition 3.3.1. A split epimorphism of monoids is an epimorphism $f : A \twoheadrightarrow M$ along with a morphism $s : M \rightarrow A$ such that $fs(m) = m$ for any $m \in M$.

Definition 3.3.2. Consider a split epimorphism (f, s) of monoids with its kernel:

$$K \rightarrowtail \longrightarrow A \begin{array}{c} \xrightarrow{f} \\ \xleftarrow{s} \end{array} \twoheadrightarrow M.$$

It is called a Schreier split epimorphism when, for any $a \in A$, there exists a unique $k \in K$ such that $a = k \cdot sf(a)$.

It is said to be a right homogeneous split epimorphism when, for any $m \in M$, the map

$$\mu_m^r : K \longrightarrow f^{-1}(m), k \longmapsto k \cdot s(m)$$

defined by multiplication on the right by $s(m)$ is bijective.

It is said to be a left homogeneous split epimorphism when, for any $m \in M$, the map

$$\mu_m^l : K \longrightarrow f^{-1}(m), k \longmapsto s(m) \cdot k$$

defined by multiplication on the left by $s(m)$ is bijective.

It is said to be a homogeneous split epimorphism when it is both right and left homogeneous.

Let us recall a key property of Schreier split epimorphisms that establishes its connection with homogeneous split epimorphisms.

Proposition 3.3.1. *Consider a split epimorphism (f, s) of monoids with its kernel K . The following are equivalent:*

1. (f, s) is a Schreier split epimorphism;
2. (f, s) is a right homogeneous split epimorphism;
3. there exists a set-theoretical map $q : A \rightarrow K$ such that $q(a) \cdot sf(a) = a$ and $q(k \cdot s(m)) = k$, for all $a \in A$, $k \in K$ and $m \in M$.

Proof. If (f, s) is a Schreier split epimorphism, we can define a map $q : A \rightarrow K$ that sends $a \in A$ to the unique $k \in K$ such that $k \cdot sf(a) = a$. Fix $m \in M$ and consider the restriction of q to $f^{-1}(m)$; then, for any $a \in f^{-1}(m)$,

$$a = q(a) \cdot sf(a) = q(a) \cdot s(m),$$

so that $\mu_m^r \circ q = 1_{f^{-1}(m)}$. Moreover, for any $k \in K$,

$$k \cdot s(m) = k \cdot sf(k \cdot s(m)),$$

so that $q(k \cdot s(m)) = k$ and $q \circ \mu_m^r = 1_K$. The set-theoretical map q just defined has the properties required in 3. and is the inverse of μ_m^r when restricted to every preimage $f^{-1}(m)$, making it a bijection.

Noting that $\{f^{-1}(m)\}_{m \in M}$ is a partition of A , if (f, s) is right homogeneous, the inverses of μ_m^r yield the unique map $q : A \rightarrow K$ that has the required properties. On the other hand, if there is such a map $q : A \rightarrow K$, then it defines the unique element in K asked in the definition of a Schreier split epimorphism. $\square$

Explicitly, a Schreier split epimorphism (f, s) can be expressed with its kernel and the set-theoretical map $q : A \rightarrow K$:

$$K \begin{array}{c} \xleftarrow{\quad q \quad} \\ \rightarrowtail \end{array} A \begin{array}{c} \xrightarrow{f} \\ \xleftarrow{s} \end{array} \twoheadrightarrow M. \quad (3.2)$$

It is immediate to show that $q(k) = k$ and $q(s(m)) = 1_A$, for every $k \in K$ and $m \in M$.

A clearer description of Schreier split epimorphisms is given by actions between monoids. In [13], it has been shown that Schreier split epimorphisms as in (3.2) corresponds exactly to actions of M on K , i.e. a monoid homomorphism $\varphi : M \rightarrow \text{End}(K)$, where $\text{End}(K)$ is the monoid of endomorphisms of K . This equivalence is given via a semidirect product construction.

Proposition 3.3.2. *Consider a Schreier split epimorphism as in (3.2). It corresponds to an action $\varphi : M \rightarrow \text{End}(K)$ defined as*

$$\varphi(m)(k) = q(s(m) \cdot k)$$

for any $m \in M$ and $k \in K$. Moreover, there is an isomorphism of split epimorphisms between (3.2) and

$$K \begin{array}{c} \xleftarrow{\pi_K} \\ \xrightarrow{\langle 1, 0 \rangle} \end{array} K \rtimes_{\varphi} M \begin{array}{c} \xleftarrow{\pi_M} \\ \xrightarrow{\langle 0, 1 \rangle} \end{array} M \quad (3.3)$$

where $\langle 1, 0 \rangle$ and $\langle 0, 1 \rangle$ are the canonical injections in the product and $K \rtimes_{\varphi} M$ is the semidirect product of K and M with respect to φ : the cartesian product $K \times M$ equipped with the operation

$$(k_1, m_1) \cdot (k_2, m_2) = (k_1 \cdot \varphi(m_1)(k_2), m_1 \cdot m_2).$$

Proof. We just provide the isomorphism $\zeta : A \rightarrow K \rtimes_{\varphi} M$: it is given by

$$\zeta : a \mapsto (q(a), f(a)),$$

with the inverse given by

$$\zeta^{-1} : (k, m) \mapsto k \cdot s(m).$$

It is just a computation to show that it is an isomorphism of split epimorphisms. $\square$

Observation. For any $m \in M$ and $k \in K$, $\varphi(m)(k)$ is the unique element in K such that

$$\varphi(m)(k) \cdot s(m) = s(m) \cdot k.$$

This property is crucial to prove that the previous maps are monoid homomorphisms. Note that $\varphi(m)(k)$ is analogous to a conjugate of k through $s(m)$.

In Proposition 3.3.1, we proved the equivalence between Schreier split epimorphisms and right homogeneous split epimorphisms. A homogeneous split epimorphism can be seen as a Schreier split epimorphism also with the defining property on the left: for any $a \in A$, there exists a unique $k \in K$ such that $a = sf(a) \cdot k$. In the equivalence with monoid actions, we obtain the following:

Proposition 3.3.3 ([4], Proposition 3.8). *A Schreier split epimorphism as in (3.2) (or equivalently in (3.3)) is a homogeneous split epimorphism if and only if the corresponding action $\varphi : M \rightarrow \text{End}(K)$ factors through the group $\text{Aut}(K)$ of automorphisms of K .*

Proof. Just note, for every $m \in M$, that

$$\varphi(m) = q \circ \mu_m^l$$

and recall that q is defined as the inverse of μ_m^r when restricted to $f^{-1}(m)$. $\square$

Proposition 3.3.4. *If M is a group, then every split epimorphism (f, s) is a homogeneous split epimorphism.*

Proof. It is sufficient to define $q(a) = a \cdot sf(a)^{-1}$ for any $a \in A$ to show that it is a Schreier split epimorphism. The dual notion of left homogeneous is given fixing $q(a) = sf(a)^{-1} \cdot a$ for any $a \in A$. $\square$

Since we will consider general surjections and not split epimorphisms, we can extend these concepts to the split epimorphism given by the kernel pair of a surjective homomorphism.

Definition 3.3.3. Consider a surjective homomorphism σ of monoids with its kernel pair:

$$M \times_N M \begin{array}{c} \xrightarrow{p_1} \\ \xleftarrow{\Delta} \\ \xrightarrow{p_2} \end{array} M \xrightarrow{\sigma} N$$

where we considered the canonical kernel pair as a subobject of the cartesian product $M \times M$ and p_1, p_2 are the projection of the element on the right and left respectively.

It is called a special Schreier surjection when (p_1, Δ) is a Schreier split epimorphism.

It is called a special homogeneous surjection when (p_1, Δ) is a homogeneous split epimorphism.

Observation. By Proposition 3.3.1, it is clear that if σ is a special homogeneous surjection, then it is a special Schreier surjection.

We now state some properties of special Schreier (resp. special homogeneous) surjections providing brief proofs. More details can be found in [3] and [4].

Theorem 3.3.5 ([4], Theorem 5.5). *Any reflexive relation*

$$R \begin{array}{c} \xrightarrow{d_1} \\ \xleftarrow{s} \\ \xrightarrow{d_2} \end{array} M$$

such that (d_1, s) is a Schreier split epimorphism is transitive. It is an equivalence relation if and only if the kernel $K[d_1]$ of d_1 is a group.

Corollary 3.3.5.1. *If a surjective homomorphism σ of monoids is a special Schreier surjection, then its kernel K is a group.*

Proof. Since the kernel pair of σ is an equivalence relation on M , the kernel $K[p_1]$ of p_1 is a group. Moreover, $K[p_1]$ is given by

$$\{(k, 1_M) \in M \times M : \sigma(k) = \sigma(1_M) = 1_N\}$$

so that it is isomorphic to K via p_2 . $\square$

Corollary 3.3.5.2. *A special Schreier (resp. special homogeneous) surjection σ which is a split epimorphism is also a Schreier (resp. homogeneous) split epimorphism. On the other hand, a Schreier (resp. homogeneous) split epimorphism (σ, s) such that its kernel is a group is a special Schreier (resp. special homogeneous) surjection.*

Proof. Fix a special Schreier surjection $\sigma : M \rightarrow N$ that is also split by $s : N \rightarrow M$. We need to show that, for any $m \in M$, there exists a unique $k \in K$ such that $k \cdot s\sigma(m) = m$. Since (p_1, Δ) is a Schreier split epimorphism and $(m, s\sigma(m)) \in M \times_N M$ for any $m \in M$, there exists a unique $(k, 1_M) \in K[p_1]$ such that

$$(m, s\sigma(m)) = (k, 1_M) \cdot (s\sigma(m), s\sigma(m)) = (k \cdot s\sigma(m), s\sigma(m)).$$

On the other hand, fix a Schreier split epimorphism (σ, s) such that its kernel is a group and consider its kernel pair $M \times_N M$. It suffices to show that for any $m_1, m_2 \in M$ such that $\sigma(m_1) = \sigma(m_2)$, there exists a unique $k \in K$ such that $k \cdot m_2 = m_1$. Since (σ, s) is a Schreier split epimorphism, there are unique $k_1, k_2 \in K$ such that $k_i \cdot s\sigma(m_i) = m_i$ for $i = 1, 2$. Thus,

$$m_1 = k_1 \cdot s\sigma(m_1) = k_1 \cdot s\sigma(m_2) = k_1 \cdot k_2^{-1} \cdot m_2.$$

□

Corollary 3.3.5.3. *A surjective homomorphism σ of monoids is a special Schreier (resp. special homogeneous) surjection if and only if its kernel pair projection p_1 is a special Schreier (resp. special homogeneous) surjection.*

Proof. If σ is a special Schreier surjection, then (p_1, Δ) is a Schreier split epimorphism and its kernel is a group since $K[p_1] \cong K$; then p_1 is a special Schreier surjection. On the other hand, if p_1 is a special Schreier surjection, (p_1, Δ) is a Schreier split epimorphism; then σ is a special Schreier surjection. □

Proposition 3.3.6 ([3], Proposition 7.1.4, 7.1.5). *Special Schreier and special homogeneous surjections are stable under products and pullbacks. Moreover, given any pullback of surjective homomorphism of monoids*

$$\begin{array}{ccc} A & \xrightarrow{g} \twoheadrightarrow & A' \\ \downarrow f & \lrcorner & \downarrow f' \\ B & \xrightarrow{h} \twoheadrightarrow & B' \end{array}$$

if f is a special Schreier (resp. special homogeneous) surjection, then f' is a special Schreier (resp. special homogeneous) surjection.

As we did with Schreier split epimorphisms, we can give a clearer description of the kernel pair of a special Schreier surjection through semidirect products.

Proposition 3.3.7. *Consider a special Schreier surjection σ with its kernel pair, the kernel $K[p_1]$ of p_1 and the map q given in Proposition 3.3.1:*

$$K[p_1] \xleftarrow{\quad q \quad} M \times_N M \begin{array}{c} \xrightarrow{p_1} \\ \xleftarrow{\Delta} \\ \xrightarrow{p_2} \end{array} M \xrightarrow{\sigma} \twoheadrightarrow N.$$

There is an isomorphism of split epimorphisms with

$$K \xleftarrow{\quad \pi_K \quad} K \rtimes_{\varphi} M \begin{array}{c} \xrightarrow{\pi_M} \\ \xleftarrow{\quad \bullet \quad} \\ \xrightarrow{\quad \bullet \quad} \end{array} M \xrightarrow{\sigma} \twoheadrightarrow N \quad (3.4)$$

where K is the kernel of σ and $\bullet$ is the multiplication on M : $(k, m) \mapsto k \cdot m$. Note that we omitted to name the canonical inclusions for the sake of clarity.

Proof. Mostly it is a consequence of Proposition 3.3.2 and the isomorphism $K[p_1] \cong K$. Let us show explicitly why p_2 corresponds to the multiplication on M : in this setting

$$q(m_1, m_2) = (k, 1_M) \text{ such that } k \cdot m_2 = m_1$$

for any $(m_1, m_2) \in M \times_N M$. Note that such a k exists and is unique because (p_1, Δ) is a Schreier split epimorphism. Now, considering $K[p_1] \cong K$, the isomorphism ζ expressed in Proposition 3.3.2 is given by

$$\zeta : (m_1, m_2) \mapsto (k, m_2) \text{ such that } k \cdot m_2 = m_1$$

with the inverse given by

$$\zeta^{-1} : (k, m) \mapsto (k \cdot m, m).$$

It is clear that

$$\bullet(k, m) = p_2 \circ \zeta^{-1}(k, m) = k \cdot m$$

□

Observation. The action $\varphi : M \rightarrow \text{End}(K)$ is given by

$$\varphi(m)(k) = q(m \cdot k, m)$$

for any $m \in M$ and $k \in K$, and it is such that

$$\varphi(m)(k) \cdot m = m \cdot k.$$

We are now ready to characterize trivial split extensions, central and normal extensions in the admissible Galois structure

$$\Gamma = (\mathbf{Mon}, \mathbf{Grp}, Gp, U, \eta, 1_{\mathbf{Grp}}, \mathcal{E}, \mathcal{F}).$$

In Definition 1.3.6, we defined the Galois groupoid of σ as the image of the groupoid $K(\sigma)$ induced by the kernel pair of σ through the left adjoint of the considered Galois structure. In doing so, it is useful to understand how a morphism of split epimorphisms works when it involves the unit η ; note that in this scenario we will consider actually homogeneous split epimorphisms since η maps into a group and by Proposition 3.3.4. We will state and prove the next result utilizing the semidirect product form (3.3), for the (equivalent) statement using the other form see [14]. This will be crucial to describe $Gp(M \times_N M)$ in a semidirect form and to gain a clearer understanding of the Galois groupoid $\text{Gal}[\sigma]$.

Lemma 3.3.8. *Consider any morphism involving η_M of homogeneous split epimorphisms and their kernels:*

$$\begin{array}{ccccc} K & \overset{\pi_K}{\overset{\leftarrow}{\dashrightarrow}} & K \rtimes_{\varphi} M & \overset{\pi_M}{\overset{\leftarrow}{\dashrightarrow}} & M \\ \downarrow \tilde{u} & & \downarrow u & & \downarrow \eta_M \\ K' & \overset{\pi_{K'}}{\overset{\leftarrow}{\dashrightarrow}} & K' \rtimes_{\psi} Gp(M) & \overset{\pi_{Gp(M)}}{\overset{\leftarrow}{\dashrightarrow}} & Gp(M) \end{array}$$

where $\tilde{u}$ is the component of u on K . It factors into the composite

$$\begin{array}{ccccc}
K & \overset{\pi_K}{\dashrightarrow} & K \rtimes_{\varphi} M & \overset{\pi_M}{\dashrightarrow} & M \\
\parallel & & \downarrow \overline{\eta_M} & & \downarrow \eta_M \\
K & \overset{\pi_K}{\dashrightarrow} & K \rtimes_{\overline{\varphi}} Gp(M) & \overset{\pi_{Gp(M)}}{\dashrightarrow} & Gp(M) \\
\downarrow \tilde{u} & & \downarrow \tilde{u} & & \parallel \\
K' & \overset{\pi_{K'}}{\dashrightarrow} & K' \rtimes_{\psi} Gp(M) & \overset{\pi_{Gp(M)}}{\dashrightarrow} & Gp(M)
\end{array}$$

where $\overline{\varphi} : Gp(M) \rightarrow \text{Aut}(K)$ is the unique group homomorphism satisfying $\varphi = \overline{\varphi} \circ \eta_M$.

Proof. Since f is homogeneous, $\varphi : M \rightarrow \text{Aut}(K)$ is a monoid homomorphism into a group. By adjointness there is a unique group homomorphism $\overline{\varphi} : Gp(M) \rightarrow \text{Aut}(K)$ such that $\varphi = \overline{\varphi} \circ \eta_M$. Note that $\overline{\varphi}$ is necessarily given by the action φ on the representatives of the classes:

$$\overline{\varphi}(\overline{[m_1][m_2]^{-1} \cdots [m_n]^{\iota(n)}}) = \varphi(m_1)\varphi(m_2)^{-1} \cdots \varphi(m_n)^{\iota(n)}$$

or

$$\overline{\varphi}(\overline{[m_1]^{-1}[m_2] \cdots [m_n]^{\iota(n)}}) = \varphi(m_1)^{-1}\varphi(m_2) \cdots \varphi(m_n)^{\iota(n)}.$$

The map $\overline{\eta_M} : K \rtimes_{\varphi} M \rightarrow K \rtimes_{\overline{\varphi}} Gp(M)$ is given by $1_K \times \eta_M$. Since $\varphi(m)(k) = \overline{\varphi}(\overline{[m]})(k)$, $1_K \times \eta_M$ is a monoid homomorphism.

The map $\tilde{u} : K \rtimes_{\overline{\varphi}} Gp(M) \rightarrow K' \rtimes_{\psi} Gp(M)$ is given by $\tilde{u} \times 1_{Gp(M)}$. This is actually a monoid homomorphism since $\tilde{u}$ is a morphism of $Gp(M)$ -actions: explicitly,

$$\tilde{u}(\overline{\varphi}(x)(k)) = \psi(x)(\tilde{u}(k))$$

for every $x \in Gp(M)$ and $k \in K$. When $x = \eta_M(m)$ is a generator, it follows directly by $u : K \rtimes_{\varphi} M \rightarrow K' \rtimes_{\psi} Gp(M)$ being a monoid homomorphism and $\varphi = \overline{\varphi} \circ \eta_M$; it extends to all $x \in Gp(M)$ since $\overline{\varphi}$ acts as φ on generators. $\square$

Theorem 3.3.9. *Consider a split epimorphism (f, s) ; then f is a trivial extension if and only if f is a special homogeneous surjection.*

Proof. By Proposition 3.3.4, $(Gp(f), Gp(s))$ is a homogeneous split epimorphism. Since the kernel of $Gp(f)$ is a group, by Corollary 3.3.5.2 $Gp(f)$ is a special homogeneous surjection. If f is a trivial extension, then

$$\begin{array}{ccc}
A & \xrightarrow{\eta_A} & Gp(A) \\
\uparrow s & \lrcorner & \uparrow Gp(s) \\
M & \xrightarrow{\eta_M} & Gp(M)
\end{array}
\quad
\begin{array}{c}
\downarrow f \\
\downarrow Gp(f)
\end{array}$$

is a pullback. Since special homogeneous surjection are stable under pullbacks, by Proposition 3.3.6, f is a special homogeneous surjection. On the other hand, if f is a special homogeneous surjection, then its kernel is a group by Corollary 3.3.5.1. Consider the morphism of special homogeneous surjections and their kernels

$$\begin{array}{ccccc}
K & \xrightarrow{\quad} & A & \begin{array}{c} \xrightarrow{f} \\ \xleftarrow{s} \end{array} & M \\
\eta_{A|_K} \downarrow & & \eta_A \downarrow & & \downarrow \eta_M \\
K' & \xrightarrow{\quad} & Gp(A) & \begin{array}{c} \xrightarrow{Gp(f)} \\ \xleftarrow{Gp(s)} \end{array} & Gp(M).
\end{array}$$

By Theorem 2.3.7 in [3], the right square is a pullback (so that f is a trivial extension) if and only if the restriction $\eta_{A|_K}$ to the kernels is an isomorphism. By Lemma 3.3.8, $\eta_A : A \rightarrow Gp(A)$ factors into $A' := K \rtimes_{\overline{\varphi}} Gp(M)$:

$$\begin{array}{ccccc}
K & \xrightarrow{\quad \iota \quad} & A & \begin{array}{c} \xrightarrow{f} \\ \xleftarrow{s} \end{array} & M \\
\parallel & & \downarrow \overline{\eta}_M & & \downarrow \eta_M \\
K & \xrightarrow{\quad \langle 1, 0 \rangle \quad} & A' & \begin{array}{c} \xrightarrow{\pi_{Gp(M)}} \\ \xleftarrow{\langle 0, 1 \rangle} \end{array} & Gp(M) \\
\eta_{A|_K} \downarrow & & \downarrow \overline{\eta}_A & & \parallel \\
K' & \xrightarrow{\quad} & Gp(A) & \begin{array}{c} \xrightarrow{Gp(f)} \\ \xleftarrow{Gp(s)} \end{array} & Gp(M)
\end{array}$$

where $\langle 1, 0 \rangle$ and $\langle 0, 1 \rangle$ are the canonical injections in the product and ι is the injection in A . Notice that we have used the same notation of Lemma 3.3.8, but $\overline{\eta}_M$ and $\overline{\eta}_A$ have a different expression since we did not express the split epimorphisms in the semidirect product form. One has $K \cong K'$ exactly when $\overline{\eta}_A : A' \rightarrow Gp(A)$ is an isomorphism. Since K is a group, A' is a group; then $\overline{\eta}_M : A \rightarrow A'$ induces a map $g : Gp(A) \rightarrow A'$ by the universal property of $Gp(A)$ such that $g \circ \eta_A = \overline{\eta}_M$. Notice that g is actually a morphism of split epimorphisms:

$$\pi_{Gp(M)} \circ g \circ \eta_A = \pi_{Gp(M)} \circ \overline{\eta}_M = \eta_M \circ f = Gp(f) \circ \eta_A$$

so that $\pi_{Gp(M)} \circ g = Gp(f)$, and

$$g \circ Gp(s) \circ \eta_M = g \circ \eta_A \circ s = \overline{\eta}_M \circ s = \langle 0, 1 \rangle \circ \eta_M$$

so that $g \circ Gp(s)$ is the inclusion $\langle 0, 1 \rangle : Gp(M) \rightarrow A'$.

Let us show that g is the inverse of $\overline{\eta}_A$:

$$\overline{\eta}_A \circ g \circ \eta_A = \overline{\eta}_A \circ \overline{\eta}_M = \eta_A$$

so that $\overline{\eta}_A \circ g = 1_{Gp(A)}$. On the other hand, it suffices to show that $g \circ \overline{\eta}_A = 1_{A'}$ in each component of A' , by Lemma 2.1.6 in [13]. One has

$$g \circ \overline{\eta}_A \circ \langle 1, 0 \rangle = g \circ \overline{\eta}_A \circ \overline{\eta}_M \circ \iota = g \circ \eta_A \circ \iota = \overline{\eta}_M \circ \iota = \langle 1, 0 \rangle$$

and

$$g \circ \overline{\eta}_A \circ \langle 0, 1 \rangle = g \circ Gp(s) = \langle 0, 1 \rangle.$$

Since $g \circ \overline{\eta}_A = 1_{A'}$ and $\overline{\eta}_A \circ g = 1_{Gp(A)}$, we get that $\overline{\eta}_A$ is an isomorphism. Thus, $\eta_{A|_K}$ is an isomorphism. $\square$

Observation. Note that we characterized trivial extensions as special homogeneous surjections only if they are split epimorphisms. Moreover, we characterized the image of a split trivial extension (f, s) as a semidirect product of groups, since

$$Gp(A) \cong K \rtimes_{\bar{\varphi}} Gp(M)$$

where K is the kernel of f and $\bar{\varphi}$ is given as in Lemma 3.3.8.

Theorem 3.3.10. *Let $\sigma : M \twoheadrightarrow N$ be a surjective homomorphism of monoids. The following are equivalent:*

1. σ is a central extension;
2. σ is a normal extension;
3. σ is a special homogeneous surjection.

Proof. Consider σ and its kernel pair

$$M \times_N M \begin{array}{c} \xrightarrow{p_1} \\ \xleftarrow{\Delta} \\ \xrightarrow{p_2} \end{array} M \xrightarrow{\sigma} N.$$

Let us first prove 2. $\iff$ 3.: By definition, σ is a normal extension if and only if p_1 is a trivial extension. Since (p_1, Δ) is a split epimorphism, by Theorem 3.3.9, p_1 is a trivial extension if and only if it is a special homogeneous surjection. By Corollary 3.3.5.3, p_1 is a special homogeneous surjection if and only if σ is a special homogeneous surjection.

By definition we get 2. $\implies$ 1.. To prove 1. $\implies$ 3., suppose that σ is a central extension. There exists p such that $p^*(\sigma)$ is a trivial extension and, by Proposition 3.2.2, $p^*(\sigma)$ is a normal extension. Applying 2. $\iff$ 3., we get that $p^*(\sigma)$ is a special homogeneous surjection, but since p is a surjective homomorphism of monoids, σ is a special homogeneous surjection by Proposition 3.3.6. $\square$

With the last theorem, we can apply the categorical Galois theorem whenever we fix a special homogeneous surjection σ . Recall that in the considered Galois structure for monoids, every extension has a monadic pullback functor, being a surjection. The next step is to characterize the Galois groupoid $Gal[\sigma]$ of a special homogeneous surjection $\sigma : M \twoheadrightarrow N$. We defined $Gal[\sigma]$ as the image of the groupoid $K(\sigma)$ induced by the kernel pair of σ through the functor $Gp : \mathbf{Mon} \rightarrow \mathbf{Grp}$. In Proposition 3.3.7, we gave an isomorphic structure of this kernel pair as a semidirect product: we can extend it to the induced groupoid $K(\sigma)$.

Corollary 3.3.10.1. *Consider a special homogeneous surjection $\sigma : M \twoheadrightarrow N$ and the groupoid $K(\sigma)$ in $\mathbf{Mon}$ described in Example 1.2.7; then it is isomorphic to*

$$(K \rtimes_{\varphi} M) \times_M (K \rtimes_{\varphi} M) \xrightarrow{m} K \rtimes_{\varphi} M \begin{array}{c} \xleftarrow{\pi_M} \\ \xrightarrow{\tau} \end{array} M$$

where $(K \rtimes_{\varphi} M) \times_M (K \rtimes_{\varphi} M)$ is the pullback of $\bullet$ along π_M , $m(k_1, m_2, k_2, m_3) = (k_1 \cdot k_2, m_3)$ and $\tau(k_1, m_2) = (k_1^{-1}, k_1 \cdot m_2)$.

Proof. By Proposition 3.3.7, it suffices to prove the actual form of the maps m and τ . Recall that the isomorphism $\zeta : M \times_N M \rightarrow K \rtimes_{\varphi} M$ is given by

$$\zeta : (m_1, m_2) \mapsto (k_1, m_2) \text{ such that } k_1 \cdot m_2 = m_1$$

and the groupoid multiplication in $K(\sigma)$ was given by the projection onto the first and the fourth factor. An element of $(K \rtimes_{\varphi} M) \times_M (K \rtimes_{\varphi} M)$ is given by (k_1, m_2, k_2, m_3) such that $m_2 = k_2 \cdot m_3$; so that

$$m(k_1, m_2, k_2, m_3) = m(\zeta(m_1, m_2), \zeta(m_2, m_3)) = \zeta(m_1, m_3) = (k_1 \cdot k_2, m_3)$$

since $k_1 \cdot k_2 \cdot m_3 = k_1 \cdot m_2 = m_1$. Analogously:

$$\tau(k_1, m_2) = \tau(\zeta(m_1, m_2)) = \zeta(m_2, m_1) = (k_1^{-1}, m_1) = (k_1^{-1}, k_1 \cdot m_2)$$

since K is a group and $k_1^{-1} \cdot m_1 = m_2$. □

Note that m and τ correspond to the multiplication and inverse on K . To fully describe the Galois groupoid $\text{Gal}[\sigma]$, it remains to apply the functor Gp . By Lemma 3.3.8 and Theorem 3.3.9 we get once again a semidirect form of $\text{Gal}[\sigma]$.

Proposition 3.3.11. *Consider $\sigma : M \rightarrow N$ a special homogeneous surjection and its Galois groupoid $\text{Gal}[\sigma] := Gp(K(\sigma))$ in **Grp** described in Definition 1.3.6; then it is isomorphic to*

$$\begin{array}{ccc} & & \xrightarrow{\pi_{Gp(M)}} \\ (K \rtimes_{\overline{\varphi}} Gp(M)) \times_{Gp(M)} (K \rtimes_{\overline{\varphi}} Gp(M)) & \xrightarrow{m} & K \rtimes_{\overline{\varphi}} Gp(M) \leftarrow Gp(M) \\ & \searrow \tau & \nearrow \bullet \end{array}$$

where $\overline{\varphi} : Gp(M) \rightarrow \text{Aut}(K)$ is the unique homomorphism satisfying $\varphi = \overline{\varphi} \circ \eta_M$ and $\bullet$ is the multiplication on $Gp(M)$, viewing K as a subset of $Gp(M)$ through η_K . Moreover, m and τ are given in the same way of Corollary 3.3.10.1.

Proof. First, note that the proof of Lemma 1.3.6 holds since σ is a normal extension, so that $Gp(K(\sigma))$ is indeed a groupoid. Since σ is a special homogeneous surjection, Lemma 3.3.8 holds and there is the isomorphism of split epimorphisms

$$Gp(K \rtimes_{\varphi} M) \cong K \rtimes_{\overline{\varphi}} Gp(M)$$

described in the proof of Theorem 3.3.9. It remains to explicitly exhibit the morphisms involved. Recall that $\overline{\eta}_M : K \rtimes_{\varphi} M \rightarrow K \rtimes_{\overline{\varphi}} Gp(M)$ is given by $1_K \times \eta_M$. Since they are group homomorphisms, it suffices to show how they act on elements of the form $(k_1, \overline{[m_2]}) = \overline{\eta}_M(k_1, m_2)$ for some $k_1 \in K$ and $m_2 \in M$, and we will do it exploiting the expressions from Corollary 3.3.10.1: one has

$$\bullet(k_1, \overline{[m_2]}) = \eta_M(k_1 \cdot m_2) = \overline{[k_1 \cdot m_2]} = \overline{[k_1]} \cdot \overline{[m_2]}$$

and

$$\tau(k_1, \overline{[m_2]}) = \overline{\eta}_M(k_1^{-1}, k_1 \cdot m_2) = (k_1^{-1}, \overline{[k_1 \cdot m_2]}) = (k_1^{-1}, \overline{[k_1]} \cdot \overline{[m_2]}).$$

Analogously, if $\overline{[m_2]} = \overline{[k_2]} \cdot \overline{[m_3]}$,

$$m(k_1, \overline{[m_2]}, k_2, \overline{[m_3]}) = \overline{\eta}_M(k_1 \cdot k_2, m_3) = (k_1 \cdot k_2, \overline{[m_3]}).$$

□

Observation. We proved that $Gal[\sigma]$ is a groupoid completely determined by the group K . Indeed, multiplication and inverse in the groupoid correspond to the structure on $Ker(\sigma)$.

To conclude, let us also describe an internal covariant presheaf (G, f, π) on the internal groupoid $Gal[\sigma]$ in **Grp**, with $f : G \rightarrowtail Gp(M)$ surjection.

Proposition 3.3.12. *Consider a special homogeneous surjection $\sigma : M \rightarrowtail N$ and its Galois groupoid $Gal[\sigma]$ in **Grp**. Fix an internal covariant presheaf (G, f, π) on $Gal[\sigma]$; then π is a group action of K on G that is compatible with the morphism $f : G \rightarrowtail Gp(M)$.*

Proof. Recall Definition 1.2.3 and consider the form of $Gal[\sigma]$ we just described in Proposition 3.3.11. First, note that K acts on $Gp(M)$ by the multiplication $\bullet : (k_1, x) \rightarrow [k_1] \cdot x$. Let us prove that

$$(K \rtimes_{\overline{\varphi}} Gp(M)) \times_{Gp(M)} G \cong K \rtimes_{\overline{\varphi} \circ f} G.$$

Indeed, $(K \rtimes_{\overline{\varphi}} Gp(M)) \times_{Gp(M)} G$ is defined as the pullback of f along $\pi_{Gp(M)}$, and for any $c, c' \in K$, $g, g' \in G$ such that $f(g) = x \in Gp(M)$ and $f(g') = x' \in Gp(M)$, it holds

$$(c, x, g)(c', x', g') = (c \cdot \overline{\varphi} \circ f(g)(c'), x \cdot x', g \cdot g').$$

With this isomorphism, $\pi : (K \rtimes_{\overline{\varphi}} Gp(M)) \times_{Gp(M)} G \rightarrow G$ is isomorphic to a map, that we also call π ,

$$\pi : K \rtimes_{\overline{\varphi} \circ f} G \longrightarrow G.$$

It is just a computation to show that the commutativity of Diagram 1.1 corresponds to the axioms for π to be a group action that is compatible with f :

$$\pi(1_M, g) = g,$$

$$\pi(k_1, \pi(k_2, g)) = \pi(k_1 \cdot k_2, g)$$

and

$$f \circ \pi(k_1, g) = \overline{[k_1]} \cdot f(g).$$

□

Observation. Note that we did something similar in Proposition 2.2.5. In that setting the Galois groupoid was a group; in this scenario, it remains a groupoid, but its structure is such that we obtain a similar result, being completely determined by a group.

To conclude, let us recall the work we have done in this chapter on the Galois structure

$$\Gamma = (\mathbf{Mon}, \mathbf{Grp}, Gp, U, \eta, 1_{\mathbf{Grp}}, \mathcal{E}, \mathcal{F}).$$

We proved that it is admissible in Proposition 3.1.1. In Theorem 3.3.10, we proved that the Categorical Galois Theorem holds if $\sigma : M \rightarrowtail N$ is a special homogeneous surjection, establishing an equivalence of categories

$$Centr(N, \sigma) \approx \mathcal{F}^{Gal[\sigma]}$$

between central extensions split by σ and internal covariant presheaves (G, f, π) on $Gal[\sigma]$ with $f \in \mathcal{F}$. Again in Theorem 3.3.10, we proved that central extensions are also normal extensions, and equivalently special homogeneous surjection. Finally, we explicitly described the Galois groupoid $Gal[\sigma]$ of σ in Proposition 3.3.11, and the internal covariant presheaves on it in Proposition 3.3.12. We proved that $Gal[\sigma]$ is completely determined by the kernel $Ker(\sigma)$ and that internal covariant presheaves on $Gal[\sigma]$ are pointed groups G on $Gp(M)$, along with a compatible action of $Ker[\sigma]$ on G .

Appendix A

Further details

A.1 Stone duality

Lemma A.1.1 ([2], 4.1.8 - 4.1.11). *Let B be a Boolean algebra. Consider, for every $b \in B$, the subsets*

$$U_b = \{F \in \text{Spec}(B) : b \in F\}$$

in $\text{Spec}(B) := \{F \subset B : F \text{ is a ultrafilter in } B\}$. Then $\{U_b\}_{b \in B}$ constitutes a basis of clopens for a topology $\mathcal{T}$ on $\text{Spec}(B)$. Moreover, the space $(\text{Spec}(B), \mathcal{T})$ is a profinite space.

Proof. We need to show that

1. for every $F \in \text{Spec}(B)$, there exists $b \in B$ such that $F \in U_b$;
2. for every $b, b' \in B$ and $F \in U_b \cap U_{b'}$, there exists $y \in B$ such that $F \in U_y \subset U_b \cap U_{b'}$;
3. for every $b \in B$, there exists $b' \in B$ such that $U_b^c = U_{b'}$.

Fix $F \in \text{Spec}(B)$; clearly $F \in U_b$ for every $b \in F$, and an (ultra)filter is never an empty set. Now, if $F \in U_b \cap U_{b'}$, then $b, b' \in F$, so that $b \wedge b' \in F$ and $F \in U_{b \wedge b'}$. It can be easily shown that $U_{b \wedge b'} = U_b \cap U_{b'}$. In this way $\{U_b\}_{b \in B}$ is a basis for a topology on $\text{Spec}(B)$. Moreover, if $b \notin F$, then $b^c \in F$, obtaining

$$U_b^c = \{F \in \text{Spec}(B) : b \notin F\} = \{F \in \text{Spec}(B) : b^c \in F\} = U_{b^c},$$

in such a way that every U_b is also closed.

Finally, for $\text{Spec}(B)$ to be a profinite space, we need to show that it is a Hausdorff compact space, since we have already shown that it has a basis of clopens. If $F \neq F'$, the maximality of F implies that $F \not\subset F'$. Choosing $b \in F \setminus F'$, we get $F \in U_b$ and $F' \notin U_b$. It remains to prove that $\text{Spec}(B)$ is compact. Consider $\text{Spec}(B) = \bigcup_{a \in A} U_a$ where $A \subset B$ and suppose we cannot extract a finite covering. Then for every finite sequence $a_1, \dots, a_n \in A$, there exists an ultrafilter F such that

$$F \notin U_{a_1} \cup \dots \cup U_{a_n} = U_{a_1 \vee \dots \vee a_n}$$

or equivalently

$$a_1 \vee \dots \vee a_n \notin F$$

or again

$$(a_1 \vee \dots \vee a_n)^c = a_1^c \wedge \dots \wedge a_n^c \in F.$$

Since $0 \notin F$, for every finite list $a_1, \dots, a_n \in A$, we have $a_1^c \wedge \dots \wedge a_n^c \neq 0$. If we take

$$G = \{b \in B : \exists a_1, \dots, a_n \in A \ a_1^c \wedge \dots \wedge a_n^c \leq b\},$$

it is just a computation to show that it is a filter and it is proper since $0 \notin G$. In particular, the set $\{a^c : a \in A\}$ is contained in G , so that it is also contained in the ultrafilter F that contains G (it exists by Zorn's lemma). We got an ultrafilter F , such that for every $a \in A$, $a^c \in F$, so $a \notin F$ contradicting that $\{U_a\}_{a \in A}$ is a covering of $\text{Spec}(B)$ since $F \not\subseteq \bigcup_{a \in A} U_a$. $\square$

Lemma A.1.2 ([2], 4.1.12, 4.1.13). *Let B be a Boolean algebra. A subset $U \subset \text{Spec}(B)$ is a clopen if and only if has the form U_b for some $b \in B$. Moreover, B is isomorphic to the set of clopens in $\text{Spec}(B)$.*

Proof. Every U_b is a clopen. Conversely, if U is a clopen, it is closed in the compact space $\text{Spec}(B)$, so it is compact. If $U = \bigcup_{i \in I} U_{b_i}$, then

$$U = U_{b_1} \cup \dots \cup U_{b_n} = U_{b_1 \vee \dots \vee b_n}.$$

The isomorphism of Boolean algebras is given by the map $b \mapsto U_b$. $\square$

Lemma A.1.3 ([2], 4.1.15). *Let X be a profinite space. The set of clopens in X is a Boolean algebra with the operations of intersection, union and complement. Moreover, X is homeomorphic to the spectrum of the Boolean algebra of its clopens.*

Proof. Trivially $\text{Clopen}(X)$ is a Boolean algebra. Moreover, given $x \in X$, we put

$$F_x = \{U \subset X : U \text{ clopen}, x \in U\}.$$

It is trivially an ultrafilter in the Boolean algebra $\text{Clopen}(X)$. We shall prove

$$\varphi : X \longrightarrow \text{Spec}(\text{Clopen}(X)), x \longmapsto F_x$$

to be a homeomorphism. If $x \neq y$, there exists a clopen U such that $x \in U$ and $y \notin U$. Thus, $U \in F_x$ and $U \notin F_y$, from which $F_x \neq F_y$ and φ is injective. If $F \subset \text{Clopen}(X)$ is an ultrafilter, consider

$$V = \bigcap_{U \in F} U.$$

Since F is a filter, a finite intersection is never empty, and since X is compact, V is non empty. Fix $x \in V$, then $x \in U$ for every $U \in F$, so that $F \subset F_x$. By maximality, $F = F_x = \varphi(x)$ and φ is surjective. An element of the basis of $\text{Spec}(\text{Clopen}(X))$ has the form U_W for some $W \in \text{Clopen}(X)$, so that

$$\varphi^{-1}(U_W) = \{x \in X : W \in F_x\} = \{x \in X : x \in W\} = W,$$

and φ is continuous. Since φ is a continuous bijection between compact Hausdorff spaces, it is an homeomorphism. $\square$

Theorem A.1.4 ([2], 4.1.16). *The functors*

$$\text{Spec} : \mathbf{Bool}^{\text{op}} \longrightarrow \mathbf{Prof}, B \mapsto \text{Spec}(B)$$

and

$$\text{Clopen} : \mathbf{Prof} \longrightarrow \mathbf{Bool}^{\text{op}}, X \mapsto \text{Clopen}(X)$$

form an equivalence of categories.

Proof. If $f : B \rightarrow B'$ is a homomorphism of Boolean algebras, we put

$$\text{Spec}(f) : \text{Spec}(B') \longrightarrow \text{Spec}(B), F \longmapsto f^{-1}(F).$$

This map is continuous since

$$\begin{aligned} \text{Spec}(f)^{-1}(U_b) &= \{F \in \text{Spec}(B') : \text{Spec}(f)(F) \in U_b\} \\ &= \{F \in \text{Spec}(B') : b \in \text{Spec}(f)(F)\} \\ &= \{F \in \text{Spec}(B') : b \in f^{-1}(F)\} \\ &= \{F \in \text{Spec}(B') : f(b) \in (F)\} = U_{f(b)} \end{aligned}$$

for $U_b \subset \text{Spec}(B)$. This defines the functor $\text{Spec} : \mathbf{Bool}^{\text{op}} \longrightarrow \mathbf{Prof}$.

On the other hand, if $g : X \rightarrow X'$ is a continuous map between profinite spaces, we put

$$\text{Clopen}(g) : \text{Clopen}(X') \longrightarrow \text{Clopen}(X), U \longmapsto g^{-1}(U)$$

that is a homomorphism of Boolean algebras. This defines the functor $\text{Clopen} : \mathbf{Prof} \longrightarrow \mathbf{Bool}^{\text{op}}$. We have already seen that the functors are mutually inverse up to isomorphism at the level of objects. Moreover, if $f : B \rightarrow B'$,

$$(\text{Clopen} \circ \text{Spec})(f) : \text{Clopen}(\text{Spec}(B)) \longrightarrow \text{Clopen}(\text{Spec}(B'))$$

is such that

$$U_b \longmapsto \text{Spec}(f)^{-1}(U_b) = U_{f(b)}.$$

If $g : X \rightarrow X'$,

$$(\text{Spec} \circ \text{Clopen})(g) : \text{Spec}(\text{Clopen}(X)) \longrightarrow \text{Spec}(\text{Clopen}(X'))$$

is such that

$$F_x \longmapsto \{W \in \text{Clopen}(X') : g^{-1}(W) \in F_x\} = F_{g(x)}.$$

□

A.2 Admissibility of the Galois structure for rings

Definition A.2.1. Let S be a ring. An ideal $I \subset S$ is regular when it is generated by its idempotent elements. Thus, when every $i \in I$ can be written

$$i = s_1 e_1 + \cdots + s_n e_n$$

with $s_i \in S$, $e_i \in I$ and $e_i^2 = e_i$.

Definition A.2.2. Let B be a Boolean algebra. A proper ideal in B is a subset $I \subset B$ such that

1. $0 \in I$;
2. $x \in I$ and $y \in I$, then $x \vee y \in I$;
3. $x \in I$ and $x \geq y$, then $y \in I$;
4. $1 \notin I$;

An maximal ideal in B is a maximal element in the poset of proper ideals in B ordered by inclusion.

Lemma A.2.1. *Let B be a Boolean algebra. The poset of filters in B is isomorphic to the poset of ideals in B . The poset of filters in B is isomorphic to the poset of open subsets of $\text{Spec}(B)$.*

Proof. The first isomorphism follows directly by Definition 2.1.2 and Definition A.2.2. The correspondence is given by

$$F \mapsto I := \{b \in B : b^c \in F\}$$

and

$$I \mapsto F := \{b \in B : b^c \in I\}.$$

Moreover, note that the basis $\{U_b\}_{b \in B}$ of clopens we defined in Lemma A.1.1 is the same of $\{\mathcal{O}_b\}_{b \in B}$, where $\mathcal{O}_b := U_b^c = U_{b^c}$. Its topology $\mathcal{T}$ is given by the subsets

$$\mathcal{O}_H := \{F \in \text{Spec}(B) : H \not\subseteq F\}$$

for every filter H in B . Indeed, for every filter $H \subseteq B$,

$$H = \bigcup_{b \in H} \uparrow b$$

where $\uparrow b$ is the filter in B given by $\{y \in B : b \leq y\}$; then

$$\mathcal{O}_H = \bigcup_{b \in H} \mathcal{O}_{\uparrow b} = \bigcup_{b \in H} \mathcal{O}_b.$$

Finally, the map

$$\text{Filter}(B) \longrightarrow \mathcal{T}, H \mapsto \mathcal{O}_H$$

is surjective by definition and injective since every filter is the intersection of the ultrafilters which contain it. It is just a computation to show that

$$H \subseteq H' \iff \mathcal{O}_H \subseteq \mathcal{O}_{H'}.$$

□

Lemma A.2.2 ([2], 4.2.10, 4.2.11). *Let S be a ring. The Pierce spectrum $\text{Sp}(S)$ of S is homeomorphic to the set of maximal regular ideals of S , provided with the topology constituted of the subsets*

$$\mathcal{O}_I = \{M : I \not\subseteq M\}$$

for every regular ideal $I \subset S$.

Proof. By Lemma A.2.1, it suffices to prove that the poset of ideals of $\text{Idemp}(S)$ is isomorphic to the poset of regular ideals of S . Fix a regular ideal $I \subset S$, then $\text{Idemp}(I)$ is clearly an ideal of the Boolean algebra $\text{Idemp}(S)$. By definition of regular ideal, distinct regular ideals have different idempotents, proving the injectivity of the correspondence. Moreover, if J is an ideal of the Boolean algebra $\text{Idemp}(S)$, then we can define I as the regular ideal generated by all elements of J ; clearly $J \subseteq \text{Idemp}(I)$. Conversely, if $e \in \text{Idemp}(I)$, then $e = \sum_{i=1}^n s_i e_i$ with $s_i \in S$ and $e_i \in J$. Since J is an ideal, $e' := e_1 \vee \dots \vee e_n \in J$ and $e_i e' = e_i \wedge e = e_i$. We obtain $e \in J$ since $e = e e' = e \wedge e'$, that is $e \leq e'$. Finally, it is obvious that the correspondence $I \mapsto \text{Idemp}(I)$ preserves and reflects the ordering. □

We now consider $Sp(S)$ as the set of maximal regular ideals of S with the topology consisting of subsets $\mathcal{O}_I$ for every regular ideal $I \subset S$ and the basis given by the subsets $\mathcal{O}_e = \{M : e \notin M\}$ for every idempotent $e \in S$.

Definition A.2.3. Let S be a ring. The Pierce structural space of a ring is the disjoint union $\coprod_M S/M$ of all quotients S/M , where M runs through all the maximal regular ideals of S ; this set is provided with the final topology for all maps

$$p_s^I : \mathcal{O}_I \longrightarrow \coprod_M S/M, N \longmapsto [s] \in S/N$$

where I runs through the regular ideals of S and s through the elements of S .

Lemma A.2.3 ([2], 4.2.16). *Given a maximal regular ideal M of S , the quotient ring S/M admits only 0 and 1 as idempotents.*

Proof. The ring S/M is the filtered colimit of the rings $S/\langle e_1, \dots, e_n \rangle$, where $\langle e_1, \dots, e_n \rangle$ is the ideal generated by some idempotents $e_i \in M$. Fix $s \in S$ such that $[s] \in S/M$ is an idempotent; then it is an idempotent in a quotient $S/\langle e_1, \dots, e_n \rangle$. Clearly, if $s \in M$, then $[s] = 0$. Suppose $s \notin M$ and fix $e' := e_1 \vee \dots \vee e_n$ and observe that $S/\langle e_1, \dots, e_n \rangle \cong S/\langle e' \rangle \cong \langle 1 - e' \rangle$. Indeed,

$$S = \langle 1 - e' \rangle \oplus \langle e' \rangle \text{ and } \langle 1 - e' \rangle \cap \langle e' \rangle = 0.$$

With the last isomorphism, there exists an element $e \in \langle 1 - e' \rangle$ such that $e^2 = e$ and $[e] = [s]$. Since $s \notin M$, $e \notin M$; then, $1 - e \in M$ by maximality. Thus, $[e] = [s] = 1$. $\square$

Theorem A.2.4 ([2], 4.3.6). *Let S be a ring. The right adjoint of the functor*

$$Sp_S : \mathbf{S-Alg}^{\text{op}} \longrightarrow \mathbf{Prof}/Sp(S), A \longmapsto (Sp(A) \rightarrow Sp(S))$$

is the functor

$$C_S : \mathbf{Prof}/Sp(S) \longrightarrow \mathbf{S-Alg}^{\text{op}}, (X, f) \longmapsto \text{Hom}((X, f), (\coprod_M S/M, p))$$

where the *Hom*-set is provided pointwise with the structure of S -algebra, via the structure of S -algebra on each fibre S/M and p is the projection of the Pierce structural space of S given by

$$p : \coprod_M S/M \longrightarrow Sp(S), [s] \in S/N \longmapsto N.$$

This right adjoint is full and faithful.

Proof. To prove the adjointness property, fix an S -algebra A ; we must prove the existence of natural bijections

$$\text{Hom}(\text{Hom}((X, f), (\coprod_M S/M, p)), A) \cong \text{Hom}((Sp(A), \alpha), (X, f))$$

where $\alpha : Sp(A) \rightarrow Sp(S)$ is the canonical morphism corresponding, by the Stone duality, to the morphism

$$\text{Idemp}(S) \longrightarrow \text{Idemp}(A), e \longmapsto e \cdot 1$$

of boolean algebras.

Given $\varphi : \text{Hom}((X, f), (\coprod_M S/M, p)) \longrightarrow A$ and a clopen $U \subseteq X$, we consider the map

$$\chi_U : X \rightarrow \coprod_M S/M, \chi_U(x) = \begin{cases} [1] \in S/f(x) & \text{if } x \in U \\ [0] \in S/f(x) & \text{if } x \notin U. \end{cases}$$

The map χ_U is idempotent and $\varphi(\chi_U)$ is an idempotent of A . This yields a homomorphism

$$\varphi' : \text{Clopen}(X) \longrightarrow \text{Idemp}(A), U \longmapsto \varphi(\chi_U).$$

This corresponds, by Stone duality, to a continuous map $\varphi'' : Sp(A) \longrightarrow X$. To prove $f \circ \varphi'' = \alpha$, it suffices to prove

$$\varphi'(f^{-1}(\mathcal{O}_{\langle e \rangle})) = \varphi(\chi_{f^{-1}(\mathcal{O}_{\langle e \rangle})}) = e \cdot 1$$

for every idempotent $e \in S$. But, for every $x \in X$,

$$e \in f(x) \iff x \notin f^{-1}(\mathcal{O}_{\langle e \rangle})$$

so that $\chi_{f^{-1}(\mathcal{O}_{\langle e \rangle})}(x) = [e] = e \cdot [1]$. Since φ is a homomorphism of S -algebras, we obtain the desired equality. Clearly, the correspondence $\varphi \longmapsto \varphi''$ is natural.

Observe that given $h : (X, f) \longrightarrow (\coprod_M S/M, p)$, the subsets $h^{-1}(p_s^{Sp(S)}(Sp(S)))$, for $s \in S$, constitute an open covering of X . Since X has a base of clopens and is compact, we can extract a finite covering of clopens. This yields clopens $U_1, \dots, U_n \subseteq X$ and elements $s_1, \dots, s_n \in S$ such that

$$h(x) = [s_i] \in S/f(x)$$

for every $x \in U_i$ and $i = 1, \dots, n$. This proves that

$$h = \sum_{i=1}^n s_i \cdot \chi_{U_i}$$

for some $s_i \in S$ and clopens U_i that are a partition of X .

Let us prove the injectivity of the correspondence $\varphi \longmapsto \varphi''$. Consider two morphisms φ and ψ of S -algebras as before, such that $\varphi'' = \psi''$. By Stone duality, $\varphi' = \psi'$ and by definition we obtain $\varphi(\chi_U) = \psi(\chi_U)$, for each clopen $U \subseteq X$. By the previous observation $\varphi(h) = \psi(h)$ for every $h \in \text{Hom}((X, f), (\coprod_M S/M, p))$.

Let us prove the surjectivity of the correspondence $\varphi \longmapsto \varphi''$. Consider $g : (Sp(A), \alpha) \longrightarrow (X, f)$ and, by Stone duality, its corresponding $\bar{g} : \text{Clopen}(X) \rightarrow \text{Idemp}(A)$. We need to define $\varphi(h) \in A$ for each $h \in \text{Hom}((X, f), (\coprod_M S/M, p))$, such that $\varphi'' = g$. By the previous observation, it is sufficient to define φ on the maps χ_U , for each clopen $U \subseteq X$; putting $\varphi(\chi_U) := \bar{g}(U)$, we obtain $\varphi' = \bar{g}$ and, by Stone duality, $\varphi'' = g$.

Finally, let us prove that this right adjoint is full and faithful. $h : (X, f) \longrightarrow (\coprod_M S/M, p)$ is idempotent if and only if $h(x)$ is idempotent in $S/f(x)$, for every $x \in X$. By Lemma A.2.3, the idempotents of $S/f(x)$ are only $[0]$ and $[1]$ so that, by the previous observation, we obtain that $h = \chi_U$, for a clopen $U \subseteq X$. By Stone duality,

$$Sp(\text{Hom}((X, f), (\coprod_M S/M, p))) \cong X.$$

□

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